A Contemporary Guide to Wargame Design

Ray Weiss

2026

Contents

A Contemporary Guide to

Wargame Design

Ray Weiss

A *Contemporary Guide to Wargame Design*

ISBN: 9798195268275

First edition.

The web edition of this book is available free at `lerugray.github.io/wargame-design-book/`.

Acknowledgements

No one designs games alone, and no one publishes them alone either. The people listed here gave me their time, their patience, and their honesty when I needed all three.

Kevin Zucker taught me how to think about operational design. He answered questions I did not yet know how to ask well, and he kept answering them long after most people would have stopped. Everything in this book about the relationship between logistics and gameplay traces back to conversations with Kevin.

Hermann Luttmann sat down at a convention and played one of my terrible early designs. He was kind about it. He did not have to be. That kind of generosity from an established designer to someone just starting out matters more than any review or award, and I have tried to pay it forward when new designers bring their work to me.

Matt Ward developed more of my published titles than I can list here without turning this into a credits page. A designer can have all the ideas in the world, but without a developer willing to grind through the details, those ideas stay on paper. Matt did the grinding.

Dean Essig gave me the opportunity to design Rostov '41 for the Standard Combat Series at Multi-Man Publishing. That project opened doors I could not have opened on my own. His series rules concept changed how I thought about game design, and his influence runs through most of the chapters in this book.

Ivan Caceres created the initial art for my games and helped establish the visual identity of Conflict Simulations LLC when I was still figuring out what that meant.

Ilya Kudriashov has contributed work across multiple titles and brought a level of craft that made my games look better than they had any right to.

Ryan Heilman formatted most of my rulebooks and has been a steady, reliable presence behind the scenes on more projects than either of us probably remembers.

Trevor Henderson worked on my early manuals when the whole operation was held together with enthusiasm and not much else.

The team at Superior POD printed my games and put up with my learning curve as a self-publisher.

I am forgetting people. I know I am. My memory is terrible and there have been plenty of others who were kind and helpful to me when I did not deserve it. If you are one of them, I owe you a drink and an apology for the oversight. Thank you.

A Note on This Printed Edition

The diagrams and illustrations referenced throughout this book — hex scale comparisons, counter anatomy, zone of control geometry, combat results tables, sequence of play flowcharts, and other visual examples — are available in the free online edition at **lerugray.github.io/wargame-design-book**. They have been omitted from this print version to keep the physical book affordable and readable. The text stands on its own; the diagrams are there when you want them.

Why Design Wargames?

For the entirety of human history, war has been humanity's greatest and most urgent crisis. From the earliest civilizations, people have built abstract systems to distill warfare down to a set of conventions that could be reproduced and studied. Chess is quite literally a simulation of medieval-era tactical combat, with pieces representing the arms available at the time. Go is an older and more abstract attempt at the same impulse, focused on encirclement, territory, and strategic planning rather than individual engagements. Both are wargames. They may be disconnected from us in the modern era, but the intent behind them was martial simulation, and that intent is what matters.

This raises a question that wargamers have been arguing about for decades: what exactly is a wargame? I have heard compelling definitions from dozens of designers and none of them agree. My own position is close to the famous line about obscenity: I know it when I see it. But if pressed, I would say a wargame is any simulation of a struggle between two or more sides engaged in martial or existential conflict, defined as such by its designer. That last part is critical. A game cannot be a wargame if the designer did not intend it as one. The definition belongs to the creator.

This means the category is broader than people assume. *Twilight Struggle* is a wargame. *Root* is a wargame. It is essentially a COIN system with animals standing in for historical factions.

The *Undaunted* series is a wargame. *Advanced Squad Leader* is a wargame. *Campaign for North Africa*, where you track individual water rations for Italian soldiers, is a wargame. All of these sit on a spectrum of complexity and abstraction, but they share the same fundamental purpose: modeling conflict in a way that gives players agency within historically or thematically grounded constraints.

What makes wargames unique among strategy games is the narrative they produce. Any game can be strategic. A Eurogame can demand hard optimization decisions. But wargames offer a kind of operational freedom that generates stories. Not scripted narratives handed to the player, but emergent ones built from the interaction of historically plausible forces, terrain, logistics, and human decision-making. When you play a good wargame, you are not just solving a puzzle. You are participating in something that feels like history, and the decisions you make create a narrative that is believable because the simulation underneath is honest about its constraints.

ASL is probably the most vivid example of this. It is a gameified version of World War II as it existed in the movies. Tactical, dramatic, full of small-unit heroics and desperate last stands. But it works because under all that drama there is a simulation engine that respects the reality of infantry combat. The stories it generates feel earned. That blend of player agency and historical plausibility producing a coherent narrative is what separates wargames from other forms of strategy gaming.

This matters beyond entertainment. After the Prussian defeat at Jena during the Napoleonic Wars, the Prussian general staff adopted Kriegsspiel, a series of sophisticated exercises in which officers role-played forces on a game board, wrote orders for a neutral referee, and resolved engagements with dice. The insights from those exercises contributed directly to the Prussian military reforms that would destroy the field armies of the French Empire within weeks during the opening campaign of the Franco-Prussian War. Today, wargame designers have been utilized by the CIA, the Pentagon, major corporations, and NATO-member militaries as a means of understanding complex strategic environments and empowering participants to develop critical thinking skills. One of my own designs, 2022: *Ukraine*, was adopted by a NATO-member military for training exercises. I mention this not to boast but because it says something about the medium itself. A small commercial wargame, designed by one person in Brooklyn, produced insights that professional analysts found useful enough to put in front of soldiers.

The reason is straightforward. War is a system. It has inputs, constraints, feedback loops, and outcomes shaped by decisions made under uncertainty. A wargame forces you to engage with that system on its own terms. You cannot play a simulation of the Eastern Front without confronting the logistical realities that determined its outcome. You cannot simulate the Franco-Prussian War without grappling with the fact that Prussian officers could act on initiative while French officers waited for orders from an emperor suffering from gout. I have compared Russian institutional failures in East Prussia in 1914 to the same failures in Ukraine in 2022. The same organizational dysfunctions, the same command rigidity, the same inability to adapt, separated by 108 years. Wargames are one of the few tools that make those patterns visible and visceral rather than academic.

The State of the Hobby

We arguably live in the golden age of wargaming, though it remains a niche hobby. The worldwide community of active wargamers is probably in the tens of thousands. Maybe more if you count the people who buy games and never play them, which is more common than anyone wants to admit. But compared to twenty or thirty years ago, the hobby is more accessible than it has ever been. VASSAL allows anyone with an internet connection to play almost any published wargame online. Board Game Geek provides a centralized community where you can find opponents, read reviews, and discover new designs. Print-on-demand services mean a designer no longer needs a five-figure print run to get a game into players' hands.

At the same time, wargaming remains a hobby for a specific kind of person. The games are expensive. The rules are long. The time investment is substantial. You are not going to impulse-buy a $90 boxed wargame the way you might grab a party game at Target. The people who buy wargames tend to be either collectors with disposable income or obsessives who find the combination of history and strategic decision-making irresistible. I am firmly in the second camp.

The hobby has also been slowly expanding into the broader world of board gaming. *Twilight Struggle* brought Eurogamers into the fold by packaging Cold War grand strategy in a card-driven system that felt familiar to people coming from *Dominion* or *Pandemic*. *Root* did the same thing by disguising a counterinsurgency simulation as an adorable woodland adventure game. These crossover successes matter because they show that the core appeal of wargaming, strategic depth grounded in thematic coherence, is not limited to grognards. The audience is wider than the hobby's reputation suggests.

For aspiring designers, this means the barriers to entry are lower than they have ever been. You do not need a publishing contract to get a game in front of players. You do not need to attend conventions to find playtesters. You do need a genuine interest in the subject you want to simulate, a willingness to do the research, and enough stubbornness to push through the parts that are tedious rather than fun. Wargame design is an academic exercise as much as a creative one, and you need some kind of drive to put yourself through it. But if you have that drive, the tools and community to support you exist right now.

Who This Book Is For

I am trying to write the book I wish I had when I was just starting out.

When I began designing wargames, I had played a fair number of them but had no idea how to build one. I knew what I liked in the games I played, but I could not reverse-engineer the decisions that produced those qualities. Why did one combat results table feel right and another feel arbitrary? Why did some games capture the character of a conflict in thirty counters while others failed to do so in three hundred? How did experienced designers decide on hex scale, turn length, movement factors? These are not questions you can answer by playing more games. You answer them by designing games, making mistakes, and paying attention to why things did or did not work.

I have been designing games for over a decade now. I own and operate Conflict Simulations LLC, a small wargame publishing company out of Brooklyn. I have designed and published nearly thirty wargames covering conflicts from the German Peasants' War of 1525 through the Russo-Ukrainian War of 2022. My designs have been published by Multi-Man Publishing, one of the major publishers in the hobby, and three of my games were nominated for Charles S. Roberts Awards, which is the closest thing wargaming has to an Oscar. I have designed games at every scale from tactical to strategic, using hex-and-counter, area-control, point-to-point, and hybrid card-driven systems.

None of which is to say I have all the answers. Every new design teaches me something I did not know, and I still make mistakes that would have been obvious to a more experienced designer. But I have made enough mistakes across enough published designs to know which ones are common, which ones are avoidable, and which ones you just have to make yourself before the lesson sticks.

This book is for anyone who wants to design a wargame and does not know where to start. That includes hobbyist wargamers who have always wanted to try their hand at design. It includes Eurogamers and board game enthusiasts curious about the wargame side of the hobby. It includes people in professional wargaming, whether defense analysts, educators, or researchers, who want to understand how commercial wargames are constructed. And it includes designers who have already started a project and gotten stuck, because getting stuck is normal and most of the solutions are knowable.

What this book will not do is make you a historian. You need to bring the interest in your subject. You need to care enough about the conflict you want to simulate to do the research honestly. What this book will give you is the framework to turn that historical knowledge into a playable game, and the practical guidance to take that game from prototype through playtesting to publication.

Several pioneers of the hobby have written about wargame design before me, most notably Jim Dunnigan and the staff of SPI. But many of those books were written thirty or forty years ago and do not account for the innovations in game design, digital playtesting tools, and publishing options that exist today. The underlying principles of wargame design have not changed, but the practice of it has. I hope this book bridges that gap.

How I Got Here

I bought *Totaler Krieg* as one of my first wargames and was immediately intimidated by it. I was not ready for that level of complexity, and it sat on my shelf for a while before I found my way into the hobby through a different door.

What eventually hooked me were John Tiller's PC wargames, specifically the *First World War Campaign* series. These are giant battalion-level simulations that cover entire campaigns in granular detail, and they were my first experience of genuinely feeling immersed in history through a game. Moving battalions through the Argonne, watching a corps-level attack develop across multiple turns, seeing the operational picture emerge from hundreds of individual unit decisions. That was the moment I understood what wargames could do. Those games informed my design sensibility in ways I am still discovering. The operational scale, the attention to order of battle, the feeling that the player is a commander making real decisions within real constraints. All of that traces back to those Tiller campaigns.

My first self-published game was *1916*, a strategic simulation of the Battle of Verdun that I put together after months of playing old GDW titles. The mechanics drew from the SPI school of design. Straightforward odds-based combat, simple movement, the kind of system where the game gets out of the way and lets the situation speak. It was rough in ways that are obvious to me now, but it worked, and seeing people play something I had designed was addictive in a way I was not prepared for.

Since then I have designed games on topics ranging from ancient Marathon to the Prigozhin mutiny of 2023. I have built modular game systems that share rules across multiple titles. I have designed grand-tactical simulations of post-Napoleonic conflicts in the age of rifles, Cold War hypotheticals, and a hybrid card-driven game on the Great Northern War that deliberately breaks the hex-and-counter format. My game *Sedan 1940* is essentially an analog version of the *Panzer Campaigns* PC series, and it is dedicated to John Tiller's memory, because without his games I would never have started designing my own.

The purpose of this book is to open up the world of wargame design and shatter the assumption that designing wargames is excessively difficult or inaccessible. It is exactly as difficult as you make it. A small, focused game with a clear thesis about its conflict can be designed, playtested, and published by one person. I know this because I have done it nearly thirty times. Here is how.

Choosing Your Subject

It is hard to say where game ideas come from. Before I was a game designer I was a musician, and the impulses that drove me to write music work in much the same way for game design. If I had to articulate it honestly, it comes down to something uncomfortable: I am going to die at some point, and I need to make sure I leave something behind that matters. I am childless except for a cat and nearly forty, so art it is. That might sound dramatic for a chapter about choosing which battle to simulate, but I think it matters. The best wargames come from designers who have something they genuinely want to say, explore, or argue about a historical event. If you are designing a game because you think it will sell or because the topic seems popular, you are going to run into a wall the first time the project gets tedious. And every project gets tedious. You need to care enough about the subject to push through the monotonous parts, because there are a lot of them.

That said, almost any situation can be made into a game. Jim Dunnigan proved this decades ago when he turned *Outdoor Survival* into a commercial success. The question is not whether a conflict can be gamed but how much work the designer has to do to make it interesting.

Easy Subjects and Hard Subjects

Some conflicts are naturally easier to turn into games than others. *1916*, my first self-published game on Verdun, was an easy subject because at the strategic level it was basically a slugfest of attrition. Back and forth across a relatively static front line with limited attempts to maneuver. The parameters were clear. Two sides grinding each other down in a confined space with well-documented forces. The design decisions almost made themselves: attrition-based combat, positional play, limited movement options. The historical situation already had built-in game tension because both sides believed they could break through if they just committed one more corps.

We Were Not Cowards, my game on Sedan 1870, was much harder and took years to develop. It was one of my earliest ideas but probably far too complex for a first attempt. The fundamental problem was that the battle was basically a foregone conclusion before it started. The French army was already encircled by the time the fighting began in earnest. Where is the game in a situation where one side has already lost? Nobody wants to play the French player if the answer is "you lose, historically." I ended up adding an approach-to-battle variant, a what-if scenario that gives the French player a chance to react before they are encircled. That created the decision space the historical battle lacked. But getting there took years of iteration on a subject I thought I understood from the start.

The lesson is not that you should avoid hard subjects. The lesson is that you should understand what you are getting into. A battle where both sides had genuine strategic options and the outcome was uncertain is going to be easier to design than one where the result was predetermined. That does not mean predetermined battles are off limits. It means you need to know going in that you will probably have to introduce counterfactual elements or find tension in places the history does not obviously provide. If nobody else has made a game on a topic, sometimes there is a reason. And sometimes the reason is just that nobody thought to try, which was the case with Sedan. I designed it partly because no one else had, and the battle still had interesting elements worth exploring between the cavalry charges and the initial envelopment. But I would not recommend it as a first project.

Scope: The Most Important Decision You Will Make Early

The most important thing an amateur designer needs to consider before anything else is scope. This connects directly to scale, which we will cover in Chapter 4, but scope is the broader question: what are you trying to represent and how much of it?

A game about all of World War II is a fundamentally different project from a game about one day at Gettysburg. Both are valid. But the WWII game requires you to abstract entire campaigns into single die rolls, while the Gettysburg game lets you model individual brigade movements across specific terrain. The smaller the scope, the less legwork is required to get the simulation working. Scales with less counter density and more abstraction are inherently easier to represent with less effort. This does not make them lesser games. Some of the best wargames ever published are small, tightly scoped designs that do one thing well.

Design that follows good principles tends to produce better systems, and having a clear idea of what you want to achieve and what you need to represent in order to achieve it is critical. Without that clarity, you will find yourself adding mechanics to cover situations that do not matter, building chrome that nobody will use, and losing track of what your game is actually about. Scope is the fence that keeps you from wandering off into the wilderness.

For my early games, I deliberately limited myself to 140 counters and a 22 by 17 inch map. This was partly to emulate the old GDW 120 series, which I admired, but it was also a production constraint. I could save on counter costs and only needed half a countersheet per game. Whatever the original motivation, the constraint turned out to be one of the best decisions I made as a young designer. With only 140 counters I could not attempt massive grand-tactical or tactical simulations. The constraint inherently limited scope and kept my early designs within reason. I had to make hard choices about what to include and what to leave out, and those choices forced me to think about what actually mattered in the conflict I was simulating.

I called this the 2140 series. The name was a reminder to myself: two hours of play, 140 counters. That framework produced some of my best early work because it demanded efficiency. When you only have 140 counters to represent an entire campaign, every counter has to justify its existence. Every mechanic has to earn its place in the rules.

My first larger game was *We Were Not Cowards*, which I released about two years after starting the company. The jump in scope was significant and I would not have been ready for it without the discipline the 2140 constraint had taught me. Even *Barbarossa Deluxe*, my *Destroy All Monsters* trilogy, was technically part of the 2140 series. It featured three individual games, one for each army group, that could be combined into a single larger simulation. Breaking the project up that way helped limit the scope of each individual design and kept the whole thing manageable. Instead of designing one enormous Eastern Front game, I designed three focused games that happened to be compatible with each other. That is a useful trick for ambitious projects: break them into modules, get each module working on its own, and then worry about integration.

When to Walk Away from a Subject

Not every idea deserves to become a finished game. I have discontinued two of my published designs because they simply did not work. 1987: On to Kaliningrad was a hypothetical World War III scenario, and 1968: Tet covered the Vietnamese Tet Offensive. Both failed for the same basic reason: I did not know the subjects well enough to make good design decisions.

The assumptions I had about how combat worked, based on my general range of interests in nineteenth century and World War era conflicts, simply did not apply to Cold War mechanized warfare or the insurgent dynamics of Vietnam. The games ended up being more fantasy than simulation. The combat systems did not reflect how those wars actually worked, the force structures did not behave realistically, and the games were broken in ways that playtesting could not fix because the problems were foundational. You cannot playtest your way out of a flawed understanding of your subject.

Part of this was my own fault for trying to design too many games at once, which I was doing at the time to try to make a living from the company. Spreading myself across too many projects meant I was not doing the research depth any of them required. The 1987 game needed me to understand NATO and Warsaw Pact doctrine, force ratios in mechanized warfare, and the specific geography of the Baltic region in ways I had not internalized. The Tet game needed me to understand counterinsurgency, irregular warfare, and the political dimensions of the conflict in Vietnam, none of which I had spent serious time studying.

The lesson for the aspiring designer is twofold. First, you need to know your subject well enough that your design decisions are grounded in something real. If you are guessing at how a conflict worked, your game will feel like guesswork. Second, do not try to design five games at once. Each design deserves your full attention during the research and initial prototyping phase. There will be time to run multiple projects later, once you have more experience and a better sense of how much attention each stage of development actually requires.

This is not meant to scare you away from unfamiliar topics. Half the fun of wargame design is learning about conflicts you knew nothing about before you started. But there is a difference between learning a subject as you design and designing before you have learned enough. The research has to lead the design, not chase it.

Finding Your First Subject

If you have never designed a wargame before, here is my practical advice on choosing a subject:

Pick a conflict you already know well. Not one you want to learn about, one you already have opinions about. You should be able to explain without looking anything up why one side won and the other lost, what the key decisions were, and what would have happened if those decisions had gone differently. That level of familiarity gives you a head start on every design decision you will face.

Pick the right level of granularity. This is more important than picking a small conflict. A common piece of advice is to start with a single battle rather than a whole war, but I actually think that can steer new designers wrong. My own mistake with *We Were Not Cowards* was not that I picked a battle. It was that I picked a battle and tried to represent it at too granular a level. Had I instead covered the first year of the Franco-Prussian War at an operational or strategic scale, as I did with a later design, it would have been far easier.

A whole war can absolutely be a first project if you approach it at the right scale. *Unconditional Surrender* covers all of World War II in Europe with extremely low counter density. It is basically a far more playable and enjoyable version of *Totaler Krieg*, the game that intimidated me off my shelf when I was starting out. The old SPI World War I games managed to cover the entire Great War in tight, manageable packages. A strategic game on the Russo-Japanese War or one of the Balkan Wars would be perfectly doable for a first-time designer. What would not be doable is trying to cover the full Crimean War across all four theaters, the Baltic, the Crimea, Moldavia, and the Caucasus, at operational scale. That is a multi-year project for an experienced designer, not a starting point. The issue is not the size of the war. It is the level of detail you are trying to represent. This ties directly into scale, which we will cover in Chapter 4.

Pick something where both sides had real choices. The best wargame subjects are situations where the outcome was uncertain and the decisions of commanders mattered. A battle where one side was doomed from the start can be made into a game, as I learned with Sedan, but it takes experience to find the tension in a predetermined situation. For your first game, make it easy on yourself. Pick a fight where either side could plausibly win.

And pick something you care about. You are going to spend months with this subject. You are going to read the same accounts multiple times, count the same regiments, argue with yourself about movement rates and combat modifiers. If the subject does not hold your interest through all of that, you will abandon the project. That is not a failure of willpower. It is a subject selection problem, and it is avoidable.

Research

Research for wargame design falls into two phases that serve different purposes, and understanding the difference early will save you a lot of wasted effort.

The first phase is secondary sources. Books written by historians, campaign studies, analytical works that give you the big picture. This is where you develop a feel for the conflict. How were armies organized? What was the prevailing doctrine? How effective were the commanders and how did their leadership style affect operations? What role did terrain, weather, logistics, and politics play in the outcome? You are not looking for specific unit strengths at this stage. You are building a mental model of the conflict that will inform every design decision you make later.

The second phase is primary sources. Staff records, after-action reports, official histories compiled from original documents. This is where you get the granular data that turns into game content: orders of battle (OOBs), unit compositions, tables of equipment, march rates, supply situations, casualty figures. These details drive concrete decisions like strength ratings, movement factors, and combat mechanics. You cannot fake this phase. A game with made-up unit values might play fine as an abstract strategy game, but it will not be a simulation, and the people who buy wargames will notice.

The two phases overlap. You will find OOB details in secondary sources and strategic insights in primary documents. But keeping the distinction in your head helps you know what you are reading for at any given point. When you pick up a campaign history, you are absorbing context. When you open a volume of staff records, you are mining data.

Where to Find Sources

The obvious starting point is books, and the best cheat code for finding them is Anna's Archive, which has an effectively endless library of digitized texts available for anyone who needs them. I mention this without further comment except to say that I have found it invaluable for research and that access to sources should not depend on how much money you can spend at Amazon.

Beyond that, here are the source types I have found most useful across nearly thirty published designs:

Secondary sources for the big picture. Campaign histories, biographies of commanders, analytical military histories. These give you the feel. Barbara Tuchman's *The Guns of August* is the kind of book where a determined person could read it and design a strategic WWI game based on that single volume. You would need more sources for a detailed operational game, but for a strategic treatment it might be enough. The point is not a specific number of books. It is reading until you feel confident you could pass a pop quiz on the subject. If you have chosen a topic you actually care about, this part should not feel like homework.

Staff records and official histories for OOBs and detailed operational data. These are the gold standard for unit-level information. Government and military archives have published multi-volume official histories for most major conflicts, and many of them have been digitized. The German staff records for the Franco-Prussian

War, for example, gave me Prussian artillery ranges and rates of fire that became the Krupp Steel rule in *We Were Not Cowards*: +1 DRM within 2 hexes, with a 6-hex range against the French 4-hex. That single data point from a staff record shaped how the entire artillery asymmetry works in the game.

Nafziger orders of battle are sometimes an acceptable starting point for OOB research, but use them with caution. I have found them inaccurate about half the time, with errors in both unit strengths and organizational details. Formations get listed that were not actually present, strengths get recorded incorrectly, and the sourcing is not always transparent enough to check. They can point you in the right direction, but treat them as a first draft to be verified against better sources, not as gospel.

Miniatures rules and scenario books are an underrated research resource. The authors of miniatures scenarios tend to do extensive historical research to get unit compositions and strengths right, often citing sources that wargame designers overlook. If someone has published a detailed miniatures scenario on the battle you are designing, their OOB work is worth examining.

Other published wargames on the same subject. This is not cheating. Wargaming is eighty percent plagiarization anyway, unless you are Volko Ruhnke or Mark Herman. You do not even need to buy or play the games. Most publishers and designers put their rules online for free, and you can examine counter manifests and component lists on Board Game Geek. Looking at how another designer synthesized the same dataset is essential context. You will not know exactly what their sources were, but you can see the choices they made and evaluate whether those choices seem right based on your own research. Sometimes you will disagree with their interpretation, and that disagreement will clarify your own design thinking.

Digital wargames can also be surprisingly useful reference material. After buying a game from Wargame Design Studio, for instance, you can look through their scenario data for OOB information, terrain analysis, and other details that I have found incredibly useful for numerous designs. These games were built by people who did serious research, and the data is right there in the scenario files.

Audiobooks are a helpful timesaver for the secondary source phase. Leave one on while you are at work, exercising, commuting, or doing anything else that keeps your hands busy but your mind free. You are not going to absorb detailed OOB data this way. Audiobooks are for building familiarity with the subject, getting a feel for the conflict, and absorbing the narrative context that will inform your design instincts. They are expensive if you are buying them outright, but library apps like Libby provide free access to a large catalog, and the time you save is real. An audiobook playing during a commute is research hours you would not otherwise have.

When Sources Disagree

This will happen. Historians disagree with each other constantly, especially about unit strengths, casualty figures, and the effectiveness of particular commanders. Two reputable sources on the same battle will give you different numbers for the same formation.

Good. When sources conflict, you have flexibility to use whatever figure best serves your system. If one source says a division had 8,000 men and another says 6,500, you can choose the number that produces the right gameplay effect for the scale you are working at. You are not writing a history paper. You are building a simulation that needs to function as a game while respecting the historical record. Disagreement among historians gives you room to make design-driven decisions without being dishonest about the history.

When sources conflict dramatically, note it. You may want to mention it in your designer's notes, and it may point to a genuine uncertainty in the historical record that your game could explore through variant setups or optional rules.

Matching Research to Scope

Research should match the scope of your design. A strategic game on an entire war does not need battalion-level OOB data. An operational game covering a single campaign does. A grand-tactical game on a specific battle might need individual regiment compositions, artillery battery assignments, and the names of brigade commanders. Where designers get stuck is researching at a level of detail that exceeds what their game actually needs, which feels productive but does not move the design forward.

For a tightly scoped strategic game, a handful of good secondary sources might be all you need. One strong campaign history, a couple of analytical works, and whatever OOB reference gives you army and corps-level compositions. You could design a playable strategic game from that. *The Great Northern War* uses leader discs and a trick-taking card system with no unit counters at all, so I needed political and diplomatic sources more than OOBs. *Marathon* needed the opposite: specific unit compositions and ancient tactical doctrine down to how much it costs a phalanx to wheel left versus right.

For an operational game, the research demands increase significantly. You need concrete sourcing for OOBs and tables of equipment. Logistics become relevant, supply lines need to be at least roughly understood, and you need to know how far units could actually move in the time frame of your turn scale. Primary sources become necessary rather than optional. This is where the staff records, official histories, and detailed OOB references earn their keep.

Most of this, honestly, comes from a natural familiarity with the subject after absorbing enough material. It is less about hitting a specific number of sources and more about reaching a point where the conflict makes sense to you as a system. You understand why things happened the way they did, and you can start to see how those dynamics would translate into game mechanics. That understanding is the real product of your research, and it arrives at different speeds depending on how much you already knew going in.

Pick something you give a shit about and the research takes care of itself. Pick something you think would be a cool game but that you have no genuine interest in studying, and you will burn out before you have enough material to design anything worth playing. I know because I have done both.

Organizing Your Research

Once you start accumulating sources, keep them organized. Build a group of books, articles, and essays that you can easily reference or pull notes from. Digital files are easier to search than physical books, but I have used both. Some designers keep detailed research binders. Others highlight and tab their books. I tend to take notes in bursts when something seems relevant to a specific design decision, but much of what I absorb ends up in my head rather than on paper, surfacing later when I am working through a mechanical problem and realize I already know the answer from something I read weeks ago.

There is no single right method. The goal is to be able to find a specific piece of information when you need it during development, because you will need it. Someone will ask during playtesting why a particular unit has a strength of 4 instead of 3, and you need to be able to point to something more convincing than "it felt right."

There is probably no such thing as too much research unless it starts to burn you out or delay the design indefinitely. But to keep things manageable, let the scope of the game set the ceiling for how deep you go. Read enough to be confident in your understanding of the conflict. Read more where the design demands specific data. And stop reading when you catch yourself researching details that your game's scale will never represent.

Scale

Scale is one of the most consequential decisions you will make as a designer, and one of the least forgiving if you get it wrong. You can recover from a bad combat results table. You can rewrite movement rules. You cannot change the fundamental scale of your game once you have started building around it. Scale determines what your counters represent, what your hexes mean, what a turn of play covers, and what decisions the player is making. Change the scale and you have a different game.

In Chapter 2 I talked about the level of detail mattering more than the size of the conflict. Scale is where that principle turns mechanical. The choices you make here constrain everything that follows.

The Four Scales

Wargames operate at one of four scales: tactical, grand-tactical, operational, and strategic. Plenty of games sit between two or shift during play, but the categories hold up as a framework for understanding what you are building and what it will demand from you.

Tactical games represent engagements at the level of squads, platoons, or companies. Hexes cover tens to hundreds of meters. Turns represent minutes. Individual weapons matter. Line of sight, cover, fire arcs, small-unit maneuver. *Advanced Squad Leader* is the canonical example and worth studying even if you never design at this scale. You can model the difference between a rifle squad and a machine gun team, between a stone building and a wooden one, between veterans and green troops. But you also have to. At the tactical level, small differences in capability compound fast, and every variable you can model is a variable you probably need to model. Designing a tactical game demands intimate familiarity with how combat worked at the squad and platoon level for your specific conflict, period, and theater. If you are guessing, your players will feel it.

Grand-tactical games sit one level up. Units are battalions or regiments. Hexes cover a few hundred meters to a kilometer. Turns represent one to a few hours. The player commands a corps or army, deciding where to commit formations, when to push an attack, when to pull back. The *Imperial Bayonets* series (my grand-tactical game series, which shares its name with but is unrelated to George Nafziger's Napoleonic reference book) works at this scale, covering battles from the post-Napoleonic era through the early twentieth century at roughly 500 to 700 meters per hex with one-hour turns. Grand-tactical design requires broad, thorough research because you are handing the player the same tools and constraints a historical commander had. How far could units march in an hour? How long did orders take to reach a subordinate? How reliable were those subordinates? What happened when an artillery bombardment disrupted a brigade's cohesion? You are modeling the organizational machinery that made armies function or break down. The research demands are real.

Operational games cover campaigns. Units are divisions or corps. Hexes represent ten to fifty kilometers. Turns cover days or weeks. The player manages the movement of armies across a theater, dealing with supply lines, strategic reserves, the broader arc of a campaign. My *Procedural Combat Series* covers this scale across four titles: *1950 Korea: The Forgotten War*, *1973*, *1995: Milosevic's Last Gamble*, and *2022: Ukraine*. PCS uses an alternating pulse system where players take turns activating individual units in a go-style back and forth. That structure keeps complexity manageable because each pulse is a discrete decision rather than a giant simultaneous movement phase. Operational games need tight scope. Detailed operational simulations will balloon into logistics tracking and variable management if you let them.

Strategic games cover entire wars or major theaters. Units might represent whole armies or army groups. Turns cover months or seasons. The player allocates resources, sets strategic priorities, manages the political and industrial dimensions of a war. Strategic games give you the most freedom to abstract. You do not need to model individual formations, weapons, or terrain features. You need to capture the dynamics that determined outcomes at the level of national strategy and political will.

Scale as Argument

Scale is an argument about what matters in the conflict you are simulating.

When I designed 1916, my first self-published game on Verdun, I chose the strategic level. Most existing games on Verdun were tactical or near-tactical, focused on individual assaults and trench lines. Those games were fine for what they were doing, but they missed why Verdun was interesting. The German plan was to bleed the French army white by attacking a position France could not afford to abandon. The operation was conceived as a strategic attrition trap. A tactical game on Verdun captures the experience of the fighting but not the logic of the battle. I was baffled that nobody had approached it at a strategic level, because the actual argument lived there.

That taught me something I have carried through every design since. Before you choose a scale, figure out what thesis your game is advancing. What are you trying to say about this conflict that existing games have not said, or have not said well? The scale should serve the argument. If your thesis is about command friction at the corps level, you need grand-tactical. If it is about national strategy and resource allocation, you need strategic. Picking a scale because it seems natural for the period, or because other games on the topic use it, risks simulating the wrong thing.

A game without a thesis will lack definition. I am sure exceptions exist, but the games I admire and the games of mine that work best are the ones where the designer knew what they were arguing and built every mechanical element, starting with scale, to support it.

Why I Recommend Starting Large

As I noted in Chapter 2, granularity matters more than physical scope. When I say start large in terms of scale, I mean operational or strategic rather than tactical. You can keep your counter count and geographic extent modest while still working at a scale where the mechanics stay manageable.

Divisions, corps, and armies are less complicated to work with than smaller units. At the strategic level you represent an entire corps with a single counter and a couple of numbers. Movement is straightforward. Combat resolves on a simple odds-based table. The game can focus on big decisions without getting buried in how individual battalions interact with terrain.

Tactical games look simpler than they are. You can picture individual soldiers. You can imagine a squad moving through a village. That concreteness feels intuitive, which is why it trips people up. Tactical design demands the most mechanical precision of any scale. You need to differentiate weapons. Cover needs to matter in specific, quantifiable ways. Fire and movement need to interact realistically. Line of sight needs rules. Morale has to function at the individual unit level. You end up with a dense web of interlocking mechanics that all need calibration against each other. A first-time designer can do this, but only if they know their subject well enough to recreate action at a small scale with confidence. Most first-time designers do not have that depth yet, and a strategic or operational game gives you room to learn the craft without drowning.

Operational sits in between, and it is doable as a first design with tight scope. The PCS system taught me where it goes wrong. Even with the alternating pulse structure keeping ground combat manageable, complexity crept in. The air combat rules are a case in point. One player on Board Game Geek drew a flowchart for the air rules in *2022: Ukraine* and demonstrated that the whole system was redundant. I was impressed by the effort, and he was right. That bloat happened because I was less familiar with modern combat than with the nineteenth century conflicts I usually design, and I compensated by adding detail where I should have been looking for cleaner abstractions. If your scope keeps expanding during an operational design, you are probably adding mechanics that belong in a different game.

Research and Scale

Scale and research are tied together, and thinking about them as a pair will save you time.

A strategic game needs breadth. You need to understand the war at the level of national strategy, major campaigns, political and economic pressures. You do not need battalion-level orders of battle. You need to know which armies existed, roughly how strong they were, what strategic options each side had. A few good secondary sources will get you there.

A grand-tactical game needs depth. You are seating the player in a commander's chair and handing them that commander's information and constraints. March rates, artillery ranges, command structures, officer competence, organizational details. For *Imperial Bayonets*, I adapted Kevin Zucker's *Library of Napoleonic Battles* for the age of rifles. The combat results table had to change. Artillery ranges and effects had to change. The way command and officer initiative interact had to change, because the armies of 1866 and 1870 did not fight the way armies fought at Austerlitz. Those changes came from research into post-Napoleonic warfare. Without that research the system would have been a Napoleonic game with different labels on the counters.

A tactical game needs precision. You are modeling weapons, terrain, and small-unit behavior at a level where approximation shows. If your machine gun has the same range as a rifle, anyone who knows the period will spot it. Tactical research means technical manuals, weapons specifications, accounts of combat at the squad and platoon level.

Operational sits in the middle. You need enough depth to model campaign-level operations credibly, enough restraint to keep tactical-level detail from creeping in. The scope discipline I talked about in Chapter 2 matters here more than anywhere.

Case Study: Scale Across the Catalog

My catalog shows how scale serves different design goals across one designer's work.

The 2140 series games, with their 140-counter constraint, pushed me toward strategic and low-density operational treatments. The constraint determined the scale as much as the subject did. With 140 counters you cannot do a grand-tactical simulation of a major battle. You can do a strategic overview of a war or a tightly focused operational game. That pushed me toward strategic designs early on, and the training was valuable. I had to identify the essential dynamics of each conflict and build mechanics around those dynamics, not around unit-level detail.

One 2140 game ended up grand-tactical. *At Villers Cotterets: Mons 1914* worked at that scale because the battle was small enough that battalion-level representation fit within 140 counters. A compact force in a confined area. Not every grand-tactical subject would have fit the format, but that one did.

Imperial Bayonets required a different design approach from the 2140 games. The research was heavier. The mechanical complexity was higher. The player had to feel like a corps commander managing formation commitment, artillery placement, and a command hierarchy. Building those systems meant understanding how the armies that fought these battles functioned as organizations, not just what happened during the fighting. That kind of research takes longer and goes deeper than what a strategic game asks of you.

The PCS games taught me about overreach. The alternating pulse structure was sound for the ground war, but layering a detailed air combat system on top of an operational framework produced something more complex than it needed to be. I was trying to do justice to a subject I was less familiar with, and the result was mechanics that added complexity without adding proportional insight. Operational design demands that you know where to stop. Not everything that exists in reality needs to exist in your game, and figuring out what to leave on the cutting room floor is half the job.

Time, Space, and the Hex

Turn length and hex size are the time and space of the wargame world. If scale determines what your game is about, these two decisions set the physical and temporal dimensions your simulation operates in. They are tied to scale, and you cannot discuss one without the other, but they deserve their own treatment because the choices you make here ripple through every other system in your game, from movement rates to combat resolution to stacking.

What a Hex Actually Represents

New designers assume that all the units in a hex are sitting inside the area that hex represents. A natural assumption, and wrong at every scale above tactical.

A counter in a hex represents the operational intent of a unit more than the unit itself. A division counter on an operational map does not mean every soldier, vehicle, and supply truck belonging to that division is crammed into a twenty-kilometer circle. The division's center of gravity is there. Its commander's attention is focused on that area. Its combat power projects from that general vicinity. The actual troops might be spread across several hexes worth of real terrain. Kevin Zucker's *Library of Napoleonic Battles* works this way. The hex scale maps to the frontage of a single brigade, around 500 meters, and a single brigade in a hex makes physical sense. But when you stack multiple brigades in the same hex, with corps-level officers increasing stacking limits, those formations are not all occupying the same 500 meters of frontage. Some are in reserve, some are behind the line, some are waiting to rotate forward. The hex represents where those formations are committed, not where every man is standing.

You could argue that a hex should represent the relative frontage of the units in the simulation. There is some logic to this. If a division in your war held a front of about fifteen kilometers, then a hex scale of fifteen kilometers means one division per hex lines up with historical force density. I find this approach limiting, and only useful for tactical simulations where the physical space a unit occupies relative to its neighbors affects gameplay directly: fields of fire, mutual support, enfilade. At higher scales, a more abstract interpretation of what it means for a unit to be in a hex gives you more design flexibility and produces games that better capture the feel of operations and strategy.

This distinction affects how you think about stacking, zones of control, and movement. If you treat a hex as a literal container, you will design stacking limits based on how many units fit. If you understand that a hex represents operational space, you can design stacking from a more useful angle.

Stacking as Command Capacity

Stacking limits express command capacity more than physical space. A stacking limit of three says a commander at this level can coordinate three formations in the same operational area.

In *Imperial Bayonets*, stacking follows organizational structure. Two combat units of any type can stack together. Three units of the same division can stack together. Five units of the same corps can stack together, but only if a leader is present. The progression tracks the chain of command: more organizational overhead on-site means more formations you can coordinate. A corps commander stacked with his units enables tighter concentration of force than an ad hoc grouping of battalions from different divisions trying to operate without unified leadership.

This models something real. You could always shove more men into a valley. The constraint on force concentration in most periods was the ability to command them coherently. Stacking limits built around command capacity force decisions about where to place leaders, which formations to group together, and whether concentration is worth the risk. Physical-space stacking limits just create a puzzle about fitting counters into hexes.

Choosing Your Hex Scale

No formula exists for choosing the right hex scale, and I would be suspicious of anyone who claimed otherwise. Start from a reference map that looks right. This comes with experience.

I usually have a mental picture of what units on a map will look like in the simulation before I settle on a hex scale. I know how many counters I want, what level of unit I am representing, and how much of the theater needs to be on the map. The hex scale falls out of those constraints more than it drives them.

For *Sedan 1940*, each hex is one kilometer. That works for a grand-tactical battle where I am representing squadrons, battalions, and regiments over a confined area with two-hour daytime turns. For *Imperial Bayonets*, hexes are 500 to 700 meters, a tighter zoom for the post-Napoleonic battles in the series where formation frontages and artillery ranges need finer resolution. For an operational game in the *Procedural Combat Series*, hexes cover ten to fifty kilometers because I am tracking divisions and corps across a whole theater over weeks.

The hex scale should serve the simulation. Give yourself enough resolution to make terrain features and force dispositions matter without burying the map in empty hexes. If you are designing your first game, pick a hex scale that works for your unit scale and map area, play a few turns, and adjust if it feels wrong. You will know. Units will bunch up where they should not, or terrain that should matter will be too small to register, or the map will stretch across the table with vast dead zones where nothing happens.

Turn Length

Turn length is tied to scale, and the relationship is straightforward. Tactical and grand-tactical games deal in minutes to hours. Operational games cover days or weeks per turn. Strategic games measure turns in months or seasons.

Turn length is less agonizing than hex scale because it functions as a sanity check against your other systems. Units should not move further in one turn than they could have moved in real life at the game's chosen time scale. If your operational game has weekly turns and a division can march from one end of France to the other in a single turn, something is broken. If your tactical game has ten-minute turns and an infantry platoon can sprint three kilometers, your movement rates are wrong.

In *Sedan 1940*, daytime turns are two hours and nighttime turns are four hours, twenty turns total. Those numbers come from the pace of the actual battle: how fast German armor moved, how fast the French could react, how long the critical engagements lasted. For *Imperial Bayonets*, one-hour turns match the tempo of post-Napoleonic battle, where formations took time to deploy, artillery preparation preceded assaults, and a corps attack developed over the course of a morning. One hour is also about the average time it took for an order dispatched on foot or horseback to reach a subordinate commander in that era. The turn length ties to the command system: each turn represents one decision cycle, the time between a commander issuing an order and that order taking effect on the field.

Turn length and hex scale together create the physics of your simulation. They determine how far a unit moves per turn, which shapes how the map plays, which shapes what options the players have. Get them close and the rest of the game has a solid foundation. Get them wrong and no amount of clever mechanics will fix the feel.

Hex Maps, Point-to-Point, and Area Control

Hex maps became the standard in wargaming for good reasons. They provide uniform movement costs, consistent adjacency, and a grid that handles directional movement well. For conflicts with defined front lines and forces that maneuver across continuous geography, hexes work.

They are not always the right choice.

For a while I had an idea for a hex-based simulation of the Peloponnesian War. I studied the conflict, started working on a design, and realized too late that hexes could not represent the dynamics of that war. There were no front lines of the kind common to modern combat. Armies marched to a location, fought a battle on flat, mostly clear land, and marched home. The geography that mattered was not terrain in the wargame sense, not hills and rivers and forests, but the political and economic connections between city-states, the naval routes that controlled trade and troop movement, the alliances that shifted over decades. A hex map failed to produce interesting choices, let alone relevant ones, compared to an area map or point-to-point design.

I got midway through the design before this clicked. I had already paid for a map and a set of cards for the prototype. The game never got published, though I hope to finish it someday. I had the map redone in an area control style. This is the kind of mistake you make when you are inexperienced: committing resources before the research tells you what the design needs. Chapter 3 again. Do your research before you commit to mechanical frameworks, because changing your mind later costs real money. In my case, somewhere between five and six hundred dollars for a map I could not use.

The Thirty Years War is another conflict poorly suited to hex maps. Decentralized warfare across a vast theater, no continuous front lines, forces operating across disconnected regions.

Point-to-point and area movement games are mechanically almost identical but look nothing alike. Both abstract geography into nodes connected by paths rather than a continuous grid. Point-to-point uses dots and lines; area movement uses bounded regions. The choice between them is aesthetic and depends on what visual language best communicates your game's geography to the player.

Both are associated with strategic games, but plenty of smaller-scale designs use area movement well. Two of my own games use point-to-point. *The Great Northern War* covers Charles XII's campaigns across the Baltic and Scandinavia, where the conflict moved between cities and political centers rather than across continuous terrain. *Prigozhin's March of Justice* follows the Wagner mutiny along a single highway corridor from Rostov to Moscow. In both cases, hexes would have imposed spatial detail that did not match how the conflicts worked. The advantage of point-to-point is that you define the geography that matters rather than imposing a uniform grid. The disadvantage is that you lose granularity. Flanking, encirclement, and detailed maneuver are harder to model on a point-to-point map.

The choice between hex, area, and point-to-point comes down to designer preference, informed by what the conflict demands. Study your subject. If front lines, terrain, and spatial maneuver are central to your thesis, hexes are probably right. If the conflict is about connections, logistics, and the political geography of a war, consider the alternatives.

Making a Playtest Map

Making a playtest map is easier than most people imagine. I have seen designers stall out here, convinced they need a polished map before they can start testing. You do not. A functional playtest map can be ugly. It needs to be readable.

Find a historical map close to your game's scale. Download GIMP, which is free. Open the historical map, create a new layer on top of it, and trace over the relevant features: coastlines, rivers, cities, major terrain. Put a background layer underneath everything and dim the original map so your traced lines are visible. Draw terrain features on their own layers. For a hex map, overlay a hex grid. GIMP has plugins for this, or you can find hex grid templates online.

At a minimum, color in hexes to note their terrain type. Green for woods, brown for rough, blue for rivers, white for clear. That is enough to start pushing cardboard and finding out if the game works. Prettying things up helps make playtesting more pleasant, but do not let map aesthetics slow down the design process. I am plenty guilty of spending too much time on playtest maps. I enjoy the cartography side of design, so I tend to work on the map before I finish rules. That keeps the process engaging for me, but it is not necessary and it is not where the real design work happens.

Point-to-point and area maps are even easier to prototype. You do not need to make judgment calls about which hexsides to align rivers with, or whether rivers run along hexsides or through the middle of hexes, a choice less common in modern games but still relevant. You draw dots and lines, or draw regions and connect them.

Other published game maps are acceptable resources for inspiration. You could test an idea using a map from another game that covers similar geography at a similar scale. The point is to get something on the table that lets you test the systems you are building. Design the map during playtesting to figure out what works and what does not. Visual polish, cartographic accuracy, graphic design: those come later.

From Playtest Map to Published Map

Once your design is finished and tested, you will want a professional map for publication. Unless you are a trained cartographic artist, this means hiring someone.

Find an artist who knows wargame cartography. This matters more than general artistic talent. For *We Were Not Cowards*, I hired Alyssa Faden, an RPG map artist. Talented, gorgeous work. The problem was that terrain did not conform to hexsides or hex spines, which created multiple elevations in a single hex. Some publishers handle this fine. Multi-Man Publishing games frequently have mixed terrain in a single hex and resolve it by having the player use the worst terrain type for movement or combat purposes. But for my game, where movement costs and ranged artillery calculations were central, mixed terrain in a hex made things too unwieldy. I had to have the entire map redone.

For most of my published games, Ilya Kudriashov has done the cartography. He understands wargame maps. He knows that the map is a playing surface first and a piece of art second. His maps are both functional and professional.

Be prepared to spend several hundred dollars for a finished map. You can bundle map work with counter graphics or box art to get a better rate. The main point: do not invest in professional art until your design is done. Get the game working on an ugly map first. Make your mistakes cheap. The Peloponnesian War map I paid for and could not use taught me that the expensive way.

Some maps come together without trouble. The map for *1914: East Prussia: The World Undone*, done by Ilya in a classic SPI style, fit the scale and the conflict. The geography of East Prussia cooperated with the hex grid. Rivers ran where hexsides fell. The terrain was distinct enough to create interesting gameplay without agonizing over hex placement. When the map works with the conflict, the process is painless. When it does not, you cannot force it.

Case Study: *We Were Not Cowards* and *The World Undone*

These two games sit at opposite ends of the map experience.

Imperial Bayonets — We Were Not Cowards: Sedan 1870 is a grand-tactical game with the fine resolution that scale demands. Every hex matters. Elevation changes, river crossings, and terrain types all affect combat and movement. The first map, though artistically accomplished, did not conform terrain to hex boundaries. A hex with two different elevation levels forced players to guess which one applied. Terrain that straddled a hexside left the rules unable to resolve what happened. A beautiful map that produces ambiguous game states is worse than an ugly map that plays without confusion, because ambiguity breeds arguments and arguments kill playtesting sessions. Ilya Kudriashov redid the entire map.

1914: *East Prussia* was the opposite experience. Ilya did the map in a classic SPI aesthetic. The lakes, forests, and road networks of East Prussia mapped onto a hex grid without friction. The scale fit the conflict, the geography cooperated with the format. No ambiguity, no redos, no arguments during playtesting about what a hex contained.

Some conflicts are easier to map than others. But the larger point is that a map is a game component, not an illustration. Design it to serve the game. Test it to find ambiguity. Fix the ambiguity before you pay for art. And when you do pay for art, hire someone who understands that a wargame map has to work before it can look good.

Physical Systems Design: The Simonsen Legacy

Everything I have said in this chapter about maps as functional tools rather than illustrations, about terrain conforming to hex geometry, about the map serving the game before it serves the eye, traces back to one person. Redmond Simonsen was the art director of SPI from 1969 to 1982, a professionally trained graphic designer (BFA from Cooper Union), and the reason wargame maps look the way they do. His full biography appears in Appendix E. What matters here is his design philosophy.

The term he coined for his approach was physical systems design, which he defined as the graphic engineering of game elements. The concept sounds simple. Every visual element in a wargame exists to make the game easier to play and the game state easier to read. The map, the counters, the player aid charts, the rulebook layout: these are not separate aesthetic concerns. They are a single integrated system, and the graphic designer's job is to engineer that system so the components serve gameplay.

Simonsen's argument about constraints is the part most relevant to working designers. He maintained that the physical components available to you, paper maps, cardboard counters, printed charts, are the single greatest limiting factor on what a game can do. Designers generate ideas faster than the physical medium can keep up. The graphic designer's job is to push those material limits, finding new ways to encode more information into the same paper and cardboard so the components do not become the bottleneck.

This plays out in specific choices that became universal standards. Rivers conforming to hexsides. Simonsen reintroduced this technique, which had been used in Avalon Hill's *D-Day* and then abandoned, and made it the permanent convention. Before this was standard, rivers could run through the middle of hexes, creating ambiguity about which hexes they affected. Forcing rivers onto hexsides eliminates that ambiguity. You are either crossing a river hexside or you are not. The mechanical payoff is direct: movement costs, combat modifiers, and supply tracing all work cleanly when terrain features align with the hex geometry the rules reference.

Counter design followed the same philosophy. The *PanzerBlitz* counter, designed by Simonsen, packed seven pieces of information onto a 5/8-inch square: offensive value, defensive value, range, movement value, unit ID, unit name, and a vehicle or weapon silhouette. Other designers adopted that density as the template. Before Simonsen professionalized counter design, wargame counters were simpler and carried less information. After *PanzerBlitz*, the expectation was that a counter should tell the player everything they needed to know about a unit at a glance, without referencing a separate chart. Player aid charts themselves became designed components rather than afterthoughts, integrated into the visual system alongside the map and counters.

One of his principles cuts against instinct and is worth remembering. The more complex the game system, the less decorated it should be. Designers who build mechanically dense games often want busy, detailed maps and counters to match the density of their rules. Simonsen's position was the opposite. As mechanical complexity increases, the visual design must get cleaner and more restrained. A complex game already demands heavy cognitive load from the player. Adding visual complexity on top of mechanical complexity makes both harder to process. Clean graphics on a complex game let the player focus on the decisions the designer built. Busy graphics on a complex game compete with the decisions for the player's attention.

The cumulative effect of Simonsen's work across over 400 titles at SPI is that the visual grammar of wargaming was established by one person over thirteen years. Hex maps with terrain conforming to hex geometry. Information-dense counters with standardized layouts. Player aids as integral components. These are things we take for granted because Simonsen made them standard. He was doing user interface design decades before that term existed in the software industry.

Simonsen laid out his philosophy in a chapter called "Graphics and Physical Systems Design" in the 1977 book *Wargame Design*, which also featured contributions from Dunnigan and other SPI staff. Kevin Zucker wrote a companion piece called "The Limits of Art" in *Fire & Movement* magazine in 1978 that explores the same ideas from a slightly different angle. Both are worth reading if you are designing your first map, or your tenth. The principles have not aged. The materials have changed, print-on-demand

and digital prototyping have replaced the production constraints Simonsen worked under, but his core insight holds. The physical components of your game are a system. Design them as a system. Engineer them so they serve the game rather than decorating it.

Orders of Battle and Unit Ratings

An order of battle (OOB) is a list of who was there and what they had. Building one is the most research-intensive part of wargame design, and the part where you will spend the most time cross-referencing sources that disagree with each other. The process is straightforward. The challenge is judgment: deciding what level of detail your game needs, what numbers to put on the counters, and when "close enough" is the right answer.

Finding Your OOB

Start with published staff histories. These are the official records kept by military organizations during and after a conflict, and they remain the most reliable primary source for unit identifications, strengths, and dispositions. For the World Wars, the U.S. Army's official histories (the "Green Books") are free online and contain detailed OOBs for every major campaign. The British official histories serve the same function. German records are patchier because of wartime destruction, but enough survived to reconstruct most formations through secondary sources.

Beyond staff histories, books are your primary tool. Osprey's "Battle Orders" series covers unit organization for dozens of campaigns. Nafziger compiled Napoleonic OOBs across multiple volumes. For specific battles, monographs will list participating units in appendices. When you find two sources that disagree on a unit's strength or composition, check which one cites primary documents and which one is citing someone else's book. The closer to the original records, the more weight you give it.

The internet has made OOB research faster but not easier. Websites like Niehorster's WWII organizational histories provide detailed tables of organization and equipment. Forums and wikis can point you toward sources you did not know existed. But online OOBs are often unsourced or copied from each other, and errors propagate. Use the internet to find leads, then verify against published sources.

Level of Representation

Your scale determines what units appear on the map. A tactical game represents battalions or companies. An operational game uses divisions or brigades. A strategic game might use corps or armies. This relationship between scale and unit level is fixed by the logic of the simulation. If your hexes represent twenty kilometers of frontage, individual battalions make no sense as game pieces. If your hexes represent five hundred meters, army-level counters make no sense either.

The gray area is in the middle. An operational game at ten-mile hexes could use brigades or divisions. The choice depends on what decisions you want the player to make. If the player should worry about committing reserves at the divisional level, use brigades. If the player's decisions are about where to commit corps, use divisions. The unit level on the counter should match the command level the player occupies.

There is a practical constraint too. More counters means more setup time, longer turns, and a bigger map. A game with 400 brigade counters plays differently from one with 80 division counters covering the same campaign. Both can be good. Know which experience you are building toward.

What Goes on the Counter

Counter design falls on a spectrum. At one end, games like *Unconditional Surrender* put almost nothing on the counters. Unit identity and combat values live on tracks and charts. The counters are markers that show position. At the other end, some tactical games print attack strength, defense strength, movement points, range, morale rating, and unit identification all on a single half-inch square.

Both approaches work. You pick based on how much information the player needs at a glance versus how clean you want the counter to look.

I tend toward less information on the counter rather than more. Koniggratz and Mars La Tour, two of my games based on Mark Herman's *Gettysburg* system, have no strength points on the counters at all. Units show movement points and an Efficiency Rating (zero, one, or two stars) that modifies an opposed die roll. There are no strength points. The player figures out combat power from the dice and DRMs, never from the counter. That is the extreme end of the spectrum, but it works for division-level games where relative quality matters more than raw numbers. If all infantry units in your game move at the same rate, you do not need to print movement points on every infantry counter. A line in the rules that says "infantry moves 4" does the same job without cluttering the counter. Print movement allowances on the counter when different units of the same type move at different speeds and the player needs to see that at a glance.

The minimum information a counter needs is identification (what unit is this?) and something that distinguishes it from other units in play. Everything else is a design choice. Attack and defense values on the counter let the player assess combat odds without consulting a chart. But those same values on a separate reference card can work if your counter art is cleaner without them. Playtest both if you are unsure. You will know which one your game wants.

Strength Points: Headcount vs. Combat Effectiveness

New designers assume that a unit's strength points should reflect its headcount. A division of 15,000 men is stronger than a division of 10,000, so it should have more strength points. This logic is reasonable and often wrong.

Headcount is one factor in combat effectiveness, and seldom the most important one. A full-strength division with no fuel cannot attack. A division at 60% strength with veteran troops, functioning logistics, and air support will destroy a full-strength division of untrained replacements who ran out of ammunition two days ago.

Dupuy spent decades quantifying this through his Quantified Judgment Model. His factors include training, morale, logistics, leadership, equipment quality, air support, and surprise. The QJM's combat effectiveness values for national armies in World War II show German forces averaging 20% higher combat effectiveness than their Western Allied opponents at equivalent strength, and higher still against Soviet forces in the early war years. Zetterling's study of Normandy 1944 confirmed this pattern through a different methodology: German units on average inflicted more casualties per soldier committed than Allied units, whether attacking or defending, even when outnumbered.

You do not need to replicate Dupuy's math. You need to understand what it tells you: a strength point should represent combat power, and combat power is a composite of many things. A German panzer division and a Soviet rifle division might both have 10,000 men, but giving them the same strength points produces a game that does not look like the Eastern Front.

How you handle this depends on your system's abstraction level. Some options:

Bake it into the strength point. The German division gets 8 strength points (SP) and the Soviet division gets 5. Simple. The combat system does not need to know why one is stronger. This works at operational and strategic scale where you want fast resolution.

Separate quality from quantity. Give both divisions similar strength points reflecting headcount, but let quality show up elsewhere. Armor attacking in open terrain gets a column shift regardless of nationality. Soviet units in a Barbarossa game might have lower movement ratings in the opening turns to reflect the organizational chaos of June 1941, or lose their first turn entirely. Quality modifiers should be subtle. A small edge repeated across many combats produces the right aggregate result without making individual battles feel predetermined.

The *Procedural Combat Series* takes this further. PCS counters have no printed combat strength at all. Each unit carries a Unit Quality Rating from A to D. At the moment of combat, both sides draw a random chit from a cup and cross-reference it against their UQR to determine how hard they hit. The UQR on the counter tells you how good a unit is. The chit tells you how its day is going.

A "D" unit occasionally draws a strong chit and fights above its weight. An "A" unit occasionally draws poorly. Over many combats the ratings sort themselves out, but any single engagement carries real uncertainty.

Use distinct ratings for distinct capabilities. Attack strength, defense strength, and movement allowance can all differ. A unit that is good on the attack might be mediocre on defense. This adds counter complexity but captures asymmetries that a single strength point cannot.

The more tactical your game, the more granularity you need. A squad-level game should distinguish between a veteran squad and a green squad in multiple ways: firepower, morale, initiative. A strategic game covering an entire war can collapse all of that into a single number and let theater-level modifiers handle the rest.

The Technology Problem

Rating hardware is where I have the least natural instinct and where I have learned the most from other designers.

When I was designing a World War I naval game, I had no idea how to rate warships. Jack Greene, who had spent years on this problem, told me the starting point was shell weight multiplied by rate of fire. That gives you a raw measure of how much steel a ship can throw at an enemy in a given time period. From there you adjust for armor penetration, fire control quality, and speed. The formula is rough but it produces relative ratings that hold up against historical engagements.

Technology ratings matter most in periods where hardware created decisive asymmetries. Ancient and medieval combat was dominated by training, morale, and numbers. The Macedonian phalanx did not beat Persian infantry because the sarissa was a superior weapon in isolation. It won because the phalanx formation, backed by years of drill and Philip's organizational reforms, created a system that Persian infantry could not answer.

After the Industrial Revolution, and more so after World War I, hardware starts to dominate. A Tiger tank and a Sherman tank are different weapons with different capabilities, and your game needs to reflect that difference. Penetration values, armor thickness, and engagement ranges become design inputs. For naval games, the differences between a dreadnought and a pre-dreadnought are so large that the ratings almost write themselves.

For land games at operational scale, I find that technology differences are better handled through combat modifiers than through individual unit ratings. German armor quality in 1941 was not superior to Soviet armor on a tank-by-tank basis. The T-34 was a better tank than anything the Germans fielded that year. The German advantage was in combined arms coordination, radio communication, logistics, and leadership. Modeling this through a technology rating on the counter misses the point. Model it through combat system modifiers, or through the command and activation systems that let German units respond faster and coordinate better.

Sedan 1940 is where I worked this out most deliberately. The game uses three armor classes encoded visually on each counter by the shape around its movement allowance. No marking means Light: foot infantry, artillery, anti-tank guns. A circle means Medium: motorized infantry, armored cavalry. A square means Heavy: tanks and aircraft. When a unit fires, the player cross-references its weapon type against the target's armor class on a national combat matrix. German and French forces use different matrices. German primary fire against Heavy targets gets a x0.5 multiplier. French primary fire against the same Heavy targets gets x1. French secondary fire against Light targets gets x2.5, reflecting a defensive doctrine built around strong anti-personnel firepower. Neither side's secondary weapons can touch Heavy armor at all.

The matrices encode how German and French forces fight, not which side has better tanks. The German player's edge comes from Luftwaffe support, more division activation chits in the cup, and an opponent saddled with reserve divisions that take column shift penalties in assault combat. The French player's edge is raw defensive firepower at close range. The technology asymmetry lives in the combat system, not on the counters, and it produces the right operational pattern: German combined arms breaking through French positions that are strong on paper but brittle under coordinated pressure.

Balancing Accuracy and Playability

A good designer finds the point where accuracy and playability reinforce each other. They operate under different constraints, but the best games satisfy both.

Mark Herman's *Gettysburg* game, published through C3I and inspired by SPI's classic *Napoleon at Waterloo*, illustrates this. Herman represented Confederate units as half-divisions and Union units as divisions. On the map, this gives the Confederacy more counters than the Union, creating a visual impression of numerical superiority that did not exist in historical terms. The CSA was outnumbered at Gettysburg.

But Herman was not making a mistake. The half-division representation captures something about how the Army of Northern Virginia operated. Lee's army was organized into smaller, more flexibly commanded formations than Meade's. The extra counters give the Confederate player more operational flexibility: more units to maneuver, more options for concentration and dispersal.

The Union player has fewer, stronger pieces that hit harder individually but offer less positional flexibility. The visual "inaccuracy" produces gameplay that reflects the actual operational character of both armies better than a strictly proportional counter mix would.

You can rationalize apparent inaccuracies in your OOB if they serve the simulation. A unit that is "too strong" on paper might produce the right combat results when it interacts with your CRT and terrain modifiers. A formation that is split into two counters when it was historically one entity might generate the right command decisions for the player. The test is whether the game reproduces historical patterns of action and result, not whether every number on every counter matches a table in an OOB reference.

The Real Test

There does not need to be rigid logic in building an OOB. I know designers who want a mathematical formula that converts historical data into strength points with mechanical precision. I have never found one that works across more than a single game.

You gather the best data available, assign numbers that reflect relative combat power, and playtest. If the 1st Panzer Division underperforms what you know it should do based on its historical record, you adjust its rating. When the game produces a front line that looks nothing like the historical campaign, check the numbers against the history. Either the OOB or the combat system is off.

Apply numbers in a way that should reproduce the effects you want to see under the constraints you have built. I have no better rule for unit ratings than that.

Case Study: *Prigozhin's March of Justice*

In June 2023, Yevgeny Prigozhin led Wagner Group mercenaries on a mutiny march from Rostov-on-Don toward Moscow. The event lasted about 36 hours, all of it playing out live on social media and television. As a simulation subject, it was unusual: a small number of identifiable units, a well-documented timeline, and an outcome that hinged on political decisions rather than combat.

The OOB was manageable. Wagner Group forces numbered roughly 8,000 men organized into a handful of identifiable columns. Russian military and security forces along the route included Rosgvardia (National Guard) units, regular army formations at scattered garrisons, and whatever could be scraped together and positioned on short notice. The total number of game pieces was small.

The rating challenge was relative quality within the Wagner contingent. These were mercenary units with combat experience in Ukraine, Syria, and Africa. By reputation, one formation was considered superior to the others. In absolute terms, as professional military forces measured against a modern army, all of them had serious limitations. The design question was whether to rate them high based on their self-image and media reputation, or to rate them as the irregular, poorly institutionalized force they were.

I split the difference. The standout unit got a higher rating than the rest, reflecting its reputation and track record. The remainder got ratings that made them competent against the poorly motivated garrison forces along the route but not dominant against organized resistance. In the design notes, I was honest: these are all subpar formations by any conventional military standard, and the ratings reflect relative capability within that band rather than comparison to professional armies.

The game's signature mechanic also solved a rating problem. Each Wagner unit sits on a Maneuver/Combat Track where the player allocates SP versus MP each turn. Push a unit toward combat power and it hits harder but moves less. Push it toward maneuver and it covers ground but fights weaker. The player decides each turn how hard a unit can hit, which captures the real tradeoff between speed and fighting capability on a forced march.

The OOB worked because the game's scope was narrow. With a dozen or so units on the map, each rating carries weight. In a game with 200 counters, a single unit being a point too strong or too weak disappears into the noise. In a game with 12 counters, it changes the outcome. Small games demand more precision in relative ratings even as they forgive more abstraction in absolute terms.

The March of Justice OOB also illustrates the value of working from well-documented events. I could watch footage of the march, read real-time reporting, cross-reference locations and timelines, and build an OOB with reasonable confidence in a few days. Compare that to reconstructing a Napoleonic corps from fragmentary archival sources that disagree on battalion strengths by 30%. The research burden scales with the age and obscurity of the conflict, and beginning designers should account for that when choosing subjects.

Sources and Further Reading

- Dupuy, Trevor N. *Numbers, Predictions and War*. Indianapolis: Bobbs-Merrill, 1979.
- Nafziger, George F. *Imperial Bayonets*. London: Greenhill Books, 1996.

- Zetterling, Niklas. *Normandy 1944: German Military Organization, Combat Power and Organizational Effectiveness*. Winnipeg: J.J. Fedorowicz, 2000.
- U.S. Army. *FM 101-10, Staff Officers' Field Manual.* Washington: Department of the Army, 1959.

Movement and Maneuver

Kevin Zucker once told me that movement is the most important part of a wargame. I have spent a decade designing games and I have not found a reason to disagree. Sieges and highly abstracted games can get away with minimal movement systems, but any simulation where maneuver matters demands careful attention. Everything about your game will be wrong if the movement system produces ahistorical results. The combat system, the order of battle, the map: none of it matters if the units can reach places they should not reach in the time the turn represents.

Movement Points and Terrain

The basic framework is familiar to anyone who has played a wargame. Units have a Movement Allowance printed on the counter. Terrain costs movement points to enter. Clear terrain costs 1 MP, rough terrain costs more, roads cost less. The designer's job is making sure those costs produce realistic movement patterns for the conflict being simulated.

Terrain effects gate movement in ways that reflect how formations moved on the ground. Some games assign different terrain costs based on unit type. Tanks pay more to enter woods than infantry does. Cavalry moves faster on open ground but suffers in mountains. These distinctions matter most at tactical and grand-tactical scales where the physical characteristics of the unit shape how it interacts with terrain. At the operational and strategic level, you can usually get away with uniform terrain costs because the time scale absorbs the differences.

The test is simple. Set up your game, move every unit at full allowance for a few turns, and measure where they end up. Compare that to where the historical forces were after the same amount of time. If your German panzer divisions are reaching Moscow a month early, your movement costs are wrong. If your Napoleonic infantry cannot keep pace with the historical advance, something needs to change. Appendix D provides historical march rate data for exactly this purpose.

Zones of Control

Zones of control are an abstraction of movement in their own right. A ZOC represents a unit's ability to control and influence the space around its center of concentration. Patrols, observation posts, and the threat of fire all project outward from a unit's position without the unit physically occupying that ground.

The most common ZOC rule stops enemy movement. A unit entering an enemy ZOC must halt. This works well for many operational games, but it can produce unrealistic results depending on the scale and era. A single battalion projecting a ZOC that halts an entire corps makes no sense at the strategic level. At the tactical level, a rigid ZOC can create impenetrable defensive lines that no historical force produced.

Several of my games handle ZOC differently depending on what the simulation needs. In the DAMOS (Destroy All Monsters Operational Series) Barbarossa trilogy, ZOC projects from most German units but only from Soviet armor and mechanized forces. Soviet foot infantry cannot freeze German movement the way a panzer division freezes theirs. This asymmetry models the German advantage in operational flexibility without touching the combat system at all. In 1950 Korea, UN ZOC does not extend into non-clear terrain, reflecting the reality that UN forces could not project control into the Korean mountains the way they could across open ground.

The DAMOS series introduces Zones of Influence that scale with unit strength. A division with 7-12 SP projects a ZOI of 2 hexes. A division with 13 or more SP projects 3 hexes. Stronger units control more space. Non-phasing units within ZOI range can attempt Reactions, rolling to move into threatened hexes or fall back. This models the OODA loop at a mechanical level. A unit with greater combat power has a longer operational reach, and defending units must decide whether to react before the threat materializes.

Koniggratz and Mars La Tour use a different approach. Units project a Zone of Influence at 2-hex range that forces any enemy in March Formation to flip to Battle Formation and stop. The ZOI does not halt movement outright. It forces a formation change. You can still move into an enemy's sphere of influence, but you arrive deployed for combat rather than strung out in a road column. This captures the historical reality that armies approaching contact had to shift from movement posture to fighting posture well before the first shots were fired.

The OODA Loop and Command Flexibility

OODA stands for observe, orient, decide, act. In military terms, it describes how quickly a force can process new information and respond. The concept is abstract, but its effects on the battlefield are concrete, and your movement system is where those effects show up.

The Franco-Prussian War is one of the clearest historical examples. The French army still carried a formidable reputation, built on the Crimea and Italy. The Prussians destroyed it in less than three months. The armies looked roughly even on paper. The difference was that the Prussian army could do several things in the time it took the French to do one.

The Prussian general staff had developed Mission Command, empowering officers at a tactical level to make decisions based on their own initiative and experience. When a French commander was killed, the unit waited for hours until someone sent new orders. French troops would not march to the sound of the guns without explicit authorization. The army's command structure depended on Napoleon III directing the whole force, and Napoleon III was no tactical genius. At Mars-la-Tour, a few Prussian corps effectively froze the entire French army because the French could not organize a response fast enough.

This translates into game mechanics through the movement and activation systems. In *We Were Not Cowards*, Prussian divisions break down into component brigades freely, while French divisions must pass an Initiative Check to do the same. Failure costs 2 MP. The Prussians get 2 March Orders per turn. The French get 1. The French have better rifles: Prussian units pay +1 MP to enter a French ZOC because of the Chassepot's superior range. The Prussians have better artillery, more activation flexibility, and a command system that lets their officers react without permission. The French player spends the game with better weapons and worse options.

The same principle works at every scale. In a card-driven game, one side might have a larger hand. In a chit-pull game, one side might have more activation chits in the cup. In a game with initiative ratings, better-trained forces check at a higher number. If one side could act faster and more flexibly than the other, your movement and activation system needs to reflect that, or the game will not feel like the conflict it represents.

Types of Movement

Road Movement

Road movement allows units to move along roadways at a faster rate than cross-country marching. At smaller scales, road movement introduces considerations about column length and vulnerability. A battalion stretched along a road in march formation occupies more linear space and presents a softer target than the same battalion deployed in a fighting position.

Several of my games model this through formation states. In *Sedan 1940*, units have Travel Mode and Deployed Mode. Travel Mode gives higher movement but lower combat values and penalties when targeted. In Koniggratz and Mars La Tour, units have March Formation (4 MP for infantry) and Battle Formation (1 MP). The shift between these states is the central tension of the game. You need March Formation to get anywhere, but arriving in March Formation near the enemy is how units get destroyed.

Imperial Bayonets uses Road March as a specific movement mode. Units pay 0.5 MP per road hex but cannot be adjacent to each other. If an enemy unit moves adjacent to a force in Road March, the force suffers temporary Demoralization. The speed benefit comes with genuine risk.

Rail Movement

Rail movement is a strategic consideration that becomes relevant at the operational and grand-tactical level, especially from the mid-nineteenth century onward. The key design question is how to handle rail movement's vulnerabilities. Units on trains are not fighting formations. They are cargo.

In 1914: East Prussia, only the German player has rail movement, reflecting the Prussian rail network's superiority over Russian infrastructure. Rail capacity increases over the first three turns as the system mobilizes. In the DAMOS Barbarossa trilogy, German supply depends on converting captured Soviet rail lines to European gauge, 1d3 hexes per turn, a slow process that constrains the entire offensive. In 2022: Ukraine, Russian logistics depend on railhead markers that must trace through friendly-controlled rail hexes back to Russia. Ukrainian airstrikes and partisan activity can push those railheads backward, stretching Russian supply lines beyond their breaking point.

Naval Movement

Naval movement tends to be non-linear, complex, and difficult to abstract well. It requires genuine expertise in how naval operations functioned to create a useful model. I avoid most naval systems for this reason. The shell weight formula Jack Greene taught me for my WWI naval game was one piece of a much larger puzzle, and I had to lean on his experience to get the rest right. If you are designing a game with naval movement, find someone who knows the subject at the level of detail your simulation requires. Do not try to wing it.

Forced March and Strategic Movement

Forced marching can be represented in several ways depending on the scale. Tactical squads might exhaust themselves for extra movement. Grand-tactical battalions of the nineteenth century might break down into road columns. The simplest approach is allowing a unit to double its movement allowance outside of enemy ZOC, a common convention for strategic movement in operational games.

Forced marching should carry a cost. Free strategic movement is simpler to implement, but forced marches wore units down, and your game should reflect that. In *Imperial Bayonets*, Road March carries the vulnerability of Demoralization if caught by the enemy. In *1916*, units that spend more than half their MP without a line of communication check the Attrition Table. In the LCOS series, Forced March gives a second MP budget equal to the original, but units must roll a straggler check. Entering an enemy ZOC during a forced march makes that roll worse.

The specific penalty matters less than having one. If a player can rush units across the map at double speed with no consequences, they will do it every turn, and your movement system becomes a teleportation mechanic.

Other Movement Restrictions

ZOC is the most common and most direct way to restrict movement, and it is a valid approach across most scales. Terrain costs are the other universal gate. Beyond those, command systems offer another way to influence where and how units move without putting hard stops on the map.

Command range is one form. In *Imperial Bayonets*, units outside their officer's Command Range must pass Initiative Checks to activate. Failing the check means the unit sits still or moves with reduced MP. The restriction lives in the command system, and it affects movement by making unsupported maneuvers unreliable rather than impossible. A player can still try to send a unit off on its own. It just might not go.

Koniggratz, Mars La Tour, and The Alma take a more explicit approach. HQ markers are placed on the map each turn, and units must move toward a hex within the HQ's range. The player picks where the HQ goes, and the units follow. You decide where the army is going. The units figure out how to get there. This forces you to think about command intent rather than individual unit optimization.

These are tools, not replacements for ZOC or terrain. A game can use all of them together. *Imperial Bayonets* has ZOC, terrain costs, and command range all operating at once, each one restricting movement in a different way for a different reason.

Stacking and Movement

Stacking limits are less about movement than about force density, but they interact with the movement system in ways worth considering. Both *Imperial Bayonets* and Zucker's *Library of Napoleonic Battles* charge +1 MP for units entering an occupied hex. This models the command overhead of consolidating forces. Coordinating the arrival of multiple formations in the same space takes time and creates friction. That 1 MP cost is small but cumulative. It discourages casual stacking and rewards players who think about the timing and sequencing of their formations.

At the tactical level, stacking limits may need to be enforced more strictly. Overstacking a hex with units that could not physically occupy that space breaks the simulation. At the operational and strategic level, stacking limits are more about command capacity than physical space, and the +1 MP approach handles it with less bookkeeping than a hard cap.

Common Mistakes

The most common movement mistake is units that are too fast for the turn scale. If your weekly turns let a Napoleonic corps march 200 miles, something is broken. Check your movement allowances against historical march rate data before you do anything else.

A related mistake is ZOC rules that create unrealistic barriers. If a single regiment can freeze an entire army in place because it projects a ZOC across a river, the ZOC is doing too much work for the game's scale. Consider making ZOC conditional: projecting only into certain terrain types, only from certain unit types, or only from units above a certain strength threshold.

The third common mistake is movement systems that are too permissive. When I was working on 1968: Tet, the Vietcong could maneuver all over the map with almost no restrictions. The simulation fell apart because unrestricted movement eliminated the friction that defined the actual conflict. The game did not work because the VC player had more options than the historical VC ever had. I shelved the project rather than try to patch it. I am not sure I had enough understanding of the subject to fix it. If your movement system is wrong, the whole game is wrong, and sometimes the honest answer is that you do not know enough about the conflict to get it right.

Figure out movement before you start building the rest of the game. Changing your movement system mid-project means changing your scale, your terrain effects, your combat results, and your scenario setups. I have done it. It is miserable. Get it right early.

Formations, Facing, and Posture

The previous chapter covered how units move across the map. This one covers how they orient themselves while doing it. Facing and formation mechanics control what a unit can see, where it can shoot, and how vulnerable it is to attack from different directions. These mechanics produce tactical decisions when they work and bookkeeping when they do not.

When Facing Matters

Facing is a mechanic inherited from miniature wargaming, where physical models on a tabletop have a literal front and back. In hex-and-counter games, facing means the counter points toward a specific hexside or hexspine, and that orientation determines its front, flanks, and rear. Attacks against the flank or rear receive bonuses. Units project ZOC only into their front hexes. The unit can only move into the hexes it faces.

This matters most for pre-twentieth century tactical combat, where formations fought in lines and the direction a unit faced determined its combat effectiveness. A Napoleonic infantry battalion in line could deliver devastating fire to its front but could do nothing if cavalry hit its flank. A Greek phalanx was a wall of shields and spears from the front and a crowd of unarmored men from behind.

Facing loses its grip as warfare modernizes. By 1866, Prussian infantry were going prone and fighting from dispersed positions thanks to the needle gun. A prone skirmish line does not have a meaningful facing the way a Napoleonic line does. By the Second World War, infantry squads could reorient in seconds. ASL, one of the most detailed tactical wargames ever made, omits unit facing. The designers knew that at their scale, a squad's orientation changed too fast to be worth tracking turn by turn.

Ask whether the historical subject demands facing, not whether your game is tactical enough to justify it. If the conflict you are simulating featured formations whose orientation relative to the enemy was a decisive factor in combat outcomes, facing belongs in your game. If it did not, facing is overhead.

Hexside vs. Hexspine Facing

When you include facing, units will point either toward a hexside or toward a hexspine. Both give six possible orientations. The choice affects how facing interacts with the hex grid.

Hexside facing means the unit points at one of the six edges of its hex. Hexspine facing means it points at one of the six vertices. Each option produces a different geometry for front, flank, and rear hexes, and each interacts with turning costs in its own way.

Hex orientation determines which option looks right. With pointy-top hexes, where a vertex points straight up and down, hexspine facing lets two opposing lines point at each other across the map. That looks like a battlefield. Hexside facing in the same grid has them staring past each other at hex edges, which looks wrong even when the mechanics work. Flat-top hexes flip the geometry: hexside facing produces the straight-across alignment and hexspine creates the offset.

I prefer hexspine facing with pointy-top hexes. *Marathon* uses this combination and opposing phalanx lines face each other without any angular awkwardness. The facing type and the hex orientation are a package deal. Pick the combination that makes your map look like the battle it represents. If it looks wrong on the map, players will feel it before they can articulate why.

Formations

Formation mechanics model the difference between how a unit moves and how it fights. Units need one posture to cover ground and a different posture to engage the enemy. Moving in fighting formation is slow. Arriving near the enemy in marching formation is fatal. The cost of switching between these states is one of the most important design levers in a wargame.

At the tactical level, this means column versus line. A battalion in column can move along roads quickly but presents a concentrated target and cannot bring its full firepower to bear. A battalion in line can deliver devastating volleys but moves at a crawl and falls apart crossing rough terrain. The historical challenge for commanders was timing the transition. Deploy too early and you waste hours covering ground in line. Deploy too late and the enemy catches you in column.

At the grand-tactical level, formation can be abstracted into Deployed and Travel modes. The mechanical idea is the same. Travel mode gives higher movement and lower combat capability. Deployed mode does the reverse. At the scale of battalions and regiments, the specific formation matters less than whether the force is configured for movement or for fighting.

In Koniggratz and Mars La Tour, units have March Formation with 4 MP and Battle Formation with 1 MP. Enemy Zones of Influence force March Formation units to flip to Battle Formation and stop. The transition is not optional. You do not choose to deploy. The proximity of the enemy forces deployment on you.

Sedan 1940 uses the same concept of Travel Mode and Deployed Mode but handles the transition differently. The player decides when to switch. Travel Mode offers higher movement but imposes penalties when engaged. The risk is voluntary: you choose to stay in Travel Mode for the speed, knowing that contact with the enemy will punish you for it.

Formation and Facing Together

Some systems tie formation state to facing orientation. In my game on *Marathon*, deployed units face a hexspine while units in march formation face a hexside. The formation a unit is in determines how it orients on the map, which determines its front, flank, and rear hexes.

I tried a more complex version of this in an early tactical variant of *Imperial Bayonets*. Units in column formation faced a hexside, which channeled their movement into the hexes ahead of them. Units in line formation faced a hexspine, sacrificing mobility for firepower across a wider front. On paper it looked elegant. In practice it was a nightmare. The constant tracking of which units faced which direction in which formation, combined with the cost of switching between states, produced a game that was more puzzle than simulation. Hermann Luttmann was kind enough to humor me by helping playtest it at a convention, but the game was not working. The formation-facing interaction created overhead without adding proportional decisions.

Combining facing and formation can work, but only when the interaction is clean enough that players internalize it rather than consult the rules every turn. *Marathon*'s system works because the formation states are simple and the facing rules are short. The early *Imperial Bayonets* variant failed because the tactical scale magnified every interaction into a separate case to manage.

An early version of that same game included a scenario for Bazeilles, a battle from the Franco-Prussian War fought almost entirely from buildings. The scenario still had facing rules. French units garrisoned in houses had front and rear hexes. It made no sense. If your battle does not involve linear formations maneuvering in the open, facing is not adding anything, no matter how tactical the scale.

The Cost of Changing Formation

Formation changes should cost something. The design decision is how much and what kind of cost.

The most common approach is spending movement points. In *Marathon*, a unit must spend its entire movement allowance to change formation. This is a steep cost that makes the decision consequential. You cannot march up to the enemy, deploy, and attack in the same activation. You march, then next turn you deploy, then you fight. The tempo of the approach is built into the formation change cost.

Other systems use partial MP costs. A unit might spend half its movement to change formation, allowing some residual movement afterward. This is more forgiving and works for games where the designer wants formation changes to be a consideration rather than a commitment.

Some games impose combat penalties instead of or in addition to MP costs. In *Marathon*, attacking a unit in march formation grants a +3 DRM to the attacker. The formation change costs your full movement allowance, but getting caught before you complete it is worse. Other systems allow opportunity fire when a unit changes formation within range of the enemy, punishing a force that reorganizes under observation.

Command systems can restrict formation changes, but be cautious about leaning too hard on this. The battalion commander or the unit's NCOs managed formation transitions. Higher command decided where the army went and when it attacked. Generals did not micromanage whether a battalion was in column or line. Unless your system models command dysfunction at the tactical level, formation changes should be a function of the unit's own capability and the situation on the ground, not something that requires orders from headquarters.

Facing and ZOC

In systems with facing, ZOC projects only into a unit's front hexes. A unit does not project a zone of control into its flank or rear hexes. Flanking forces can move through those hexes without stopping.

One useful exception is skirmishers. Light troops historically operated in loose formations, moving and firing in any direction without maintaining a fixed front. In *Marathon*, skirmisher units treat all six adjacent hexes as front hexes. They project ZOC in every direction, can move and fire freely, and have no flank or rear to exploit. This matches how light troops operated compared to heavier line formations.

Flank and rear attacks receive bonuses in most systems that use facing. *Marathon* gives +1 DRM for attacks into an enemy's flank and +2 DRM for attacks into the rear. These modifiers are small enough that flanking is an advantage rather than an automatic kill, but large enough that players work hard to protect their flanks and exploit the enemy's.

Facing also affects retreat. A unit that must retreat will fall back through its rear hexes. If enemy units occupy or project ZOC into those hexes, the retreating unit may be eliminated. Encirclement emerges from the facing and ZOC rules working together, without requiring a separate encirclement mechanic.

The Rightward Drift

Marathon includes an asymmetric turning cost for phalanx units. Turning to the right costs 1 MP. Turning to the left costs 2 MP. This models a phenomenon described by classical scholars: because each man in a hoplite phalanx drifted to the right, trying to stay behind his neighbor's shield, the entire formation had a natural tendency to slide rightward during movement.

I could have modeled this by forcing phalanx units to shift one hex to the right every other turn of movement. That would have been historically precise and mechanically miserable. The asymmetric turning cost captures the same idea. The phalanx turns more naturally to the right because the formation's structure favors it. Players feel the constraint without tracking lateral drift hex by hex.

The phenomenon is real and tactically significant. A direct simulation would produce accurate results and terrible gameplay. The abstracted version communicates the same constraint through movement point costs, a mechanism players already understand, and requires no additional rules.

The Digital Advantage

Many facing and formation mechanics that are too fiddly for cardboard work well in digital implementations. The Wargame Design Studio series handles facing, formation changes, and their interactions with a level of detail that would be unbearable on a tabletop. The computer tracks orientation, calculates modifiers, and resolves formation changes. The player makes decisions. The software handles bookkeeping.

If you find yourself designing a facing or formation system that requires constant reference to charts and edge cases, ask whether you are designing a tabletop game or a computer game. The two media have different tolerances for mechanical complexity, and a system that shines in one can be unbearable in the other.

Choosing What to Include

If you are designing a game and wondering whether to include facing, formations, or both, the decision follows from your scale and subject. Here is how I think about it.

Tactical, ancient or medieval: Facing and formations both. These conflicts were defined by the orientation and posture of the fighting unit. A hoplite phalanx, a Roman maniple, a medieval pike block. Facing determines who can fight and where. Formation determines whether the unit can move or hold. *Marathon* uses both, and the interaction between them is the game.

Tactical, horse-and-musket era: Facing and formations both, but watch the complexity. Napoleonic infantry had column, line, and square, each with distinct capabilities. The design temptation is to model all three plus transitions between them. That can work, but test whether the transitions produce decisions or just bookkeeping. If players spend more time managing formation changes than maneuvering, cut a formation state.

Tactical, twentieth century onward: Formations rarely. Facing rarely. Modern infantry reorients in seconds. ASL omits facing for this reason. If your twentieth century tactical game includes facing, you need a strong argument for why the constraint existed in your specific situation.

Grand-tactical: Formations yes, facing usually no. The distinction between march posture and combat posture is the core design lever at this scale. A brigade in travel mode versus deployed mode creates the right decisions. Facing a brigade counter at a hexspine does not, because the regiments within it are oriented in different directions depending on the tactical situation. Koniggratz, Mars La Tour, and *Sedan* 1940 all use formation states without facing at this scale.

Operational and strategic: Neither. Divisions and corps reorient continuously. Formation state and facing at these scales add management burden without producing decisions. If you need to model the difference between a force on the march and a force in defensive positions at operational scale, handle it through movement modes (strategic movement vs. normal movement) rather than formation counters.

When to Leave It Out

Facing should not be included at the grand-tactical scale unless your game operates at a particularly small scale within that category. A brigade does not have a meaningful front and rear at five miles per hex. The individual regiments within it are oriented in various directions depending on the tactical situation, and the brigade's overall posture is better modeled through formation state than through a counter pointing at a hexspine.

At the operational and strategic level, facing is almost never appropriate. Divisions, corps, and armies orient and reorient continuously. Facing at these scales adds a management burden that produces no interesting decisions.

Even at the tactical level, facing is not automatic. If the conflict you are simulating involved forces that could reorient fast, such as modern infantry or guerrilla fighters, facing models a constraint that did not exist. Include it when the historical subject demands it.

Case Study: *Marathon* and the Cost of Getting It Wrong

Marathon went through three versions of its facing and formation system before I found the one that worked.

The first version used hexside facing for all units. Phalanx lines faced each other at awkward angles on the map. Flanking attacks required counting hexsides and consulting a chart to determine whether the attacking hex qualified as flank or rear. Players stopped to argue about geometry. The facing rules were technically correct and produced miserable play.

The second version switched to hexspine facing, which solved the visual alignment problem. Opposing phalanx lines pointed straight at each other. But I kept a complex formation system with three states: march column, open order, and close order. Each state had different movement costs, different combat modifiers, and different facing rules. Changing between them required spending movement points and passing a command check. Playtesting revealed that players spent more time managing formation transitions than making tactical decisions. The three-state system modeled historical reality and buried it under procedure.

The third version, the one that shipped, uses hexspine facing with two formation states: march and deployed. March formation faces a hexside, deployed faces a hexspine. Changing formation costs your full movement allowance. Getting caught in march formation by the enemy gives the attacker a +3 DRM. The asymmetric turning cost (1 MP right, 2 MP left) captures the rightward drift of hoplite lines. The skirmisher exception (ZOC in all six hexes, no flank or rear) models light troops.

The system works because each rule produces a decision. Face the phalanx toward the enemy or angle it to protect a flank. Deploy now and fight next turn or gamble on one more turn of marching. Commit skirmishers to screen the approach or hold them back for flank protection. The rules are short enough that players internalize them after two turns and stop consulting the rulebook.

The lesson from the failed versions is that historical accuracy in formation mechanics does not guarantee good design. The three-state system was more historically precise than the two-state system. It was also worse. Precision that the player cannot act on is wasted complexity. The two-state system captures what mattered about ancient formation warfare: the tension between mobility and combat readiness, the vulnerability of units caught unprepared, and the physical constraints of maintaining a formation under pressure. It leaves out what did not produce decisions at the game's scale.

If your formation system needs a chart, it is too complex. If players manage it without thinking about it, you have found the right level of abstraction.

Combat Resolution

Combat resolution is the mechanical heart of most wargames. Players maneuver their forces, position for advantage, commit to an attack, and then read the results off a table or resolve a procedure. The CRT determines whether a flanking maneuver pays off and whether a defensive position holds. The rest of the game's systems feed into this moment or flow from it. A combat system that produces historical results makes the whole game feel right. One that does not will undermine the best map, the best order of battle, and the best sequence of play you can write.

This is also the part of design I find most creative. Building a CRT is an exercise in controlled probability. You are deciding what can happen, how often it should happen, and what it costs both sides. Few resources explain how to do this well. Richard Berg wrote a piece on the subject in SPI's Staff Study #2 that remains one of the only serious treatments I know of. Part of the reason more does not exist is that there is no single correct method. The other reason is that designers who are good at it learned by studying other people's tables and iterating in spreadsheets until the results looked right.

The Odds-Based CRT

The odds-based combat results table has been the standard approach since the early days of SPI and Avalon Hill. The mechanic is straightforward. Total the attacking strength points, total the defending strength points, and create a ratio. If 12 SP attack 4 SP, the ratio is 3:1. If 9 SP attack 6 SP, the ratio is 1.5:1, which most classic systems round down to 1:1 in favor of the defender. Some publishers round up at the half, including MMP's Standard Combat Series. *Rostov '41*, which I designed for SCS, uses this convention. Both approaches work. Rounding down is more traditional and gives the defender a slight edge in borderline situations. Rounding up is more forgiving to the attacker. It is a matter of preference. Find the column on the CRT that matches the ratio, roll a die, and cross-reference the result.

The system scales well. A 3:1 attack means the same thing whether you are pushing three divisions against one or thirty battalions against ten. The table does not distinguish between absolute numbers, only the relationship between forces. This makes odds-based tables suitable for scales from tactical firefights to strategic theater-level campaigns.

The columns on a typical CRT run from low odds on the left to high odds on the right. A 1:2 attack is desperate and will fail most of the time. A 3:1 attack is strong and will succeed most of the time. The results shift from bad for the attacker on the left to bad for the defender on the right. At low odds, the attacker risks elimination or retreat. At high odds, the defender faces the same. The middle columns produce uncertain outcomes where both sides might take losses.

If you want to understand how CRTs work, study the classics. SPI's *The Marne*, *Soldiers*, and *Battle for Germany* all have CRTs worth examining. Those tables will teach you more about combat resolution than reading about it in the abstract.

Differential CRTs

The alternative to odds-based resolution is the differential table. Instead of creating a ratio, you subtract the defender's strength from the attacker's strength and use the resulting number as the column. If 10 SP attack 6 SP, the differential is +4. If 6 SP attack 10 SP, the differential is -4.

Differential tables eliminate the rounding that odds-based tables require, and they handle low-strength combat situations with more precision. Two units with 1 SP each fighting on an odds-based table produce results identical to two 20 SP stacks fighting at the same 1:1 ratio. On a differential table, the absolute size of the forces can matter, which some designers prefer.

The weakness is that differential tables can produce strange results at the extremes. The gap between +2 and +4 feels the same on the table, but the tactical situations producing those numbers can be different in ways the resolution does not capture. Hermann Luttmann uses differential tables in several of his designs and makes them work well. One trick I have borrowed from his approach is applying column shifts to the differential table based on traditional force ratios. If the attacker has a 3:1 advantage, shift the column right by two. You get the precision of a differential calculation with the intuitive scaling of odds-based ratios. *Prigozhin's March of Justice* uses this hybrid approach, and the table produces results that feel right across a wide range of combat situations.

I use differential tables in some of my more strategic games and sometimes in the context of tactical assaults within a larger operational system. For most designs, I prefer odds-based resolution for its simplicity and the half century of proven track record behind it.

Dice Resolution

The die you choose shapes what your combat system can do. A single d6 gives flat probability. Each number from 1 to 6 has the same chance of appearing. A 2d6 roll produces a bell curve where 7 is the most common result and 2 or 12 are rare. This is not a cosmetic distinction. It changes how players experience combat and how you as the designer can tune outcomes.

Flat dice produce volatile results. On a 1d6, the chance of rolling a 1 is the same as rolling a 6. Extreme results show up often enough that players cannot rely on favorable odds to guarantee outcomes. I prefer 1d6 or 1d10 for games where I want combat to feel unpredictable. A Napoleonic battle where a 3:1 attack can still collapse captures something true about warfare. Commanders plan, commit forces, and sometimes the plan falls apart on contact.

A bell curve gives you finer control over outcomes. On 2d6, rolling a 7 happens about 17% of the time, but rolling a 2 happens less than 3% of the time. Middle results dominate. Small modifiers produce larger effects near the center of the curve because each +1 or −1 shifts a bigger slice of probability than it would on flat dice. I use bell curves in games where I want player agency to carry more weight than randomness. If you build a good plan and achieve a favorable ratio, a 2d6 table pays that off more consistently.

ASL uses 2d6 for all resolution, which fits a game that functions more as a competitive tactical exercise than a strict simulation. Skillful play produces consistent results on a bell curve. There is no wrong answer. Use flat dice when you want chaos and consequence. Use bell curves when you want the better plan to win more often.

Combat Results

The results on your CRT determine what combat does to the units involved. The most common result types are step losses, retreats, exchanges, and eliminations. Each simulates something different, and the mix you choose sets the tone for the game.

Step losses reduce a unit's combat effectiveness. In many games, flipping a counter to its reduced side represents this. A step loss does not mean casualties in the literal sense. It represents the degradation of a formation's ability to fight: fatigue, disorganization, loss of key leaders, ammunition expenditure. Combat efficiency breaks down before a unit is destroyed. Using step losses as your primary result produces attritional games where forces wear down over time.

Retreats move the losing unit backward without destroying it. This is often the most common result at lower odds ratios. A 1:1 attack that produces a defender retreat is a success for the attacker without being catastrophic for the defender. In my World Undone series covering the opening campaigns of 1914, eliminations and reductions are rare. Forces rubberband back and forth as a result of battles, which matches what happened in those campaigns. The massive armies of 1914 fought, fell back, regrouped, and fought again. They did not annihilate each other in single engagements.

Exchanges require both sides to take losses. One common version eliminates the defender and requires the attacker to lose strength points equal to half the defender's total. Another version gives each side a single step loss. I use exchanges to introduce variance into CRTs so the results do not map too neatly along expected parameters. An exchange at 3:1 odds makes the attacker pay for a victory they expected to win clean. Players cannot treat high-odds attacks as automatic and free when exchanges appear in those columns.

Eliminations remove a unit from the game. These should be rare at most scales and in most conflicts. The threat of elimination gives teeth to the combat system, but if most fights can destroy a unit, the game becomes a knife fight where one bad roll ends a player's position. Calibrate how bloody your CRT is by looking at historical casualty rates for the conflict you are simulating. If the battle you are modeling produced 10% casualties, your CRT should not be eliminating a quarter of the forces engaged per turn.

Scale matters for result magnitudes. A defender retreating three hexes makes sense at tactical scale where each hex is a few hundred meters. At strategic scale where hexes are fifty miles across, a three-hex retreat represents an army falling back 150 miles from one engagement. Match your result types and magnitudes to your hex and turn scale.

Column Shifts and Die Roll Modifiers

Column shifts and DRMs both adjust combat outcomes, but they work at different scales of effect and serve different design purposes.

A column shift moves the attack to a different odds column on the CRT. A one-column right shift turns a 2:1 into a 3:1. This changes the entire probability distribution, not a single number on the die. Column shifts should represent major tactical realities. Mountain terrain granting a two-column left shift in the defender's favor should feel significant when it applies. I keep column shifts scoped tight in my designs. Granting two shifts is acceptable for significant terrain like mountains or fortified cities, but stacking three or four shifts starts to undermine the odds calculation that the CRT is built on.

DRMs add or subtract from the die roll. A +1 DRM shifts the result one row on the table. This is a finer adjustment than a column shift and suits situations where the modifier represents an advantage without transforming the tactical picture. Combined arms bonuses, leader effects, flanking attacks, disruption penalties: these work well as DRMs. They add granularity to the combat system without the sledgehammer effect of changing the odds column.

Terrain and weather benefit the defender in most conflicts across most eras. A unit dug in on a ridge or defending a fortified city should be harder to dislodge, and your modifiers should reflect that. Positioning, combined arms, and concentration of force benefit the attacker. These are the tools an attacking player uses to create favorable odds, and the modifier system should reward good use of them.

The interaction between column shifts and DRMs gives your combat system its tactical texture. A player who maneuvers a flanking force into position to earn a +1 DRM while avoiding an attack across a river that would cost -1 DRM is making the kind of decision your game should reward.

Artillery and Support Fire

How you handle artillery depends on scale. At tactical and grand-tactical scales, artillery units often appear as separate counters with their own combat mechanics. Bombardment tables and fire missions are distinct systems because at that scale, how a commander uses their guns is a separate decision from how they commit their infantry.

At operational and strategic scales, a division's combat strength includes its organic artillery. Separate artillery mechanics at these scales add complexity without adding decisions that matter at the level of abstraction you are working at. Exceptions exist. If a particular conflict featured artillery as a decisive independent arm, as heavy artillery was at Verdun, a separate system may be warranted.

Some games split the difference. *Imperial Bayonets* includes bombardment as a separate phase before combat, allowing artillery to disrupt defenders or create obligations that shape the main combat phase. PCS incorporates air support as a modifier within the overall combat system rather than a standalone resolution. The question you should ask is whether artillery use represents a separate decision in the conflict you are simulating, or whether it was one component of an integrated assault.

Making Combat Match the Conflict

A Napoleonic battle should not feel like a World War I battle, and neither should feel like a modern armored engagement. The CRT is your primary tool for creating that distinction.

Napoleonic combat produced modest casualties in most engagements but generated catastrophic results when a formation broke. A CRT for a Napoleonic game should have retreat results dominating the middle columns with step losses appearing at extreme odds and devastating exchanges scattered through the table. Battles hinged on whether a formation held or ran, not on the physical destruction of the force.

World War I combat on the Western Front was attritional. Attacks produced casualties on both sides regardless of who held the ground at the end. A CRT for the Western Front should feature mutual losses as common results, with defender retreats being rare and hard-won. The attacker should expect to pay for each hex gained.

Modern maneuver warfare produces rapid, decisive results when one side achieves tactical surprise or technological superiority, but bogs down into attritional fighting when forces are matched. A CRT for a modern conflict might use high variance: devastating results at favorable odds, stalemates at parity.

Research is the only way to know how a particular conflict should feel. You cannot design a CRT for a war you have not studied. The casualty data and the campaign narratives tell you what combat looked like, and your table needs to reflect that.

Diceless Combat

Not all combat resolution uses dice. Chit pull and card-based systems produce different kinds of uncertainty.

In a chit pull combat system, the randomness comes from drawing a value from a cup rather than rolling a die. PCS uses this approach. Each unit draws a combat chit at the moment of engagement, and the drawn value represents the unit's performance in that fight. A good unit can draw a low chit and a mediocre unit can draw a high one. The randomness lives inside the unit's capability rather than in an external die roll applied to the resolution.

Card-based combat, as used in games like *Up Front* and *Combat Commander*, resolves combat through card draws or card play. The deck functions as both randomizer and resource, because the cards you spend on combat are cards you cannot spend on other actions. This built-in tension between spending and saving does not exist in dice-based systems.

These are different tools for different design goals. Diceless combat tends to produce games where the resolution mechanic is more integrated with the rest of the game's systems, and where uncertainty distributes across the game in ways that dice cannot replicate.

Designing Your First CRT

Your first CRT should be inspired by one you like. Find a published game with a combat system that appeals to you and use its CRT as a starting point. This is how designers in this hobby have learned for decades. Eighty percent of wargame design is building on the work of those who came before you, and the CRT is no exception.

My process is simple. I open a spreadsheet, write out the odds ratios as columns, and start filling in results. I iterate through different combinations until the table looks right to me. There is no formula. I am asking myself questions as I work: what should happen at 1:1? How bad should a 1:2 attack be? At what ratio does the defender start getting destroyed? How often should exchanges appear? I fill in results, look at the overall pattern, adjust, and repeat.

Start simple. A basic odds-based CRT with retreat and step loss results is enough. You can add column shifts for terrain later, DRMs for modifiers later, exchanges and special results later. Get the basic shape right first. Does a 3:1 attack succeed most of the time? Does a 1:1 attack feel like a coin flip? Does a 1:3 attack feel desperate? If the answers feel right, your CRT is working.

To make this concrete, here is a simple 1d6 CRT for a hypothetical operational game. Seven columns, five result types.

1d6	1:2	1:1	2:1	3:1	4:1	5:1	6:1
1	AE	AR2	AR1	EX	DR1	DR1	DR2
2	AR2	AR1	EX	DR1	DR1	DR2	DR2
3	AR1	EX	DR1	DR1	DR2	DR2	DE

1d6	1:2	1:1	2:1	3:1	4:1	5:1	6:1
4	AR1	DR1	DR1	DR2	DR2	DE	DE
5	EX	DR1	DR2	DR2	DE	DE	DE
6	DR1	DR2	DE	DE	DE	DE	DE

AE = Attacker Eliminated. AR = Attacker Retreat (number of hexes). EX = Exchange (defender eliminated, attacker loses half the defender's SP). DR = Defender Retreat. DE = Defender Eliminated.

Three design decisions in this table worth examining. First, exchanges appear in the lower odds columns (1:2 through 2:1) but not the higher ones. At 1:2, the exchange sits on a roll of 5, meaning the attacker can still hurt the defender even at bad odds, but they pay for it. At 3:1, the exchange sits on a roll of 1, the worst result for a strong attack. Above 3:1, exchanges disappear because an attacker who has assembled overwhelming force should not be bleeding for it. Where you place exchanges determines how punishing combat feels at each odds level.

Second, the 1:1 column has no Attacker Eliminated result. A 1:1 attack is risky (you can retreat two hexes on a roll of 1) but it will not destroy your force. In many operational campaigns, commanders accepted 1:1 attacks when positioning demanded it. If 1:1 carried an elimination risk, players would never attack without a 2:1 advantage, and the game would stall. Removing AE from the 1:1 column encourages players to commit to borderline attacks, which produces more dynamic play.

Third, look at the high-odds columns. At 5:1 and 6:1, Defender Eliminated shows up on rolls of 4, 5, and 6 (or 3 through 6 at 6:1). A player who masses enough force to achieve 6:1 should expect to destroy the defender most of the time. If the high-odds columns are too gentle, there is no incentive to concentrate force. If they are too harsh, small units become disposable and players stop caring about individual formations. The balance between those extremes is where your game's personality lives.

This table is a starting point. Playtesting will tell you whether the 3:1 column is too bloody, whether retreats are too common at 1:1, whether the high-odds columns feel decisive enough. Adjust column by column until the results match the conflict you are simulating.

Expect the CRT to change more than any other component during development. Playtests will reveal problems your spreadsheet cannot. Results that seemed reasonable on paper will produce strange outcomes on the map. Adjust, playtest again, adjust again.

Once you are comfortable with established systems and understand what makes them work, begin to experiment. Introduce mechanics from games you admire. Combine approaches. Try a differential table with ratio column shifts. Try a dice pool system. The proven systems exist because they work, but the best games in this hobby have come from designers who understood the conventions well enough to deviate from them with purpose.

Case Study: The *Procedural Combat Series*

PCS uses a combat system that shares little with my other designs. There is no printed CRT. Combat is differential-based, resolved through a procedural sequence rather than a single table lookup.

The core philosophy is unit quality over quantity. Combat interactions come down to one lead unit engaging one target defending unit. Adjacent friendly units provide support bonuses, but the fight itself is between two formations. This reflects how modern combat works at the operational level, where an engagement is dominated by the units in contact and supported by those nearby, rather than a mass of forces attacking at once.

Each unit draws a random combat chit at the moment of engagement. The chit value, combined with combat support adjustments from adjacent units, terrain, air support, and movement points spent approaching the target, produces a final combat strength for each side. Both sides then roll a d8 and add it to their combat strength. The difference between the two totals is the combat differential, which the players use to determine losses through a loss ratio.

The random chit draw gives PCS its identity. A unit's combat strength is not a fixed number printed on the counter. It varies from engagement to engagement. A player can commit forces to an attack and not know how effective they will be until the chit comes out of the cup. This models the fog and friction of modern combat, where the outcome of an engagement depends on factors no commander can predict or control.

Unit Quality Ratings differentiate between elite, regular, and poor formations. A higher-quality unit receives bonuses to its combat support adjustments. Quality matters more than mass in PCS, which matches how modern combined-arms warfare works in practice. A well-trained brigade with air support and favorable terrain can defeat a larger but less capable force.

I designed this system because traditional CRTs handle modern subjects poorly. An odds-based table treats combat as a contest of mass, which fits conflicts where massing force at the point of contact was the primary tactical challenge. Modern combat rewards positioning, quality, and combined arms integration. PCS makes those factors the core inputs to resolution rather than secondary modifiers on a ratio-based table.

Sequence of Play

A sequence of play is an algorithm. You write a set of ordered instructions that allow at least one or more human beings to run the simulation you have designed. A player opens your rulebook and reads "Phase 1: Weather Determination. Phase 2: Supply Check. Phase 3: Movement. Phase 4: Combat. Phase 5: Exploitation." That player is reading a program written in natural language. The sequence tells them what to do, in what order, and when control passes from one player to the other. If the algorithm is clear, the game runs. If it is ambiguous or poorly ordered, players will spend more time arguing about procedure than making decisions about the battle.

The sequence of play is one of the more useful contributions wargaming has made to game design as a whole. Board games benefit from clear turn structure, but wargames formalized it into something closer to a flowchart. A good SOP lays out how the game works in practical terms. New players read it to understand what they are going to do on their turn. Experienced players reference it when a timing question comes up mid-game. Every other system in your design, movement, combat, supply, command, plugs into the sequence of play. You build the SOP and hang everything else off it.

IGO-UGO

The most common sequence structure in wargaming is IGO-UGO: one player completes their entire turn, then the other player completes theirs. The phasing player moves all their units, resolves all their combats, handles administrative tasks, and passes control. The non-phasing player watches, may roll dice when required, and waits.

IGO-UGO has dominated the hobby since SPI and Avalon Hill established the conventions, for good reason. It is easy to learn, easy to teach, and easy to develop for. You as the designer know when each system activates because the sequence is linear and predictable. Movement happens in the movement phase. Combat happens in the combat phase. No ambiguity about timing. For the player, IGO-UGO provides a complete picture of the situation before they commit to action. You see where every enemy unit sits, calculate your odds, and execute your plan.

Most of my games use IGO-UGO to some degree. *The World Undone* series, *Imperial Bayonets*, the LCOS games, all of them give one player a full turn before the other player acts. I keep coming back to IGO-UGO because it works. The structure is transparent, the timing is unambiguous, and I spend my design effort on the systems rather than the meta-structure holding them together.

The weakness is passivity. The non-phasing player has nothing to do for extended stretches. In a complex game with a large order of battle, one player's turn can take thirty minutes or more. The other player sits there. Competitive players use that time to study the board. Casual players disengage. This is a real problem, and it is one reason designers have explored alternatives.

The other weakness is predictability. The phasing player knows they will complete all their movement before any combat occurs.

They know no enemy unit will interrupt their plan. That certainty is unrealistic. Real commanders did not get to move every unit on the battlefield before the enemy responded. But the abstraction works because the turn represents a block of time during which both sides were acting at once, and IGO-UGO is a practical way to resolve that simultaneity in a sequential format.

Impulse Activation

Impulse activation breaks the turn into smaller segments. Instead of one player moving everything and then fighting, each player activates a portion of their force, a single unit or a formation or a stack, and then control passes. The turn becomes a series of alternating impulses until both players have activated everything or chosen to stop.

The benefit is engagement. Neither player sits idle for long. You activate a unit, I activate a unit. The back-and-forth keeps both players involved. In an IGO-UGO game, you plan and then execute. In an impulse game, the situation changes between each of your activations, so you are recalculating after every opponent move.

Impulse activation also introduces cost. If activating a unit costs something, action points, command tokens, a card from your hand, then you face decisions about what to activate and in what order. Do you move that formation now or save the activation for a reaction later? IGO-UGO does not ask that question. Every unit activates for free during its phase.

The design challenge with impulse systems is pacing. If each impulse involves a single unit, the game might crawl. If impulses involve entire formations, you are back to something close to IGO-UGO with extra steps. A tactical game where individual squads activate one at a time feels right. An operational game where individual divisions activate one at a time gets tedious if there are forty divisions on the map (PCS is guilty of this sometimes). Find the right granularity for your subject.

Chit Pull

Chit pull deserves more attention from designers than it gets. The basic mechanic: place activation markers for each eligible formation into an opaque cup. Draw one at random. That formation activates. Draw the next. Continue until the cup is empty.

Chit pull simulates the unpredictability of conflict better than any other structural mechanic I have used. You cannot rely on a specific formation to activate when you need it. Your plan requires the 2nd Panzer Division to move before the enemy infantry digs in, but its chit might be the last one drawn. That uncertainty changes decision-making at every level. You cannot build a plan that depends on precise sequencing because you do not control the sequence. You have to build plans that work across a range of activation orders, which is closer to what real commanders face.

In *Sedan 1940*, I built a non-linear chit-pull driven sequence of play. Your activation chit gets pulled and you can move, fire, assault, enter overwatch, or do whatever else the rules allow. No fixed phase structure within the activation. The game was inspired by John Tiller's Panzer Campaigns PC series, which I admired for the fluidity of its activations, but I combined that fluidity with chit pull to add the unpredictability that a deterministic PC game could not provide. The 1940 campaign in France was fast, disorienting, and depended on which commanders seized initiative at the right moment. The activation system produces that tempo without scenario rules having to force it on top of providing a hook for replayability and solitaire players.

I would encourage designers to explore chit pull across many different systems and subjects. At tactical scale, squad-level activation creates the chaos of small-unit combat. At operational scale, formation-level activation models the friction of coordinating a multi-corps offensive. At strategic scale, theater-level activation captures the competing priorities of a high command that cannot do everything at once. The mechanic is adaptable and intuitive. It produces uncertainty that makes games replayable without adding rules overhead.

A Note on Card-Driven Games

Card-driven games are sometimes discussed as a separate category of activation system. That framing is misleading. A CDG can be IGO-UGO, impulse-based, or use a form of chit pull depending on how the cards function. The cards are a resource and randomization layer that sits on top of one of the activation structures discussed above. Mark Herman's We the People and its descendants use cards for both operations and events, but the underlying turn structure varies from game to game. CDGs deserve their own chapter, and I will get to that later.

Reactions and Opportunity Actions

The most effective way to solve the passivity problem in IGO-UGO games is to give the non-phasing player things to do during the phasing player's turn.

Reactions, opportunity fire, interceptions, and reserve activations all serve this purpose. They break the normal sequence of play in controlled ways. The defending player responds to threats as they develop rather than waiting for their own turn. Real defenders did not sit motionless while the enemy maneuvered. They fired at approaching troops, repositioned reserves, and attempted to disrupt attacks before they materialized.

In *Mons 1914*, the non-phasing player can conduct opportunity fire against units moving within range. The BEF's rifle fire in August 1914 was so rapid and disciplined that a popular trope is that German commanders believed they were facing machine guns (though that could just be a bit of historical propaganda by the British). Opportunity fire models that capability without giving the British player a full turn of action out of sequence. The German player moves and the British player shoots at them as they move. Simple, historical, the British player stays engaged during the German turn and vice-versa.

DAMOS handles reactions differently. Non-phasing units within a Zone of Influence can attempt reaction movement when enemy units approach. The reacting player rolls against the unit's quality to see if the reaction succeeds. Better-trained formations react faster and more reliably. A Soviet rifle division in 1941 is unlikely to react in time. A German panzer division almost certainly will. The same mechanic applied to different quality ratings produces that asymmetry.

Great Northern War includes interception mechanics where forces can attempt to intercept enemy movement. In the LCOS series, units can intercept when enemies move into a EZOC. These are narrow responses to specific triggers, not full activations. The non-phasing player acts only in defined circumstances, which keeps the SOP predictable while adding meaningful interaction.

Kevin Zucker instilled in me a healthy skepticism of chrome, and that skepticism applies to reaction systems as much as anything else. A reaction mechanic should solve a specific problem. If your game has a passivity issue, reactions help. If your simulation needs to model a defender's ability to respond to threats, reactions help. But adding reactions because they seem like a good idea adds complexity to every single turn of the game. The non-phasing player now evaluates triggers after every move. The phasing player anticipates reactions when planning movement. If the reaction mechanic does not pay for its complexity with better gameplay or better simulation, cut it.

Timing and Phase Order

The order in which phases occur within a turn is not arbitrary. It is a design decision with consequences for how the game plays.

Consider supply. If you check supply at the beginning of the turn before movement, units out of supply suffer penalties before the player has a chance to reconnect them. Supply disruption becomes more punishing. If you check supply at the end of the turn after movement, the player can maneuver to restore supply lines before the check occurs. Disruption becomes a temporary inconvenience. Neither approach is wrong, but they produce different games. The timing of the supply check determines how much weight supply carries as a strategic concern.

The same logic applies to recovery from disruption. If disrupted units recover at the start of their turn, disruption is a one-turn penalty. If they recover at the end of their turn, they spend a full turn impaired. If recovery requires skipping movement or combat, disruption becomes a serious operational cost. I included disruption in *Imperial Bayonets* and the timing of recovery was critical. French units hit by Krupp artillery fire needed to spend time recovering before they could act again. If recovery had been instant at the start of the turn, disruption would have been meaningless. The whole point was modeling the shock of breechloading artillery on troops that had never experienced it.

Zucker was skeptical I should include disruption in *Imperial Bayonets* at all. His argument was that the system should handle the effects of artillery holistically, that if the combat results and modifiers were designed right, you would not need a separate disruption state. I disagreed. Disruption was one of the easier lightweight ways to model the specific effect of French troops getting shattered by Krupp artillery at ranges they could not answer. The Chassepot outranged the needle gun, but it could not answer Krupp steel. A simple disruption status captured that without building a separate artillery subsystem. I still think I was right, but Zucker's instinct to question mechanical additions was sound. Every phase you add to the sequence of play is a phase that runs every single turn. If it does not carry its weight, remove it.

Reinforcement timing matters too. If reinforcements arrive at the start of the turn, they are available for the full turn's movement and combat. If they arrive at the end, they sit on the map but cannot act until next turn, giving the opponent a chance to react to their arrival. Reinforcements that arrived and threw themselves into the battle should appear at the start. Reinforcements that arrived in the rear and needed time to deploy could appear at the end, but I recommend sticking with one implementation.

Weather determination belongs at the start of the turn, before any other phase. Weather affects movement costs, combat modifiers, and sometimes supply. Players need to know the weather before they make decisions. Putting weather determination anywhere else creates situations where a player commits to a plan and then discovers the weather invalidates it, which is frustrating rather than interesting. Weather, reinforcements, supply: these are administrative phases. They handle the historical and logistical context of the turn. They are the infrastructure that makes the simulation work.

The SOP as Design Output

Your sequence of play will emerge from the game you are building. You design a movement system, a combat system, a supply system, and the SOP is the structure that tells the player when to use each one. The SOP is an output of the design process, not something you decide in advance.

But timing matters, and the order you choose has downstream effects that are easy to miss. I have found it useful to write out the full sequence of play early in development, even before all the systems are finalized, and revisit it as the design evolves. Moving a phase from one position to another can change how the entire game feels. Swap the order of combat and movement, letting players fight first and then advance rather than moving into position and then fighting, and you have a different game. Both structures exist in published designs. The choice between them should be deliberate.

The test for whether your SOP is working is simple. Set up a game turn and walk through it. Does the sequence feel logical? Does each phase lead naturally into the next? Are there phases where nothing happens most turns? If a phase is irrelevant in most turns, consider making it conditional rather than mandatory. “Check supply if any unit is not in range of a supply source” is better than “Check supply for every unit every turn” if most units are always in supply.

Administrative Phases vs. Chrome

There is a distinction worth drawing between administrative phases and chrome, because the line between them affects how you build your SOP.

Administrative phases handle the historical infrastructure of the conflict. Weather determination, reinforcement arrival, supply checks, victory point tallying. These are the framework within which tactical and operational decisions occur. An Eastern Front game without supply is not simulating the Eastern Front. A Napoleonic game without reinforcement schedules ignores the arrival of forces that shaped the battle's outcome. These phases belong in the SOP because the simulation requires them.

Chrome is mechanical granularity that adds detail without adding decisions. Morale recovery rolls for every unit every turn. Detailed ammunition tracking. Disruption states that exist because the designer thought they were interesting rather than because the simulation demanded them. Chrome has a place in wargame design. I have used it when it solves a specific problem, as disruption does in *Imperial Bayonets*. But every piece of chrome you add to the SOP runs every turn, and players feel the weight of it.

Ask about any phase in your SOP: does this phase produce a decision or a result that matters? If a weather roll changes movement costs and combat modifiers in ways that force the player to adapt their plan, the weather phase carries its weight. If a weather roll produces "clear" ninety percent of the time and the other ten percent changes nothing meaningful, the phase is dead weight. Cut it or make it matter.

Building a Non-Linear SOP

Most wargame SOPs are linear. Phase 1 leads to Phase 2 leads to Phase 3. This is the default because it is the easiest structure to explain and execute. Non-linear SOPs exist, and they produce experiences that linear structures cannot.

Sedan 1940 is the most radical example in my own work. No fixed phase order within an activation. A formation's chit gets drawn and the activating player chooses what to do: move units, conduct fire, launch assaults, place units in overwatch, or any combination. The "sequence" is whatever the player decides based on the tactical situation. The 1940 campaign in France was characterized by commanders at every level making rapid decisions about what to prioritize in the moment. Guderian did not wait for a movement phase before ordering an assault. He saw an opportunity and acted. The non-linear SOP lets the player do the same.

The tradeoff is cognitive load. A linear SOP tells the player what to do next. A non-linear SOP asks them to decide what to do next. For *Sedan 1940*, that cognitive load is the game. The decisions about what to do with your activation are the most interesting choices the player makes. A non-linear SOP applied to a game where the interesting decisions live elsewhere, in combat planning or strategic resource allocation, would add complexity without adding fun.

Writing It Down

The physical presentation of the SOP matters more than designers recognize. A player aid card with the complete sequence of play, formatted clearly, is essential for any game with more than a handful of phases. Players will reference it every turn for the first several games and periodically after that.

Write the SOP in clear, direct language. Be precise about what is mandatory and what is optional. "Move eligible units" tells the player this is the movement phase. "Units may be moved" tells them movement is optional, which matters if you do not want to force every unit to act. "Roll for weather" is better than "Weather is determined" because it tells the player who does what. The SOP is an instruction manual for running the simulation, and word choice in instructions carries mechanical weight. "Move all units" and "units may be moved" are different rules, not different phrasings of the same rule.

Number every phase and sub-phase. Use consistent formatting. If phases have conditional triggers, state the conditions explicitly. "If any unit is in an enemy ZOC at the start of this phase, resolve ZOC attrition" is clear. "ZOC attrition may occur" is not.

Every sequence of play is an algorithm that allows human beings to run the model you have designed. You are writing a program in natural language. The quality of that program determines whether players can execute your design or spend the evening flipping through the rulebook looking for clarification. The clearer and more well-designed the SOP, the easier time players will have running the game. Treat it with the same rigor you bring to your CRT or your movement system.

Common SOP Patterns

To make this concrete, here are the structural patterns I see most often and have used in my own work.

The Classic Wargame Turn: 1. Weather/Administrative Phase 2. Reinforcement Phase 3. Supply Phase 4. Movement Phase 5. Combat Phase 6. Exploitation/Mechanized Movement Phase 7. End Phase (victory check, marker removal)

This is the SPI/AH standard. Thousands of games use some variant of it. The structure has survived because it maps onto how military operations unfold: logistics first, then maneuver, then combat, then exploitation of results.

The Double-Impulse Turn: 1. First Player Movement 2. First Player Combat 3. Second Player Movement 4. Second Player Combat

This is IGO-UGO with both players acting within the same turn. Many of my games use this structure. It keeps the simplicity of IGO-UGO while ensuring both sides act within each game turn rather than alternating full turns.

The Chit-Pull Turn: 1. Administrative Phase (weather, supply, reinforcements) 2. Activation Phase (draw chits until cup is empty; each activation allows movement and/or combat) 3. End Phase

This is the *Sedan 1940* pattern. The administrative work happens once at the top, and then the game is pure activation until the turn ends. The simplicity of the outer structure compensates for the complexity of the non-linear activations within it.

The Reaction-Enhanced IGO-UGO Turn: 1. Phasing Player Movement (non-phasing player may conduct opportunity fire/reactions) 2. Phasing Player Combat 3. Non-Phasing Player Reaction Phase (limited movement and/or combat for reserve units) 4. Swap phasing player and repeat

This is close to what *Mons 1914* and Verdun use. The reaction opportunities break the passivity of pure IGO-UGO without abandoning the structure entirely.

These are templates, not prescriptions. Your game's subject and your design goals will tell you which pattern to start with. Playtesting will tell you how to modify it.

Supply, Morale, and Other Subsystems

Every wargame has a mechanical core: movement, combat, sequence of play. Those systems carry the game. But the subsystems surrounding that core determine whether the simulation feels like the conflict it represents or like a puzzle with military graphics. A game about the Eastern Front without supply pressure is not a game about the Eastern Front. A Napoleonic battle without morale collapse is missing the mechanism that decided most engagements of that era. Supply, morale, command, weather: these are the connective tissue between your core mechanics and the historical reality you are trying to model.

I have heard designers say that most games get supply wrong. I disagree. Supply is one of the more straightforward things to model within a reasonable degree of plausibility. It does not require exotic mechanics or elaborate subsystems. What it requires is that the designer understand what supply is doing in the simulation and implement it at the right level of abstraction for the game's scale. The same principle applies to morale, command, and every other subsystem discussed in this chapter. Include what the simulation demands. Abstract what it does not. Cut what adds complexity without adding decisions.

Supply

Supply is a gate. Its purpose in your design is to force players to operate within the logistical constraints that real commanders faced. At tactical scale, supply rules are rarely necessary. Turns representing five to thirty minutes of combat do not create situations where units run out of ammunition or food. The logistical infrastructure supporting a tactical engagement exists below the resolution of the game, handled by quartermasters and supply sergeants the player never sees. Supply becomes a design concern at grand-tactical scale and above, where turns represent hours or days and the question of whether a formation can sustain operations over that timeframe becomes relevant.

A player who can attack anywhere on the map with full strength regardless of where their supply lines run is not making the decisions that the historical commander made. Supply forces the player to think about where their forces are relative to their rear area, which roads and railheads matter, and what happens when those lines get cut. If your game is about a conflict where logistics shaped the campaign, supply needs to be in the design. If logistics were not a significant constraint, abstract their effect into other systems and move on.

The simplest supply model, and the one I use most often, is the supply trace. A unit traces a line of hexes from its position back to a supply source: a map edge, a city, or a depot. If the trace is unblocked by enemy units or their zones of control, the unit is in supply. If the trace is blocked, the unit is out of supply and suffers penalties. The penalties vary by game: reduced combat strength, restricted movement, inability to attack, eventual elimination if unsupplied for multiple turns.

The supply trace works because it creates a spatial problem. The player does not just think about where to attack. They think about the path between their attacking units and the source keeping them fed and armed. Cutting an opponent's supply line becomes a viable strategy, sometimes more effective than direct assault. An armored spearhead that drives deep into enemy territory and severs a supply line can paralyze an entire sector of the front without firing a shot. That kind of operational decision is exactly what supply mechanics should produce.

Zones of control interact with supply in ways that need careful handling. In most of my games, an enemy ZOC blocks supply trace unless the hex is occupied by a friendly unit. This is important. If ZOCs block supply unconditionally, a single enemy unit can cut supply to an entire army by sitting on a road hex, which is unrealistic at most scales. Friendly units in the hex negate the ZOC for supply purposes because in reality, a garrison or rear-area force would keep the road open. This exception matters more than designers sometimes realize.

At a granular level, supply shortages affect different unit types differently. A World War II armored division that cannot trace supply might be immobile. No fuel means the tanks do not move. An infantry division in the same situation can still march and fight, with reduced effectiveness because their ammunition and rations are running low. If your game operates at a scale where this distinction matters, your supply rules should reflect it. A blanket "out of supply units have halved strength" is adequate for many games, but if you are designing an operational treatment of a mechanized campaign, differentiating between fuel-dependent and foot-mobile formations adds a meaningful layer of decision-making.

The timing of the supply check within your sequence of play matters, as I discussed in the previous chapter. Checking supply at the start of the turn before movement means a player cannot maneuver out of a supply crisis before it takes effect. Checking at the end means disruption is temporary and correctable. Neither is wrong, but the choice determines how punishing supply problems are, and that should match the historical weight logistics carried in your conflict.

Some games track supply with more granularity than a simple trace, and some of those games are better for it. Der Weltkrieg, the massive World War I strategic series published by Schroeder Publishing & Wargames, tracks supply by subtracting supply points for attacks and defense. That sounds tedious, and in a less careful system it would be. But Der Weltkrieg is operating at a scale and complexity level where that granularity pays off. The supply point expenditure forces players to husband their resources and choose where to commit offensive effort with real economic weight behind the decision. The system works exceptionally well for what it is trying to do.

DAMOS, my operational series, takes a modular approach to supply that I think illustrates how supply mechanics should be tailored to the theater being simulated. The supply system is not standardized across the series. Each title implements supply rules specific to its conflict, because the logistical challenges of Barbarossa have almost nothing in common with the logistical challenges of North Africa.

In the Barbarossa titles, supply revolves around rail gauge conversion. The Germans must convert captured Soviet broad-gauge rail to European standard gauge before they can use it. Conversion extends 1d3 hexes per turn along the rail line. Meanwhile, the German non-rail supply trace is only 3 hexes from the nearest railhead, reduced to 2 hexes in winter. The Russians trace 5 hexes with no winter reduction. That asymmetry is the entire logistical story of Barbarossa in mechanical form. The further the Germans advance, the longer and more fragile their supply lines become. The pace of the rail conversion determines how fast the advance can sustain itself. Every hex gained stretches the line thinner. The player feels the culminating point approaching without the rules having to announce it.

In the North Africa titles, supply works differently. Physical Supply Unit counters represent logistics assets. These counters operate in two modes: mobile mode, truck columns that can only supply adjacent units, and deployed mode, established fuel dumps that project supply within a 4-hex radius. The Rommel marker can exempt one force from line of communication requirements per turn, modeling Rommel's historical tendency to outrun his logistics through sheer audacity. Random events introduce fuel shortages, sandstorms, and convoy arrivals that directly impact the logistical situation. The desert war was defined by the supply line running along the coast road, and the supply system makes that road the spine of the entire game.

The key teaching point from the DAMOS approach is that supply mechanics should be designed for the specific conflict you are simulating, not imported wholesale from another game. A generic supply trace works for many subjects. But when logistics are central to the historical narrative, a tailored system does more to put the player in the commander's seat than any amount of combat chrome.

If logistics do not have realistic impact in your game, that invites skepticism from informed players. The effect is similar to what happens when movement rates are wrong: the simulation loses legitimacy. A game about the 1941 Eastern Front where German panzers can operate 200 miles ahead of their railheads without consequence is not simulating Barbarossa. It is wearing Barbarossa's uniform while ignoring everything that made the campaign what it was. Supply is a subtle mechanic, often working in the background, but the constraints it provides are essential to giving the player the correct decision space.

Morale

The importance of morale in your design depends on scale. At tactical and grand-tactical scales, morale is often the decisive mechanical factor. Firefights at squad and company level are won or lost based on which side breaks first. The physical destruction of a force matters less than its willingness to keep fighting. At operational and strategic scales, morale becomes abstracted into other systems or modeled through broader mechanisms that represent organizational cohesion rather than individual courage.

The most common tactical morale mechanic is the morale check. A unit has a morale rating printed on its counter. When triggered by incoming fire, casualties, or proximity to a threat, the unit rolls a die. Equal to or below the morale rating means the unit holds. Above means it falters. The consequences of failure vary: disruption, suppression, retreat, rout, or some combination depending on how badly the check was failed.

This is a proven mechanic because it does two things at once. It introduces uncertainty into combat beyond what the CRT provides, and it differentiates unit quality in a way that matters every time fire is exchanged. Elite troops with a morale rating of 8 on a d10 shrug off situations that send conscripts with a rating of 4 running. That distinction creates asymmetric gameplay without requiring separate rules for different troop types. The morale rating does the work.

At grand-tactical scale, morale often operates at the formation level rather than the individual unit level. A corps or division accumulates losses until it crosses a threshold, at which point the entire formation becomes demoralized. This is closer to how organizational breakdown actually works at that scale. A corps does not collapse because one regiment breaks. It collapses because cumulative losses erode the organization's ability to function as a coordinated fighting force.

Imperial Bayonets uses this approach. A corps becomes Demoralized when its infantry strength point losses reach 40% of its starting total. The effects are severe: initiative reduced by 1, combat strength halved, no advance after combat, no construction of improved positions, no zone of control projection. The corps is not destroyed, but it is combat-ineffective until it can rebuild past the threshold. That recovery process carries its own risk. Demoralized units that roll a 5 or 6 on their initiative check during the recovery phase are permanently eliminated, not just reduced but removed from the game entirely. This models the historical reality that shattered formations attempting to reconstitute under pressure sometimes disintegrated completely. The 40% threshold and the permanent elimination risk together create a system where players must manage their corps' exposure carefully. Committing a corps too aggressively risks pushing it past the threshold into a state that is difficult and dangerous to recover from.

Not every army in *Imperial Bayonets* uses the same morale structure. Italy and Piedmont-Sardinia track army-wide morale rather than per-corps morale because they lack corps-level organization. The morale mechanic adapts to the force structure of the belligerent, which is how subsystems should work: the implementation follows the history, not the other way around.

Rout is a related but distinct concept from morale failure. A disrupted or broken unit retreats in some order. A routed unit flees. The distinction matters most at tactical scale and in earlier historical periods where all-out routs were common. Ancient and medieval battles often ended with one side's army dissolving into a panicked mass fleeing the field, suffering more casualties in the rout than in the battle itself. *Marathon*, one of my ancient tactical designs, includes rout movement and rout status that trigger when units accumulate enough disruption points. The transition from fighting to running captures something essential about ancient warfare that a simple retreat result on a CRT cannot.

At later periods and larger scales, rout becomes less common as a mechanic. Napoleonic armies routed, but the phenomenon was more localized. Individual battalions broke, not entire wings of an army at once (with notable exceptions like Jena). By the twentieth century, units that collapsed under pressure were more likely to surrender or disperse than to flee en masse. ASL is a notable exception: it includes rout movement for a World War II tactical game, but ASL is an exceptionally granular simulation that models squad-level behavior in detail most tactical games do not attempt. The complexity of ASL's rout rules is part of why the system has a reputation for being among the most difficult rulesets in the hobby. Most WWII tactical designs abstract that behavior into retreat results on the CRT. Match your morale mechanics to the period, the scale, and the level of granularity you are willing to support.

At strategic scales, morale can be rationalized into a small strength point bonus or a relevant variable rather than a separate subsystem. A veteran division on the Eastern Front with higher morale than its opponent does not need a morale check mechanic. Its advantage can be built into its combat rating or applied as a column shift. The designer's job is to decide at what point morale stops being a separate system and starts being a component of existing systems. There is no universal answer. It depends on the game.

Command and Control

Command systems follow the same scaling principle as morale. At lower scales, command is literal: units need to be within range of their headquarters to function at full effectiveness, and the mechanics model the physical communication links between commanders and their subordinates. At higher scales, command becomes abstracted into limits on how many actions a player can take or how many formations can be activated.

Sedan 1940 has one of the more detailed command systems in my designs. Each headquarters unit has a Radio Value. At the start of the turn, each HQ makes a Radio Communications Check by rolling 1d6 under its Radio Value. A failed check means the HQ is Out of Command. All units under an Out of Command HQ lose indirect fire capability and cannot receive air support. The radio represents the link between the off-map corps headquarters and the divisional HQs on the map. A single bad radio roll grounds an entire division's long-range artillery coordination. Units outside their HQ's Command Range suffer similar penalties regardless of the radio check: no indirect fire, a leftward column shift in assaults, no prepared assaults.

This system works for *Sedan 1940* because the 1940 campaign was defined by command and control. The French collapse was as much about communication breakdown and decision paralysis as it was about German tactical superiority. Gamelin's headquarters had no radio. Orders took days to reach the front. The German advantage was not just better tanks but faster decision-making at every echelon. The command system makes the player feel that asymmetry every turn.

At operational and strategic scales, I tend to implement command through HQ units that provide bonuses to attached or in-range units. An HQ within range provides a combat bonus, a movement bonus, or enables capabilities that units cannot access on their own. Units outside HQ range suffer penalties or lose access to certain actions. This is a lighter touch than the *Sedan 1940* system, but it creates the same fundamental pressure: keep your forces within the command structure or pay a price.

Activation rolls and order delays make the most sense at tactical and grand-tactical scales, where individual commanders are making decisions that directly affect the units under them. A battalion commander freezing under fire or a brigade commander misinterpreting an order are tactical events. At strategic scale, these become abstractions. A theater commander's indecision is better modeled as a limit on the number of offensive operations that can be launched per turn than as a die roll per HQ.

Like supply, command is a logical in-system gate, but it tends to be less subtle in its effects. A unit that is out of command suffers obvious, immediate penalties. A unit that is out of supply might function normally for a turn or two before the effects bite. That difference in immediacy matters for design. A bad radio roll forces the player to react now. A severed supply line gives them a turn or two to fix the problem before it becomes critical. Both produce tension, but of different kinds.

One practical concern: if your design already carries a complex supply system, think carefully before layering an equally complex command system on top of it. Players have a finite tolerance for administrative overhead. A game where every turn requires tracing supply for every unit and checking command range for every unit and rolling radio checks for every HQ is a game where the administrative phases take longer than the movement and combat phases. That is a problem. If supply is the more important logistical story for your conflict, keep command simple. If command breakdown is the central theme, you can afford to abstract supply. Some games carry both with full weight, but those games are designed for a specific audience that accepts that overhead as part of the experience.

Engineering, Bridging, and Fortification

Engineering capabilities (bridge construction and demolition, field fortifications, obstacle clearance) appear in wargames when the conflict demands them. These are not universal subsystems like supply or morale. They are tools the designer reaches for when a specific historical situation requires them.

The question to ask is whether engineering creates a meaningful decision for the player. If a river crossing is a major obstacle that shaped the historical campaign, then bridging rules that allow the player to build, protect, and destroy bridges add a real decision. If rivers are minor terrain features that slow movement by one hex, a separate bridging subsystem is overhead without payoff.

The same logic applies to fortifications. If the conflict you are simulating featured significant defensive construction, like the Western Front in 1914–1918, then rules for building and occupying fortified positions matter. If defensive positions were improvised and temporary, the effect can be abstracted into a terrain modifier or a combat status like "dug in" without a separate construction mechanic.

Some designers handle engineering by abstracting its function into basic rules rather than representing individual pioneer or engineer units. Infantry units that can provide their own river-crossing assistance, at some cost in movement or combat effectiveness, capture the engineering function without requiring separate counters and separate rules. The player gets the decision, do I spend the extra movement points to cross the river or find another way, without the mechanical overhead of managing engineer assets. This approach works well at scales where individual engineer battalions are below the resolution of the game.

When you do include engineering as a distinct system, keep it tightly scoped. Engineer units should have clear capabilities: they build bridges in X turns, they demolish bridges as a combat action, they clear obstacles at a cost of Y movement points. Do not let engineering become an open-ended subsystem where players can construct arbitrary fortifications anywhere on the map unless the conflict specifically revolved around that capability.

Weather and Environmental Effects

Weather belongs in your game if it affected the campaign you are simulating. Russian winter, Mediterranean storms, monsoon seasons, North African sandstorms: these are not flavor. They shaped operational decisions and constrained what armies could do. A game about Barbarossa where winter has no mechanical effect is ignoring the single most important environmental factor of the campaign.

The standard implementation is a weather table rolled at the start of each turn. The result modifies movement costs, combat effectiveness, and sometimes supply. Clear weather means normal operations. Rain or mud increases movement costs and may penalize attackers. Snow or extreme cold can restrict mechanized movement, reduce combat effectiveness, and shorten supply range. The weather table should be calibrated to the historical climate of the theater. A game set in the Sahara should not have the same weather table as a game set in Norway.

Weather is most useful when it forces the player to adapt rather than just imposing penalties. If rain turns roads to mud and doubles movement costs, the player has to decide whether to push forward slowly or wait for better weather. If winter reduces supply range, the player has to consolidate their position before the season changes. Those are real decisions driven by environmental constraints, and they mirror the decisions historical commanders faced.

The DAMOS North Africa titles integrate weather through random events rather than a separate weather phase. Sandstorms arrive without warning and shut down operations. This approach suits a theater where weather was episodic rather than seasonal. You cannot predict when a sandstorm will hit, so you cannot plan around it with certainty. That unpredictability matches the historical record and produces interesting decisions about risk.

Political Constraints

Some conflicts cannot be simulated without modeling the political context that constrained military action. Coalition warfare imposes limits on how allied forces can be used. National morale creates pressure to achieve objectives within a timeframe. Political events can force offensives or mandate withdrawals that make no military sense.

These constraints are specific to the conflict, and they should be implemented as scenario rules or special rules rather than core systems. A game about the Italian front in 1866 might require the Italian player to launch attacks against fortified positions because the political situation demanded offensive action regardless of military logic. A game about coalition operations might restrict which allied units can stack together or operate in the same sector.

Political constraints should create the correct decision space for the player. If the historical commander had to balance military logic against political pressure, the player should face the same tension. If the political situation did not constrain military operations, skip the political rules.

Integration and Restraint

Every subsystem discussed in this chapter can improve your simulation or burden it with overhead that distracts from the core gameplay. The decision about what to include is about what creates meaningful decisions for the player within the scope of the conflict you are simulating.

Supply, morale, and command are the three subsystems that appear most often in wargames because they represent the three universal constraints on military action: armies need to be fed, soldiers need to be willing to fight, and commanders need to communicate with their forces. Most games need at least one of these modeled explicitly. Many need two. Few need all three at full complexity.

The subsystems you include should reinforce each other and support the game's central theme. If your game is about the logistical challenge of sustaining an offensive deep into enemy territory, supply should be the dominant subsystem. Morale and command can be abstracted. If your game is about the breakdown of an army's fighting spirit under relentless pressure, morale should be the dominant subsystem. If your game is about the communication failures that led to a military disaster, command should take center stage.

Kevin Zucker's skepticism of chrome applies here more than anywhere else in the design process. Every subsystem you add runs every turn. Every supply check, every morale roll, every command range measurement adds time to the game and complexity to the rules. Include what the simulation demands and what produces interesting decisions. Abstract everything else. The player should spend their time maneuvering and fighting, not administering.

I have seen designs collapse under the weight of subsystems that did not justify their presence. A game that takes twenty minutes of administrative phases before the player gets to move a single unit has a subsystem problem. The subsystems exist to serve the simulation. The simulation does not exist to showcase the subsystems. Keep that hierarchy clear in your design.

Victory Conditions

Victory conditions define what the game is about. You can build a CRT that produces historical casualty rates, a map that captures every ridge and river crossing, and a sequence of play that models command friction at every echelon. None of it matters if the players do not know what they are trying to accomplish or if the objectives you give them bear no relationship to the historical situation. Victory conditions are the contract between the designer and the player. They tell the player what to optimize for, and in doing so, they shape every decision the player makes from the first turn to the last.

I have seen designers treat victory conditions as an afterthought, something bolted onto the game after the systems are built. That is backwards. Victory conditions should be among the earliest design decisions because they determine what the rest of the game needs to support. A game where victory depends on territorial control needs a map where territory matters and a combat system where taking and holding ground produces results. A game where victory depends on inflicting casualties needs a combat system with enough granularity to differentiate between a pyrrhic win and a clean one. The systems serve the victory conditions. If you design the systems first and then try to figure out what "winning" means, you will end up with objectives that do not connect to the game the player has been playing for the last three hours.

What Victory Conditions Do

Victory conditions do three things in your design. They give the player an objective to pursue, they create a framework for evaluating how well the player performed, and they make an argument about what mattered in the historical conflict.

The third function is the one designers overlook most often. When you write victory conditions for a game about Gettysburg, you are deciding what mattered at Gettysburg. If the Confederate player wins by occupying Cemetery Hill, you are arguing that the battle hinged on terrain. If the Confederate player wins by destroying a certain number of Union strength points, you are arguing that the battle hinged on attrition. If the Confederate player wins by exiting units off the map toward Washington, you are arguing that the battle was about Lee's strategic objective of threatening the capital. These are different arguments, and they produce different games. The victory conditions do not just end the game. They tell the player what the game was about.

This is why victory conditions need to come from your research, not from generic design templates. "Control X hexes" is a structure, not a victory condition. The historical context tells you which hexes matter and why. "Destroy X strength points" is a structure. The historical context tells you what level of losses would have changed the outcome. Your victory conditions should reflect what the historical commanders were trying to accomplish, adjusted where necessary to produce a playable game.

Territorial Victory Conditions

The simplest and most common type. One or both players need to control specific locations on the map by the end of the game or by a certain turn. Cities, crossroads, bridges, ports, supply centers.

Territorial victory works well when the historical conflict was about controlling ground. Most operational games use territorial victory because campaigns are fought over geography. The German player in a Barbarossa game needs to capture Moscow, or Leningrad, or the Caucasus oil fields. The Allied player in a Normandy game needs to secure the beachhead and break out. These objectives mirror the historical mission.

The risk with territorial victory is that it can reduce the game to a race toward a hex, ignoring the historical dynamics that made reaching that hex difficult. If the only thing that matters is whether the German player sits on the Moscow hex at the end of turn 12, then every German decision reduces to "what gets me closer to Moscow fastest?" The historical campaign was more complicated than that. Army Group Center had to protect its flanks, maintain supply lines, deal with Soviet counterattacks, and manage the deteriorating condition of its panzer divisions. Your victory conditions should reward the player for handling those problems, not just for reaching a hex.

One solution is to require the player to hold territory with adequate force rather than just touching it with a single unit. If Moscow must be controlled by at least three divisions in supply, the German player cannot strip the front bare to rush a token force into the city. Another solution is to include negative conditions: the German player loses points for Soviet-controlled hexes behind their front line, representing the historical threat of partisan activity and bypassed garrisons.

Point-Based Victory Conditions

Victory points allow you to weight multiple objectives and create a composite score that reflects overall performance. The player earns points for capturing territory, destroying enemy units, achieving specific objectives by specific turns, or meeting political conditions. The player with the most points at the end wins, or the player who crosses a VP threshold triggers an automatic victory.

Point-based systems give you more granularity than territorial victory. You can assign 5 VP to a critical city and 1 VP to a secondary crossroads. You can award VP for destroying an elite formation and no VP for destroying a garrison unit. You can penalize a player for losing their own elite units. The VP schedule becomes a priority list that guides the player's decisions.

The danger of VP systems is complexity. A game with thirty different VP triggers scattered across the map and the turn track asks the player to hold too many objectives in their head at once. The best VP systems are simple enough that a player can evaluate their position at a glance. Am I ahead or behind? What is the most valuable objective I can reach with the forces I have? If the player needs a calculator to figure out whether they are winning, the VP system is too complicated.

VP systems also create a temptation for the designer to fix balance problems by adjusting point values rather than fixing the underlying mechanics. If the German player wins too often, the instinct is to increase the VP cost of their objectives or decrease the VP value of their conquests. This can work, but it can also produce victory conditions that feel arbitrary. If Moscow is worth 10 VP in one version and 15 VP in the next because the playtesting showed the Soviets needed help, the player can smell the artificial adjustment. When possible, fix the game systems rather than the

VP schedule. Victory conditions should reflect the history, not the playtest results.

Automatic Victory

Some games include a condition that ends the game immediately when triggered. If the German player captures Moscow before turn 8, the game ends in a decisive German victory. If the Soviet player retakes Smolensk, the German player loses immediately.

Automatic victory works when the historical situation had a clear breaking point. If a certain event would have ended the campaign, the game should recognize that. Automatic victory also creates tension by giving both players a live threat to worry about throughout the game. The Soviet player cannot ignore the Moscow axis even if they are winning everywhere else, because a German breakthrough triggers an instant loss.

Use automatic victory sparingly. A game that ends on turn 3 because one player achieved a surprise breakthrough is historically interesting but gives the losing player a short, frustrating experience. Pair automatic victory with a requirement that is difficult to achieve, so it functions as a constant threat rather than a likely outcome.

I distinguish between automatic victory and sudden death. Automatic victory ends the game when a strategic condition is met. Sudden death ends the game when one player's position becomes irretrievable. Sudden death exists to prevent a game from dragging on after the outcome is determined. If the Soviet player in a Barbarossa game loses Moscow, Leningrad, and the Caucasus by turn 6, continuing to play until turn 20 is not interesting for anyone. A sudden death condition is a mercy rule, not a victory condition. Short games with fixed turn limits do not need one.

Long games benefit from one because the gap between "the game is decided" and "the game is over" can stretch for hours.

Scaled Victory

Most real conflicts did not end in total victory or total defeat. The results fell on a spectrum, and scaling your victory conditions to match that spectrum adds granularity that a binary win/lose structure cannot provide.

A typical scaled system uses four or five tiers: decisive victory, strategic victory, marginal victory, draw, and marginal defeat (with the inverse for the other player). Each tier corresponds to a range of VP totals or a combination of territorial and point conditions. A decisive victory represents the best plausible outcome for that side. A marginal victory represents a modest improvement over the historical result. A draw represents roughly the historical outcome. A marginal defeat is worse than history but not catastrophic.

This structure does several things at once. It gives the losing side something to play for. Even if the French player in a Franco-Prussian War game cannot win outright, they can aim for a marginal defeat instead of a catastrophic one, and the difference between those outcomes should feel meaningful. It creates historical insight by showing the player how close the actual result was to alternative outcomes. And it produces replayability because the game generates different narratives depending on whether the result lands at the top or bottom of the spectrum.

The key to scaled victory is calibrating the tiers against the historical record. The historical outcome should fall somewhere in the middle of the range, not at one extreme. If the historical result maps to a "decisive defeat" for one side, the designer has created a game where that side needs to outperform history by a wide margin just to reach a draw. That is a balance problem disguised as a victory condition problem.

Designing for Lopsided History

Some conflicts were so one-sided that the historical loser had no realistic path to victory. Sedan 1870 is one of these. The French army was strategically outmaneuvered before the battle began. MacMahon marched into a trap. The Prussians surrounded Sedan and bombarded the French into submission. Napoleon III surrendered with over 100,000 troops. As a military engagement, the outcome was close to predetermined.

I designed a game on this battle, *We Were Not Cowards*: Sedan 1870, published in the *Imperial Bayonets* series. The design problem was immediate. If the victory conditions required the French player to win the battle, nobody would play the French. If the victory conditions were so generous that the French won half the time, the game would lie about what happened. Both options fail.

The solution is to reframe what victory means for the disadvantaged side. The French at Sedan could not win the battle. But they could have fought better. They could have inflicted more casualties on the Prussians. They could have held longer. They could have organized a breakout attempt that, even if it failed, would have preserved more of the army than capitulation did. The French player in my game is not trying to defeat the Prussian army. They are trying to perform better than MacMahon did, which is a lower bar but still a meaningful one.

Scaled victory is essential for these situations. The French player is aiming for a marginal defeat instead of the historical catastrophe. The Prussian player is trying to match or exceed the historical result. Both players have something to play for because both players are measured against history rather than against an abstract win/lose binary. The French player who holds out two turns longer than MacMahon did and inflicts 30% more casualties has achieved something worth achieving, even though the French army still surrenders at the end.

This approach teaches the player something about the conflict that a binary result cannot. It answers the question "could the French have done better?" and lets the player explore how. Wargames are the only medium that lets you inhabit a historical decision space and test alternatives against the same constraints the real commanders faced. Victory conditions are where you build the door into that space.

Victory Conditions and Alternate History

Good victory conditions create alternate history as a byproduct of play. They do not force it. If the Allied player in a D-Day game secures the beachhead by turn 4 and breaks out by turn 8, they have created an alternate history where the Normandy campaign went faster than it did. They did not set out to create alternate history. They set out to win the game, and the victory conditions guided them toward decisions that produced a plausible alternative to what happened.

This is different from a game that instructs the player to "explore what-if scenarios." The what-if emerges from the interaction between the player's decisions and the game's systems. Victory conditions establish the framework. The player fills it in.

When you are designing victory conditions, ask yourself: if a skilled player achieves a decisive victory, does the resulting narrative make historical sense? If the German player captures Moscow in a Barbarossa game, is the path they took to get there plausible? Did they have to solve the same problems the historical commanders faced, or did the victory conditions let them bypass those problems? If the path to victory requires historically unrealistic behavior, your victory conditions need work.

Connecting Victory to Your Systems

Victory conditions interact with every system in your game. The CRT determines how fast territory changes hands and how fast casualties accumulate. The supply system determines which advances are sustainable. The morale system determines when an army collapses. The sequence of play determines how much the players can accomplish in a turn.

If your victory conditions require the attacker to capture a city by turn 6, the movement rates and combat results need to make that achievable but uncertain. If the math says the attacker cannot possibly reach the city before turn 8, the victory condition is broken regardless of how historically correct it is. If the math says the attacker reaches the city on turn 3 without serious opposition, the victory condition is too generous.

Playtesting (covered in Chapter 16) is where you calibrate this. Run the game multiple times and track where the final VP totals land relative to your scaled tiers. If the results cluster at one extreme, either the systems or the victory conditions need adjustment. The goal is a spread of results centered roughly on the historical outcome, with decisive results in either direction possible but uncommon.

Case Study: *Rostov '41* and the SCS Victory Structure

Rostov '41, my design for MMP's Standard Combat Series, covers the first German drive on Rostov-on-Don in November 1941 and the Soviet counteroffensive that retook the city, the first time the Wehrmacht was forced to give up a major objective on the Eastern Front.

The SCS uses a straightforward VP system. Players earn VP for controlling specific hexes at the end of the game. The challenge was that the historical outcome was a German failure. The Germans took Rostov on November 21 and were driven out by November 29. If I set the victory conditions to reward the German player for holding Rostov, the game would punish historical play. If I set them to reward the Soviet player for retaking Rostov, the German player would have no reason to take it in the first place.

The solution was scaled territorial victory tied to the timing of control. The German player earns VP for every turn they hold Rostov and for territorial gains east of their start line. The Soviet player earns VP for retaking Rostov and for the speed of the recapture. Both players have objectives that pull in different directions. The German player wants to grab territory and hold it as long as possible. The Soviet player wants to conserve strength in the early turns and then counterattack at the right moment. The historical outcome, German capture followed by Soviet recapture, produces a result in the middle of the VP range. A German player who holds Rostov for two extra turns improves their position. A Soviet player who retakes it a turn early improves theirs.

The VP structure also had to account for the broader operational context. The German drive on Rostov was part of Army Group South's autumn offensive. The Soviet counteroffensive was Timoshenko's first successful large-scale operation. The VP awards reflect these stakes: Rostov itself is worth points, but the overall territorial balance matters too. A German player who takes Rostov but loses ground everywhere else has won a battle and lost a campaign.

This is the kind of design problem victory conditions present. The history gives you the parameters. The scaled VP system lets you create a game where both players are trying to outperform history, where the historical outcome is a plausible result rather than a guaranteed one, and where the VP tiers map to historically meaningful differences in performance.

Summary

Victory conditions belong at the start of the design process, not the end. They define what every other system needs to accomplish. Build them from the historical record. Scale them to create a spectrum of outcomes rather than a binary win/lose. Give the disadvantaged side something meaningful to play for. Connect them to your systems so that the path to victory requires historically plausible behavior. Test them through repeated play to verify that the results cluster around the historical outcome with room for both sides to improve on it.

The best victory conditions teach the player something about the conflict. They answer the question "what would it have taken to change the outcome?" and let the player test their answer against a system built from the same constraints the real commanders faced.

Naval, Air, and Specialized Systems

I have designed nearly thirty wargames, and almost all of them are land games. Hex grids, ground formations, supply lines running back to railheads and depots. That is where my experience lives. I have opinions about CRT design that I can defend with data from dozens of development cycles. I have built supply systems tailored to specific theaters and iterated on them across an entire series. I can talk about morale, command, and movement from the position of someone who has made every mistake available and learned from most of them.

Naval and air warfare are different. I have studied them, played games that simulate them, and attempted to design in those spaces. But I do not have the depth of experience in naval and air game design that I have in land combat. This chapter is honest about that. I am not going to pretend I have the same authority here that I have in the preceding eleven chapters. What I can do is orient you toward the design problems that naval and air systems present, point you at the games and designers who have solved those problems well, and share the lessons I learned from my own limited forays into these domains. If your subject demands a naval or air system, this chapter will help you understand what you are getting into. It will not hand you a finished framework. You will need to do additional research, study the games I reference, and in some cases find collaborators who know more than you do.

That last point is the most important thing in this chapter.

Naval Warfare

Naval wargame design operates under a different set of assumptions than land combat. The fundamental dynamics are not the same. On land, terrain dominates. Rivers, mountains, forests, and cities shape movement, restrict avenues of approach, and provide defensive advantages. At sea, the terrain is featureless. There are no ridges to defend, no chokepoints to block, no forests to hide in. What matters is detection, range, speed, and firepower. Naval combat is more an exercise in modeling technical and physical systems than the more ephemeral concerns that drive land warfare: morale, cohesion, command breakdown, the willingness of troops to hold a position under fire.

Ships do not rout. A warship that comes under fire does not break and flee the way an infantry battalion might. It fights until it cannot fight anymore, or it maneuvers to disengage. This changes the texture of the game. In a land game, you are asking whether a formation will hold or collapse. In a naval game, you are asking about positioning. Maneuver is the critical element. Getting your ships into a favorable firing position while denying that position to your opponent is the core decision loop. The combat itself, once ships are in range and bearing, tends to be more deterministic than land combat. Guns fire, shells land, damage accumulates. The uncertainty comes from whether you can achieve the positional advantage in the first place.

The era you are simulating determines almost everything about your naval design. Ancient naval combat, where triremes rammed each other and marines boarded enemy vessels, shares almost nothing mechanically with a World War II carrier engagement fought at ranges of hundreds of miles. Age of sail combat, with its dependence on wind direction and broadside gunnery, demands a different movement and firing system than a modern submarine game where detection is the primary challenge. Even within the twentieth century, the jump from World War I dreadnought engagements to World War II carrier warfare represents a fundamental shift in what the game needs to model. When you choose a naval subject, the era constrains your design more rigidly than it does in land games. The technology defines the tactics to a degree that has no parallel on land.

Hex grids work for tactical naval games where individual ship positioning matters. Players need to track bearing and range for gunnery, and hexes provide the spatial framework for that. At operational and strategic scales, areas work better than hexes for sea movement. The open ocean is vast and homogeneous. Putting a fine hex grid over the North Atlantic creates an overwhelming number of movement options without adding meaningful decisions. Sea zones or areas reduce the spatial problem to its essentials: where are your forces, where are the enemy's forces, and where do the shipping lanes run?

Avalanche Press's *Great War at Sea* series handles the operational-to-tactical transition in a way worth studying. Players plot orders on a strategic map, move forces simultaneously, and then shift to a tactical battle map when opposing forces intercept each other. The dual-map approach separates the two core problems of naval warfare: the search problem, finding the enemy across a vast ocean, and the engagement problem, fighting effectively once contact is made. That separation is clean and reflects how naval operations work. A fleet commander's first challenge is always knowing where the enemy is. The battle itself comes second.

Jack Greene used staggered movement phases in several of his naval designs, including *Royal Navy*. The initiative player moves after the non-initiative player, allowing them to react to the opponent's positioning. This is an attempt to achieve some of the benefits of simultaneous movement, where both sides are adjusting to each other in real time, without the bookkeeping burden of written orders. It produces a reasonable asymmetry: the player with initiative gets the last word on positioning, which is a significant advantage in a domain where maneuver determines everything.

Greene was one of the more experienced naval wargame designers in the hobby, and I learned from him directly when I attempted a standalone World War I naval game early in my design work. I came into the project assuming I knew how to rate warships for a game. My approach was straightforward: multiply a gun's rate of fire by the shell size to produce a firepower rating. Greene corrected me. The right formula was shell weight multiplied by rate of fire. Shell size, the diameter of the projectile, is not what determines the destructive effect of a hit. Shell weight is. A 12-inch shell and a 14-inch shell may not differ much in diameter, but the heavier shell carries more explosive filler and more kinetic energy at impact. The distinction matters when you are trying to rate the relative firepower of different warships. Greene's formula became the basis for every ship rating in the game.

I shelved that project. I was fascinated by World War I naval warfare, had read about Jutland and the Dreadnought arms race, and wanted to build a game around those engagements. But as the design progressed, I recognized that my understanding of the subject was not deep enough to produce a compelling standalone game. I could model the gunnery. I could build the ship ratings. But the larger design, the operational framework, the detection mechanics, the way formations maneuvered at fleet scale, required a level of expertise in naval warfare that I did not have. Rather than publish something mediocre, I put it aside. That was the right decision.

Greene passed away recently. His contributions to naval wargaming were substantial and spanned decades of design work. Games like *Royal Navy* and the naval components he contributed to various projects demonstrate a designer who understood his subject with the kind of depth that produces good simulations. The hobby is poorer for his absence.

Air Warfare

Air power in wargames exists on a spectrum from full abstraction to standalone simulation, and most land wargames stay toward the abstract end for good reason.

The most common approach in land games is to treat air power as a support mechanic. Air units provide column shifts or DRMs during combat, interdict movement in certain hexes, or fly supply missions. The player decides where to allocate a limited air budget each turn, and the effects integrate into the existing ground combat systems. This approach works because it captures the strategic decision, where to commit air assets, without requiring the player to resolve individual air missions through a separate combat system. Most ground commanders did not control the details of how air strikes were executed. They requested support and it either arrived or it did not. A simple allocation mechanic mirrors that experience.

The danger comes when a land game designer tries to model air operations with too much fidelity. I know this from personal experience. The *Procedural Combat Series*, my operational system, includes an air warfare module that attempts to model a realistic menu of air operations: air commitment to mission boxes, air-to-air combat resolved in two stages, surface-to-air missile fire, on-call recycling, ground support, interdiction, suppression of enemy air defenses, strategic bombing, airmobile operations with their own interception procedures. Each of these individually makes sense. Air-to-air combat happens. SAMs fire at attacking aircraft. Suppression of air defenses is a real mission type. No single rule in the system is unreasonable.

The problem is cumulative. The same air-to-air combat and SAM fire procedures recur inside six or seven different mission contexts. A player resolving a single air strike walks through a procedural chain that touches half a dozen subsystems. A BGG user named rzymek had to create a flowchart to make the air strike resolution sequence comprehensible.

When your players need flowcharts to navigate a support mechanic within a ground game, the system has become too heavy for its role. Air operations in PCS should have been simplified. The procedural weight is excessive for what is, at its core, a modifier on ground combat. I would design it differently today.

The lesson is not that air power should always be simple. The lesson is that the complexity of your air system needs to be proportional to its role in the game. If air power is a support element in a ground war game, it should take minutes to resolve, not tens of minutes. If air power is the central subject of the game, then the complexity is justified because the player is there specifically to engage with those decisions.

Wing Leader by Lee Brimmicombe-Wood, published by GMT, is an example of what happens when a designer builds an air combat game from first principles rather than trying to adapt ground-game conventions to the sky. The most striking design choice is the map orientation. Instead of the traditional top-down hex map, *Wing Leader* uses a side-view perspective: the X-axis represents the raid path and the Y-axis represents altitude. This is not a gimmick. It reflects a genuine insight about what WWII air combat was about. The tactical problem for fighter controllers and squadron leaders was not navigating a vast horizontal space. It was managing the raid axis and altitude. Where is the bomber stream heading, and can my fighters get above it in time to make an effective pass? The side-view map captures that problem directly.

Wing Leader operates at squadron and flight scale, not individual aircraft. It models formation management and attrition rather than individual dogfights. Units have tally and spotted states that represent situational awareness, whether a formation knows the enemy is there. Cohesion checks model formation breakdown under the stress of combat. Brimmicombe-Wood recognized that WWII air combat, at the scale that mattered operationally, was about whether formations held together and executed their missions, not about individual aces performing aerobatic maneuvers. That insight drove every design decision in the game, and the result is one of the more innovative wargames published in recent years.

DVG has built entire game series around air combat using card-based and tableau systems rather than hex maps, demonstrating that the design space for air warfare is broader than many designers assume. GMT's catalog includes *Downtown*, *Bloody April*, and other titles that find different mechanical solutions to the problem of simulating air operations. If you are designing an air combat game or a game with a significant air component, study these titles. They represent a range of approaches, and understanding what each one chose to model and what it chose to abstract will inform your own design decisions.

When Your Subject Demands Something Different

This chapter is not about ships and aircraft. It is about recognizing when your subject requires mechanics that fall outside the standard hex-and-counter land combat model, and having the honesty to respond to that.

Some conflicts do not fit the conventional wargame framework. A game about submarine warfare in the Battle of the Atlantic is about detection, convoy routing, and attrition over time. Trying to force that into a traditional CRT-driven hex game will produce something that misses what made the campaign unique. A game about the strategic bombing campaign over Germany needs to model industrial targeting, fighter defense allocation, weather over target areas, and the cumulative degradation of infrastructure. These are not problems that standard ground combat mechanics were built to solve.

When you encounter a subject that demands specialized mechanics, you have three options. First, you can do the research and develop the expertise to build those mechanics yourself. This takes time, and it requires genuine humility about what you do not know. Reading one book about carrier warfare does not qualify you to design a carrier combat system. You need to study the existing games in that space, understand why designers made the choices they did, and build your own understanding of the tactical and operational realities before you start writing rules.

Second, you can collaborate with someone who has the expertise you lack. This is what I did with Jack Greene on the naval project. I brought the game design experience. He brought the naval warfare knowledge. The combination produced better ship ratings than either of us would have generated alone. Collaboration is underused in wargame design. Designers tend to work alone, treating the entire project as a solo endeavor. But if your subject spans domains, and many interesting subjects do, finding a collaborator with complementary expertise is not a weakness. It is good design practice.

Third, you can recognize that the subject is not the right fit for you and set it aside. I did this with the World War I naval game. I was interested in the subject, I had done real work on it, and I walked away because I knew the result would not meet my own standards. That is not failure. It is judgment. A designer who publishes a mediocre game about a subject they do not understand well enough has done more damage to their reputation than a designer who never published it at all.

Hybrid Systems and Integrated Designs

Many wargames need to model more than one domain without being a full simulation of each. A game about the Pacific War needs naval, air, and ground components. A game about the Normandy invasion needs amphibious operations, naval gunfire support, airborne drops, and air interdiction alongside the ground campaign. These are integration problems, and they are among the hardest challenges in wargame design.

The temptation is to build a complete subsystem for each domain. Full naval rules, full air rules, full ground rules, all bolted together. The result is usually a monster. The player spends more time switching between subsystems than making strategic decisions. The rules expand to cover the interactions between domains, which multiply faster than designers expect. What happens when an air unit attacks a naval unit? What happens when naval gunfire supports a ground assault against a coastal defense? Each interaction requires its own resolution procedure, and the administrative weight piles up.

The better approach, for most designs, is to identify which domain is the primary focus of your game and build that system at full resolution, then abstract the other domains into support mechanics that feed into the primary system. A game about the ground campaign in Normandy can abstract naval support into a bombardment table and air power into interdiction zones without losing what makes the game interesting. The player's decisions are about ground maneuver. The naval and air elements create constraints and opportunities within that primary framework.

Games that try to give equal weight to all domains, the true multi-domain simulations, can work, but they require careful integration and a willingness to simplify each individual domain more than a standalone treatment would. The Pacific War games that succeed do so by finding an abstraction level where naval, air, and ground operations can all be resolved with roughly the same mechanical complexity. If your naval system takes thirty minutes to resolve a battle and your ground system takes five, the game is unbalanced regardless of how accurate each system is individually.

Studying Outside Your Comfort Zone

I have spent this entire book telling you to study existing games as part of your design process. That advice applies with even more force when you are working outside your area of expertise. If you are a land game designer attempting a naval game, do not start by designing. Start by playing. Play *Great War at Sea*. Play *Royal Navy*. Play *War at Sea*, the old Avalon Hill classic, and then play a modern treatment of the same subject. Understand what changed between those designs and why. Read the designer notes. Study the ship ratings and figure out how they were derived. Then start designing.

The same applies to air games. Play *Wing Leader*. Play one of the DVG air combat games. Look at how *Downtown* handles the integration of air defense networks and strike packages. Understand the design space before you try to occupy it.

And be honest with yourself about whether you understand the subject well enough to design a good game about it. There is no shame in recognizing the limits of your expertise. The shame is in ignoring those limits and publishing something that demonstrates your ignorance to everyone who plays it. Ask for help. Find collaborators. Study the work of designers who know more than you do. That is not a concession. That is how good design works, in this domain and in every other.

Card-Driven Games

I do not design card-driven games as a rule. Personal preference, no considered judgment about the format behind it. I have never been drawn to cards as a platform for mechanics, and I cannot point to a specific reason for that. Some designers gravitate toward certain tools and away from others. Cards have never been the tool I reach for first. I have designed two card-based games, studied a number of others, and I recognize that card-driven design represents one of the more significant innovations in wargaming over the past thirty years. I respect the format, I have worked in it, and I think more designers should understand what it offers even if it is not their default approach.

Cards are a viable and often elegant platform for wargame mechanics. My avoidance of them says more about my habits than about the format. I should experiment with cards more often than I do. I am doing so now with a current design on the Siege of Anabaptist Münster, though that project is more of a state-of-siege and euro hybrid than a traditional wargame. If you are a hex-and-counter designer who has never considered cards, you are leaving design space unexplored.

The Dual-Use Card

Every card in your hand can be played for its event, a specific historical occurrence or capability, or for its operations value, a number that determines how many units you can activate or how far they can move. You cannot do both. Play the card for the event and you lose the operations. Play it for operations and the event does not happen. That forced choice drives the format.

Mark Herman formalized this mechanic with *We the People* in 1993 and refined it through *For the People*, *Empire of the Sun*, and subsequent designs. Herman has written about his design philosophy in several places, most notably in his article "My Philosophy Behind Card Driven Games" published in C3i Magazine. His core argument is worth understanding because it explains why the dual-use card works as a simulation tool and not just a game mechanic.

Herman's position is that the most important variable in a wargame is uncertainty. His study of warfare convinced him that real combat is chaos because senior decision-makers operate with imperfect information. The more successful commanders could function in an environment where they could not plan with confidence. The hand of operations cards, in Herman's framing, is his attempt to simulate that uncertainty at the decision-maker level. You can look at the map and see where enemy forces are. You cannot know what your opponent intends to do with those forces, and you cannot guarantee that you will have the resources to execute your own plans. Your hand constrains you. You might need to reinforce the south, but you do not have the right cards for it this turn. That frustration is the simulation working.

Herman has reported receiving letters from players of *For the People* who said they loved the game but had to stop playing for a while because they could not handle the stress. Every decision carries weight. You never have enough cards to do everything you want, and every card played for operations is an event that did not happen. That pressure is the experience Herman designs for, and it maps to the resource constraints real commanders face. You are managing scarcity, priorities, and the constant awareness that doing one thing means not doing another.

The dual-use card also brings what Herman calls "soft factors" into the simulation without heavy rules overhead. Politics, logistics, guerrilla warfare, diplomatic crises. Modeling these through traditional hex-and-counter subsystems buries the player in additional rules. An event card handles them in a sentence or two of card text. The political complexity of the American Civil War fits on a card in your hand instead of filling a twelve-page political subsystem in the rulebook.

Cards and Activation Structure

In Chapter 10, I discussed IGO-UGO, impulse activation, and chit pull as the fundamental structures for organizing a game turn, and I noted that card-driven games are sometimes misunderstood as a separate category of activation. They are not. The card is the resource that fuels whatever activation system the designer chose.

Paths of Glory, one of the landmark CDGs, uses an alternating card play structure that resembles impulse activation. Each player plays one card, resolves its effects, and then the other player plays a card. *Twilight Struggle* works similarly, with players alternating card plays within a round. *For the People* layers card play onto a more traditional operational framework. The card system sits on top of the turn structure. It does not replace it.

Cards are compatible with any activation framework. You can build a CDG with IGO-UGO underneath. You can build one with chit pull, where the cards determine what you do once your formation is activated but not when it activates. You can build one with simultaneous play, where both players commit cards face-down and reveal them together. You are adding resource management, uncertainty, and narrative on top of whatever structural foundation you have already chosen.

Deck Construction as Design

Building a card deck is designing the narrative arc of your game. The deck becomes a second design space alongside the map, and that is where CDG design separates most from traditional hex-and-counter work.

When you build an order of battle (OOB) for a hex game, you are defining what forces are present and what they can do. When you build a card deck, you are defining what can happen and when. Event distribution across the deck determines pacing. A deck front-loaded with powerful events creates an explosive early game. A deck where the strongest events are buried deep creates a slow build toward a climax. The number of cards, the hand size, and the draw rate all interact to control how much of the narrative a player sees in any given game and how quickly the situation evolves.

The practical process of building a deck resembles building an OOB. Open a spreadsheet. Number your cards. Start naming events based on your research. You will want a mix of historical events, things that happened during the conflict, and historically plausible events that represent capabilities or situations that could have occurred. The plausible ones tend to be more tactical in nature: a surprise ambush, a forced march, a retreat in good order, a supply crisis. The historical ones anchor the game to its subject and give the deck its identity. A *Paths of Glory* deck feels like

World War I because the events reference specific campaigns, offensives, and political developments from that war.

You will iterate on the deck multiple times. The first version will have balance problems, pacing problems, and events that seemed interesting in concept but do not produce good gameplay. This is normal. Deck construction is a development process, not a one-pass exercise. Playtest, adjust card values, add events, remove events, change when cards enter the game. The deck is as much a designed artifact as the map or the CRT, and it requires the same iterative refinement.

A CRT and OOB alone cannot tell the story of a conflict the way a deck can. A player who draws the “Brusilov Offensive” card in *Paths of Glory* is experiencing a narrative moment that hex-and-counter chrome cannot replicate. The event is a historical reference point that connects the player to the subject. CDGs draw in players who might never engage with wargames through any other format, and the narrative quality of the deck is a large part of why.

The Ops-Event Economy and Hand Management

The central tension in most CDGs is hand management. Your hand is your set of options for the turn, and managing it well is the difference between competent play and flailing.

Hand size is a critical design parameter. A large hand gives players more choices but reduces the pressure of any individual card play. A small hand forces harder decisions because each card represents a larger fraction of your available resources. Most CDGs settle on hand sizes between five and nine cards, which is large enough to provide meaningful choice but small enough that every card matters.

Discard mechanics interact with hand management in ways designers often overlook. If players discard unused cards at the end of a round, the hand refreshes completely and long-term planning is limited to the current round. If players retain cards across rounds, they can save powerful events for the right moment, but holding cards means not playing them, which has its own cost. Mandatory events, cards that must be played when drawn regardless of whether the player wants to, add another layer. They force players to deal with situations they did not choose, which is realistic but can feel punishing if the mandatory events are too powerful or too frequent.

The balance between events and operations values across the deck shapes the overall feel of the game. If most cards have strong events, players face agonizing choices every turn. If most cards are operationally useful but have weak events, the event system becomes a background feature rather than the driving force. The best CDGs create a distribution where a handful of cards in each hand present genuine dilemmas: the event is powerful and historically important, but the operations value is exactly what you need to execute your plan. Neither choice is wrong. Both cost you something.

What CDGs Do Well

CDGs excel at several things that traditional hex-and-counter games struggle with.

Narrative. The card deck creates a story that unfolds differently each game. Players remember specific card plays the way they remember specific dice rolls in other games, but card plays carry historical context that dice do not. Drawing the right event at the right moment creates stories that feel earned rather than random.

Asymmetry. Each side can have its own set of events, its own operations values, its own card distribution. The deck itself can be asymmetric, with one side's cards weighted toward events and the other's toward operations. Hex-and-counter games achieve asymmetry through the OOB and the map. CDGs can build it into the core mechanics.

Replayability is built into the format. Because the deck is shuffled, every game presents a different subset of events in a different order. A strategy that works when you draw your best events early may fail when those events are buried at the bottom of the deck. This variability extends the life of a CDG well beyond what most hex-and-counter games achieve with fixed setups.

Resource decisions in CDGs are multi-dimensional. You are deciding where to attack, whether to attack at all or play the event instead, whether to hold a powerful card for later or spend its operations value now. Those layered decisions keep experienced players engaged across dozens of plays.

Accessibility. Cards are more approachable than the non-linear freeform format of most hex-and-counter wargames. A new player looking at a hex map covered in cardboard counters faces a steep learning curve to understand the game state. A new player holding a hand of cards with event descriptions and operations values written on them has an easier entry point. The cards tell you what you can do. The map shows you where.

I have a friend who does not play wargames. He has no interest in combat results tables, zone of control rules, or supply line tracing. I could never get him to sit through a hex-and-counter game. *Paths of Glory* is one of the only wargames I could ever get him to play, and he enjoyed it. I bought him a copy. CDGs bridge the gap for non-wargamers in a way that traditional designs rarely can.

What CDGs Struggle With

CDGs are not without problems, and being honest about those problems is part of understanding when to use the format.

Simulation precision is the primary trade-off. A hex-and-counter game can model a specific engagement at a specific scale with high fidelity. The OOB can reflect the exact units present. The CRT can be calibrated to historical casualty rates. The map can represent terrain at a resolution that captures the tactical nuances of the battlefield. CDGs, by their nature, abstract many of these details into card effects and operations values. The simulation is broader but less precise. If your design goal is to model the exact tactical dynamics of a specific battle, a CDG may not be the right tool.

Consistency across plays is another concern. The variable card draw that makes CDGs replayable also means that individual games can produce outcomes that diverge from history in ways that feel wrong. A one-sided card draw can give one player a dominant position with no historical basis. The same mechanic that creates interesting variability can also create outlier games. Designers mitigate this through card distribution, mandatory events, and deck structure, but the tension between replayability and historical plausibility is baked into the format.

Card counting is a subtler problem. In any game with a fixed deck, a player who memorizes the card distribution gains an advantage. If you know what cards have been played, you can deduce what remains in the deck and calculate the probability that your opponent holds specific events. In a game like *Paths of Glory* with a large deck, this advantage is modest. In a game with a smaller deck, card counting can become a dominant strategy that undermines the uncertainty the format exists to create. If a player can predict what is in their opponent's hand based on what has been played, the fog of war Herman designed into the system evaporates. Deck size and card turnover rate are your primary tools for managing this. Larger decks with partial draws make counting impractical. Smaller decks where both players see the full contents each game are more vulnerable.

Beyond Ops and Events: Other Card Frameworks

The dual-use ops-and-events model dominates the CDG landscape, but other designers have explored card frameworks that serve different purposes.

Dan Verssen Games has built entire game series around card-based systems that bear little resemblance to the Herman model. A CDG like *Paths of Glory* asks "what do I do with this card?" The tension sits in the dual-use choice on each individual card. DVG's designs ask "how do I manage all of these systems?" The tension sits in the interaction between multiple single-purpose card subsystems.

The Leader series, *Hornet Leader*, *Phantom Leader*, and their successors, uses cards to build a campaign management system. Pilot cards function like character sheets, tracking experience, stress levels, and individual combat capabilities across multiple missions. Target cards present objectives with specific defensive layouts. Weapon cards have realistic characteristics: weight, range, altitude restrictions that constrain loadout decisions. Three separate event decks trigger at different phases of each mission. You manage pilot fatigue against target urgency against declining resources over the course of a campaign. No single card presents a dual-use dilemma. The dilemmas come from the interactions between card systems.

Warfighter, another DVG series, goes further. Location cards replace the hex map, forming a linear path from mission start to objective that you build from your hand. Action cards represent tactical options, and your hand size is tied to your soldiers' health. Take casualties and you lose tactical flexibility, an elegant abstraction of combat degradation. Hostile cards spawn onto locations based on terrain type, creating encounters that vary with every play. The central resource is the body count economy: bank kills toward your victory condition or spend them to activate abilities. The game simulates squad-level combat through interlocking subsystems, and the ops-event economy is nowhere in sight.

Pavlov's House, designed by David Thompson, uses cards in yet another way. Soviet action cards function as menus: draw four, pick three, and each card offers two options of which you choose one. The Wehrmacht side is card-driven as an AI opponent. You draw a card and execute what it says, no decisions required. Fog of war cards, dead draws, accumulate as the game progresses, modeling escalating chaos through a mechanical deterioration of the Soviet player's action economy. Three different card systems run at the same time, each serving a distinct purpose.

Cards can replace maps, generate enemies, manage campaigns, simulate AI opponents, and model resource degradation without ever presenting the player with a "play for the event or play for operations" choice. If you are considering cards for your design, study these alternatives alongside the Herman model. The right card framework depends on what you are trying to simulate.

The COIN Series: Seeded Decks and Shared Cards

Volko Ruhnke's COIN series, published by GMT Games starting with *Andean Abyss* in 2012, represents the most radical rethinking of the card-driven format since Herman's original *We the People*. Ruhnke, a retired CIA national security analyst, built a system that keeps cards at the center of gameplay but throws out hand management and private decks. The result is a framework that handles four or more factions in asymmetric conflict, a problem that traditional CDGs were never built to solve.

Ruhnke's key insight came from his experience with both *Wilderness War*, his own traditional CDG, and Labyrinth, his two-player CDG on the War on Terror. He recognized that hand management "distracted from player focus on the situation on the map." In a game like *Paths of Glory*, a significant portion of your mental energy goes into optimizing your hand: which cards to save, which to play for ops versus events, how to time your big plays. Ruhnke wanted players focused on the board state, the political situation, the factional dynamics. He replaced private hands with a single shared deck, revealed one card at a time. Nobody holds cards. Nobody chooses which card comes next.

Seeded Deck Construction

The COIN deck is built through a process that guarantees structural pacing without dictating exact card order. In *Andean Abyss*, the deck contains 72 event cards and 4 Propaganda Cards. The setup procedure builds three stacks of 16 cards, each with one of the first three Propaganda Cards buried among event cards, with the fourth Propaganda Card seeded into the remaining event cards. Each stack is shuffled separately, then the stacks are placed on top of each other to form a single draw deck.

The result: Propaganda Cards appear at roughly regular intervals. You know one is coming within each period of 15 or so cards, but you do not know if it is the next card or several cards away. As more event cards are played within a period, the probability that the next card is a Propaganda Card increases. That escalating tension is itself a mechanic. Players grow more cautious or more desperate as a period wears on, depending on whether they are ahead or behind.

Propaganda rounds are the only time victory is checked, the only time factions collect income, and the only time the eligibility system resets. They function as scoring phases, payroll, and hard resets rolled into one. The seeded construction means you cannot predict exactly when these pivotal moments arrive, but you know the window. Later volumes refined the approach further, with some sorting the Propaganda Card into the lower third of each period pile to guarantee a minimum number of events play out before the reset.

Compare this to *Paths of Glory*, which uses epoch-gated sub-decks. Each player's 55-card deck is divided into Mobilization, Limited War, and Total War subsets. Cards enter the game as the war escalates. *Twilight Struggle* does something similar with Early, Mid, and Late War decks shuffled in at fixed points. Both approaches gate which cards can appear based on game progression. COIN's seeded deck gates when structural events happen within a period of otherwise random card play. The distinction matters for designers: epoch-gating controls the narrative by controlling available content; seeded construction controls the narrative by controlling pacing.

The Eligibility System

The mechanical core of COIN is the eligibility system, which replaces hand management as the primary source of strategic tension.

All factions start the game eligible. A card is revealed from the deck. Faction symbols across the top show the turn order for that specific card. Only eligible factions can act. The first eligible faction in the card's order chooses what to do: execute an operation from their faction-specific menu, execute an operation plus a special activity, play the card's event, or pass. The second eligible faction then acts, but their options depend on what the first eligible faction chose. If the first faction took an operation plus a special activity, the second faction can only take a limited operation or the event. If the first faction took the event, the second faction gets full operational freedom.

The critical consequence: factions that acted become ineligible for the next card. Factions that passed remain eligible. Act now and you sit out next turn. Pass and you keep your options open for whatever comes next. Players can see the next card on top of the deck at all times, which means they know what is coming and have to weigh the current opportunity against the next one.

This creates a rhythm that traditional CDGs do not have. In *Paths of Glory*, you play a card every turn and draw a new hand each round. The decision is always about which card to play. In COIN, the decision is about whether to act at all. Passing is a real strategic choice, not a concession. You receive resources for passing, you remain eligible for the next card, and you might position yourself to be first eligible on a card whose event is exactly what you need.

Dual Events and Multi-Faction Design

Every COIN event card carries two events: one unshaded, which tends to favor counterinsurgent factions, and one shaded, which tends to favor insurgents. Whichever faction chooses to execute the event selects which text to activate. A government player can grab a card and use the insurgent-favoring event to deny it to the insurgent, executing it at a less damaging time or in a less damaging way. An insurgent can grab a government-friendly event to prevent its effect. The dual-event system means every card is both an opportunity and a threat, and grabbing the event means giving up your operation for that card.

This solves a problem that has limited CDGs since their inception: scaling beyond two players. Most CDGs are two-player games because hand management with four players creates enormous downtime. COIN handles four factions through the combination of shared deck, eligibility limiter, and faction-specific operation menus. Only two factions act per card, keeping turns fast. The other two factions watch, but they are watching because they are positioning for the next card, not because they have nothing to do.

Each faction has its own menu of operations and special activities, and these menus are where COIN expresses its asymmetry. In *Andean Abyss*, the Government trains troops, patrols, sweeps, and assaults. FARC rallies guerrillas, marches, attacks, and spreads terror. The Cartels cultivate coca, process drugs, and bribe officials. The card tells you whether you can act. What you do comes from your faction's unique capabilities.

What COIN Teaches a Designer

The COIN series has expanded well beyond Ruhnke's original vision of modern insurgency games. More than a dozen volumes and counting cover subjects from the Gallic Wars to the American Revolution to Indian independence to a fictional Mars colonization conflict. Volumes across wildly different subjects demonstrate that if you express asymmetry through faction-specific capabilities rather than through the core engine, the same structural framework can model conflicts across centuries and continents.

The Irregular Conflicts Series, a GMT spin-off, has taken COIN-inspired mechanics even further from traditional wargaming, into medieval England with *A Gest of Robin Hood* and urban development conflict with *Cross Bronx Expressway*. Ruhnke's observation that the system "succeeded beyond my ambitions, particularly because my original vision for the series was way off" is a useful reminder that a well-designed framework can outgrow the designer's initial concept.

For your own work, the COIN series demonstrates several principles worth internalizing. A shared deck can produce deeper decisions than private hands when paired with the right activation system. Seeded deck construction creates narrative arc without scripting. Two factions acting per card keeps multi-player games moving without killing the experience through downtime. And combat can be stripped down to near-deterministic resolution when the real conflict is about positioning, initiative, and political control. The early COIN games like *Andean Abyss* use fixed removal rates with no dice at all. Later volumes like *Liberty or Death* reintroduced dice in specific contexts where uncertainty served the simulation. The design question is where the interesting decisions in your conflict live, and whether randomized combat adds to those decisions or distracts from them.

Deck-Building and Euro Influences

Deck-building originated in the Eurogame space and has made its way into wargaming. Players start with a basic deck and acquire new cards during play, adding capabilities and shaping their deck's composition through purchasing decisions. Dominion popularized the mechanic, and designers in every genre have since adapted it.

Mark McLaughlin's *Hitler's Reich*, co-designed with Fred Schachter, uses a card-based system where players buy event cards from a common market and integrate them into their hands. The game abstracts World War II at the strategic level into a series of contested conflicts resolved through card play, with the acquired events providing advantages in future contests. *Hitler's Reich* was one of the inspirations for my own Great Northern War design. I borrowed the concept of hand size as a player resource from McLaughlin and Schachter, treating it as an abstraction of national morale and manpower rather than a mechanical limit on card draw.

Deck-building introduces management mechanics that traditional CDGs do not have. Choosing which cards to add to your deck is a strategic decision layered on top of the operational decisions about how to play the cards you hold. It opens up questions about deck efficiency, card synergy, and long-term planning that create a different kind of engagement than pure hand-management. The risk is overhead. Every deck-building mechanic adds decisions to the game, and if those decisions do not connect to the simulation, they become administrative weight. The deck-building choices need to feel like strategic decisions about the war, not an optimization puzzle detached from the subject matter.

Cards Allow for Granular Subsystems

Cards provide a clean framework for subsystems that would be clunky to implement through traditional rules. Hand size, replacements, holding cards across turns, discard costs, mandatory plays. These mechanics emerge from the card format without needing dedicated rules sections. A hex-and-counter game would require specific overhead for each one.

Hand size as a variable can model national morale, industrial capacity, or command efficiency depending on how you frame it. In *The Great Northern War*, hand size represents the combined morale and manpower of each coalition. Losing control of objectives reduces your hand size, which means you have fewer options each turn, which means you are less capable of recovering those objectives. The feedback loop is intuitive and does not require any rules beyond "adjust hand size when objectives change hands." A hex-and-counter game modeling the same dynamic would need a separate morale track, rules for how morale affects unit performance, and probably a table to resolve the interaction. The card format handles it without additional rules.

Replacements and reinforcements can work through card acquisition. Drawing new cards or gaining access to previously unavailable cards mirrors the arrival of fresh forces or new capabilities. Holding cards across turns models reserves and strategic patience. Discarding cards as a cost models attrition and resource expenditure. These mechanics feel natural within a card game and would feel bolted-on in a traditional design.

The key, as with any subsystem, is restraint. Cards allow for experimentation with granular mechanics, but the designer's job is to use that flexibility without creating excess overhead for the players. Every additional card interaction is a rule the player must track. If your card game requires players to manage hand size, discard costs, mandatory events, card purchasing, deck reshuffling, and three types of special card abilities, you have probably overbuilt the system. The elegance of cards is that they package complexity into a familiar, manageable format. Do not squander that elegance by stacking too many subsystems on top of each other.

Case Study: *The Great Northern War*

The Great Northern War is the most developed card-based game I have designed.

The inspiration was Fred Serval's A *Very Civil Whist*, a game that struck me because it used trick-taking, a mechanic with centuries of history in card games, as the foundation for a conflict simulation. After studying that design, I built a print-and-play game on the Thirty Years War called *Pike and Piquet* as practice, a learning exercise to test whether trick-taking could support wargame decisions. *Pike and Piquet* is available on the Conflict Simulations website, but its primary purpose was preparation for the Great Northern War design.

I wanted to figure out whether trick-taking, with all the deception built into the mechanic, could work for a wargame format. Trick-taking is about information asymmetry. You know your hand. You do not know your opponent's hand. When you lead a card, you are making a bet about what your opponent can respond with. When you respond, you are revealing information about your hand that your opponent will use against you in future tricks. The bluffing, the card reading, the calculated risks. These are the same cognitive processes that military deception involves, in a different context.

My approach was to treat each trick interaction as a contest that determines which player gets to act. The trick mechanic drives the action economy: win a trick and you activate a leader, attempt a mobilization, or gain a decision card. Lose and your opponent draws from the deck instead. The trick is the currency of the game, not a separate minigame bolted onto a wargame. Every action in the game flows through the trick-taking system.

Hand size in GNW represents national morale and manpower, not a mechanical limit. The Swedish player starts with a hand size of seven, the Anti-Swedish player with six. Losing control of objectives reduces your hand size, which directly reduces your options each turn. I borrowed the concept from Mark McLaughlin and Fred Schachter's *Hitler's Reich*, which treated hand size as a strategic resource. I reduced the number of dice in combat resolution from three to two compared to *Hitler's Reich* because the card values in GNW have a narrower range, and the math needed to stay tight.

Decision cards, set aside from the main deck at the start of the game, function as rule-breaking events. Traitors let you bypass siege procedures. Scorched Earth lets you destroy enemy supply infrastructure. Peter I Modernizes Russia represents the historical transformation of the Russian military, granting the Anti-Swedish player increased hand size at the cost of temporarily freezing all Russian leaders. These events introduce historical elements without requiring additional rules sections. The card does what it says, and the player decides when to use it. Some decision cards are more generic than others. Reconnaissance and Force March are capabilities available to any army of the period, while Stanislaw and the Crimean Revolt are specific to this conflict. That mix of generic and specific events is deliberate. The generic ones provide tactical flexibility. The specific ones anchor the game to its historical subject.

The game encourages historical behavior through incentives rather than scripting. Instead of forcing the Swedish player to attack Warsaw and Cracow, the design rewards doing so by granting access to the high-value 10 cards. The historical path is strategically attractive. Nobody mandates it. I carry this philosophy into all my work, but it fits a card-driven format with particular ease because you can build incentive structures into the card economy itself.

The Ottoman Empire was the hardest design problem in GNW. The Ottomans were a wild card whose intervention could have changed the war's trajectory. Modeling the specific historical events would have required scripting the Swedish player into suffering massive losses in specific locations, railroading gameplay in ways I do not enjoy as a designer or a player. I treated the Ottomans as a strategic gamble accessed through the Diplomatic

Appeal track and a dice-based bribery mechanic. The abstraction captures the uncertainty and potential impact of Ottoman intervention without forcing players down a historical script.

The Great Northern War is more of a game than a simulation, but I designed the abstractions so that the history shows through. The game ends through attrition, with each side attempting to wear the other down in morale and resources, which mirrors the actual outcome of the war. Sweden's male population was devastated by the conflict, and the resulting exhaustion was a primary reason Sweden sued for peace after Charles XII's death. Sweden would never again be a major factor in European affairs. That historical arc emerges from the game's mechanics without scenario rules imposing it.

When to Use Cards

Cards work better for some jobs than for others.

They fit strategic and operational scale, where abstracting individual unit actions into resource allocation decisions makes sense. An army-level commander does not direct each battalion. He allocates resources, sets priorities, and reacts to events beyond his control. The CDG hand maps to that experience. At tactical scale, where individual unit actions are the core of the simulation, cards can still work. Warfighter and the DVG tactical games prove that. But the design challenge is different. You are using cards to generate tactical situations rather than to model strategic resource allocation.

Cards fit subjects that involve competing priorities, resource constraints, and events that disrupt planning. A conflict where both sides had clear plans and executed them without interference does not call for a CDG. A conflict full of political crises, logistical failures, and agonizing trade-offs does.

Cards fit designs where replayability matters. If your design goal is a precise simulation of a specific battle that plays out the same way each time, a fixed-setup hex game will serve you better. If you want a game that tells a different story each play while remaining anchored to its historical subject, cards provide that variability as a structural feature.

And cards fit designs where accessibility matters. If you want a wargame that a non-wargamer might sit down and play, the card format lowers the barrier to entry without sacrificing depth. The decisions in *Twilight Struggle* are as deep as anything in a hex game, but the entry point is "look at your cards and pick one to play."

Whatever card framework you choose, ops and events, tableau management, trick-taking, deck-building, or something of your own, the cards have to be good. They need a thoughtful mix of all the mechanics they touch. They need to create interesting decisions, not administrative procedures. They need to serve the simulation, not decorate a game that would work without them. If your cards are not doing real work, you do not need cards. If they are, they will open design space that hexes and counters alone cannot reach.

From Concept to First Prototype

The preceding thirteen chapters covered the anatomy of wargames in detail. Scale, time and space, orders of battle (OOBs), movement, combat, sequences of play, subsystems, specialized domains, card systems. You now have a vocabulary for the components that make up a wargame and a sense of the design decisions each component demands. This chapter is about sitting down and building one.

I cannot give you a formalized process for this. Every designer I know works differently, and every designer I know has tried to formalize their own process at some point and failed. You are going to start with whatever excites you most about your design, and that starting point will be different for every project and every designer. Some designers begin with the map because visualizing the theater is what orients them. Others start with the OOB because they want to see the forces before making any other decisions. Some start with a single mechanic, a combat system or an activation structure, and build everything else around it. All of these are valid entry points. None of them is the correct one.

I can walk through the minimum set of things you need before you can push cardboard, and then be honest about how I work, including the parts of my process that are mistakes.

The Minimum Viable Prototype

Before you can play a single turn of your game, you need five things.

A map. It does not need to be beautiful. It does not need to be finished. It needs hexes, terrain, and enough geographic accuracy that the spatial relationships between forces hold up. A hand-drawn map on graph paper works. A quick map thrown together in GIMP works. If you are doing area control or point-to-point, you need locations and the connections between them. The map will change. Do not invest significant time in making it look good at this stage.

Counters. You need physical pieces to push around the map. Blank counters are available online from various hobby suppliers. The old-school technique, and one that still works, is to print your counter sheet on paper, glue it to a piece of cardboard, and cut them out. This produces functional double-sided counters for almost no cost. If you are testing digitally through VASSAL or Tabletop Simulator, you can skip physical counters, but I find that pushing real cardboard teaches you things about your game that digital testing does not. Moving pieces, stacking them, flipping them, picking through a pile to find the one you need. These interactions tell you whether your game has too many pieces, whether the information on the counters is readable, whether the stacking limits feel right.

A sparse rulebook. Not a finished rulebook. Not a comprehensive rulebook. A document that contains the sequence of play, the core movement and combat procedures, and enough explanation that you can run a turn without stopping to invent rules on the fly. Write this as a reference, not as a teaching document. You are the only person who needs to understand it at this point. It will be incomplete. You will fill in gaps as you play. That is the purpose of the exercise.

Charts. Whatever your game needs to resolve its procedures. A CRT, a terrain effects chart, a turn record track, a combat modifiers summary. Print these on separate sheets so you can reference them during play without flipping through the rulebook.

Dice, cards, or whatever randomization your game uses. If your game uses a card deck, you will need at least a rough version of that deck. Index cards with handwritten descriptions are sufficient for early testing.

That is the minimum. Map, counters, sparse rules, charts, and randomizers. Everything else is refinement. Do not write a full rulebook before you have played a single turn. The rulebook exists to document what you have tested, not to prescribe what you are about to test. I have watched designers spend months writing detailed rulebooks for games they have never played, and in every case the first playtest invalidated half of what they wrote.

The Design Document

I have never made a design document for any of my analog games. This is a mistake.

I have designed close to thirty published wargames without a formal GDD, and I got away with it because I tend to start projects with strong opinions about what the game should feel like and what it should not try to be. Those opinions live in my head, and they survive the development process because I am the one making every decision. But there have been projects where scope crept in slowly, where a mechanic that seemed like a natural extension of the design turned into a subsystem that dragged the game in a direction I did not intend. A design document would have caught those moments earlier.

I started using design documents for digital projects I am working on, and the difference is noticeable. A written thesis, a clear statement of what the game is trying to do, keeps the project from expanding past its boundaries. It does not need to be elaborate. A page is enough. Your thesis boiled down to its mechanical representation: what is the central question the player is answering, what scale and scope serve that question, what systems are necessary and which are not. Write it before you build anything. Refer to it when you are tempted to add a subsystem. If the subsystem does not serve the thesis, it does not belong in the game.

I did not do this for most of my career, and I paid for it in extended development cycles and features that got cut after consuming weeks of design work. You can get away without a design document if you have strong enough instincts and you are willing to rework things when they drift. But having one is better than not having one. I am certain of that now.

How I Work

My process is not the recommended process. It is the process I have settled into after a decade of designing games, and it has strengths and weaknesses that I am aware of but have not fully corrected.

I tend to start with the map and the OOB. I want to see the theater and the forces before I make mechanical decisions. Looking at a map of the campaign area and seeing where the armies were positioned, where the terrain features create natural defensive lines, where the roads and rail lines run: that orients me. The

OOB does the same thing from the other direction. Seeing the specific formations involved, their relative strengths, how they were organized: that tells me what scale feels right and what level of detail the game needs.

Once I have a rough map and a rough OOB, I start thinking about systems. I tend to have strong ideas about what the core mechanic should be, some notion of how activation should work or what the combat system needs to capture. Those ideas come from the research, from the specific qualities of the conflict that make it worth simulating. I develop those systems, test them against the map and OOB, and then write the framework around them. Rules that do not make sense get cut. Systems that add overhead without adding decisions get cut. I prune after building, rather than designing a minimal framework from the start.

This approach has a disadvantage that I should be honest about. If I design the OOB and then later change a mechanical rule that affects how unit ratings work, the entire OOB needs reworking. This has happened to me more than I would like.

We Were Not Cowards, my *Imperial Bayonets* series game on the Battle of Sedan in 1870, went through four or five complete reworkings of the strength and initiative ratings over the course of development. That game took years to finish, in large part because the combat system, which I believed was well designed and did not want to change, was not producing the outcomes I wanted with the original unit ratings. The system was sound. The numbers feeding into it were not. Each time I adjusted the numbers, I had to recalculate every unit in the game, cross-reference against the historical data, playtest the new values, and determine whether the results matched what I was looking for. Multiply that by four or five iterations and you understand why the project took as long as it did.

A designer who writes rules first and builds the OOB to fit those rules avoids this problem. The trade-off is that their OOB may feel forced, shaped by mechanical requirements rather than historical research. Neither approach is wrong. Both have costs. I am comfortable with my approach because reworking OOBs is something I know how to do, but I would not claim it is the more efficient method.

Tools

You do not need expensive software to design a wargame. The tools that matter are the ones that let you create, iterate, and share your work without friction.

GIMP is an open-source image editor that does everything Photoshop does for the purposes of wargame design, and it costs nothing. I use it for map drafts, counter sheets, and charts. The learning curve is real but manageable, and for a designer who will be creating and revising visual assets across the entire life of a project, learning GIMP is time well spent.

Google Sheets is where OOBs live, where CRTs get built, and where combat math gets tested. Spreadsheets are the backbone of wargame design whether you realize it at the start or not. Every unit rating, every probability calculation, every strength point total will end up in a spreadsheet at some point. Google Sheets has the advantage of being free, accessible from any machine, easy to share with playtesters and collaborators, and easy to hand off to an artist when the time comes. If your OOB is in a Google Sheet, your counter artist can work directly from the source data instead of translating your handwritten notes.

VASSAL is valuable for playtesting and prototyping, but it requires specialized knowledge to build modules. If you are comfortable learning the module editor, VASSAL lets you playtest your game digitally, share the module with remote playtesters, and iterate on the map and counters without reprinting physical components. The barrier to entry is not trivial. Building a VASSAL module for even a simple game takes time, and debugging module behavior can consume hours. But if your playtesting network is geographically dispersed, and in 2026 most playtesting networks are, VASSAL is worth the investment.

Tabletop Simulator fills a similar role to VASSAL with a different set of trade-offs. It is more accessible to non-wargamers and provides a three-dimensional environment that some playtesters find more intuitive. The downside is that it is a commercial product, your playtesters need to own copies, and the scripting required for automated game functions is more complex than VASSAL's module tools.

Basecamp or similar project management tools become useful when you are working with a group. If you have a developer, an artist, and playtesters all contributing to the same project, you need a shared space for files, discussions, and task tracking. Basecamp is what I use. It is a paid tool, but there are free alternatives. The specific tool matters less than having a system. When multiple people are working on the same game, files get lost, conversations happen in scattered email threads, and nobody is sure which version of the rulebook is current. A shared project space prevents that.

Pen and paper. I keep a notepad next to me whenever I am playtesting. Ideas come up during play that you will forget if you do not write them down. A mechanic that feels clunky, a unit that seems too strong, a terrain feature that does not work the way you expected. Write it down in the moment. Review your notes after the session. This is the simplest tool on the list and one of the most important.

Print and Play

When your prototype reaches the point where other people need to play it, print and play is the easiest distribution format.

A print-and-play kit is a set of files that a playtester can download, print on their own printer, and assemble into a playable game. The map prints on standard paper, possibly taped together if it spans multiple sheets. The counters print on a sheet that the playtester cuts out and mounts on cardboard. The rules and charts print as separate documents. Everything a playtester needs to play the game fits in a folder of PDFs.

This is simpler than manufacturing and mailing physical playtest kits, and it scales to any number of playtesters. You email the files, they print and assemble, and you are testing. The trade-off is that your playtesters need to do the assembly work, which means you need to include clear instructions for how to put the kit together. Which pages to print, how to mount the counters, how to assemble the map if it spans multiple sheets. Do not assume any of this is obvious. Playtesters who receive a pile of unmarked PDFs with no instructions will not playtest your game. They will set it aside and forget about it.

Keep your print-and-play files organized and version-numbered. When you update the counter sheet because you changed unit ratings, the old version needs to be clearly superseded. Playtesters working from outdated components will give you feedback based on values you have already changed, which wastes their time and yours.

Conceptual Work vs Physical Prototype

There is a distinction between the conceptual design of a game and the physical prototype, and it is worth being explicit about the difference because new designers sometimes conflate them.

The conceptual work, deciding on scale, working out the core systems, researching the OOB, drafting the sequence of play, thinking through how the mechanics interact, can happen in a week. Not a finished design, but enough of a framework that you know what game you are building and what its core systems look like. This is desk work. Reading, thinking, scribbling in a notebook, building a spreadsheet. It does not require any physical components.

The physical prototype, a map you can lay on a table, counters you can push, charts you can reference, takes longer. Weeks to months, depending on your skills and the complexity of the game. If you are comfortable with image editing, counter layout, and the physical craft of cutting and mounting components, you can move faster. If those skills are new to you, expect a learning curve. The prototype does not need to be beautiful, but it needs to be functional, and "functional" means readable counters, a legible map, and charts that contain the right information. Building all of that takes time.

Do not let the prototype phase intimidate you into staying in the conceptual phase forever. A game that exists only as notes and spreadsheets is not a game. It is an idea. The idea might be brilliant, but you will not know until you play it. Push yourself to build the prototype even if it looks rough. You will learn more from one play session with ugly cardboard than from another month of conceptual refinement.

Case Study: *Prigozhin's March of Justice*

Prigozhin's March, discussed in detail in Chapter 6, illustrates what happens when the subject itself is small enough to compress the entire design process. The event lasted thirty-six hours. The forces were a handful of units on a linear route. The central question was obvious from day one. I had a working prototype within days, not weeks.

What made this project different from my usual workflow was the speed of iteration. The game was small enough that I could play a full session in a short sitting, identify a problem, adjust a value, and play again immediately. That rapid cycle is difficult to sustain on a larger game, but it worked here because the scope allowed it. Two or three turns told me whether a change improved things or made them worse. I also playtested early and heavily, something I am usually reluctant to do with larger designs where I want the systems more developed before I subject them to outside evaluation.

If you are designing your first game, Prigozhin's March is the model. A game with a narrow geographic scope, a limited number of units, and a compressed time frame will teach you more about design per hour of effort than an ambitious operational game covering an entire campaign.

Playtesting

I hate playtesting. This is an ironic admission from someone who spent years as a musician and had no problem listening to his own material on repeat, picking it apart, reworking arrangements, and playing the same song dozens of times to get it right. Playtesting a wargame should feel the same. It is the same process: you built something, you evaluate it, you revise. But for whatever reason, I find playtesting my own designs tedious in a way that recording my own music never was. I avoid it more than I should, and I have paid for that avoidance with published games that shipped with problems playtesting would have caught.

I mention this because if you find yourself reluctant to playtest, you are not alone. It is a common trait among designers, and it is a dangerous one. The games I am most proud of are the ones that received the most testing. The games I regret are the ones where I convinced myself the design was solid enough to skip the step.

Why Playtesting Cannot Be Skipped

In Chapter 2, I discussed discontinuing 1987: *On to Kaliningrad* and 1968: *Tet* because the foundational research was flawed. Those games had problems that playtesting alone could not fix. But most games do not fail for foundational reasons. Most games fail because individual systems do not interact the way the designer expected, because unit ratings produce the wrong outcomes at the table, because a mechanic that sounded elegant in concept turns into bookkeeping in practice. Playtesting catches these problems. Nothing else does.

Your prototype is a hypothesis. You believe the combat system will produce historical casualty rates. You believe the movement values will create realistic operational tempos. You believe the supply rules will force the same kinds of logistical decisions that constrained the historical commanders. Playtesting is the experiment that tests those beliefs. Some will hold up. Many will not. That is normal and expected. The designer who treats a failed playtest as a setback is missing the point. A failed playtest is the design process working correctly.

Testing Systems in Isolation

My compromise with my own reluctance to playtest is that I test individual systems in isolation before committing them to the full design. This is faster, less tedious, and catches a large category of problems early.

If I am building a CRT, I set it up with a handful of representative units and run through combat scenarios. I pick attack-to-defense ratios across the full range the game will produce and roll through them repeatedly, tracking the results. Do the casualty rates feel right for the conflict? Are there dead spots on the table where the results are always the same regardless of the ratio? Are there cliff edges where a small change in odds produces a wildly different outcome? You can answer these questions with a CRT, some counters, and twenty minutes of rolling dice. You do not need the full game set up.

The same applies to any subsystem. If you are considering an initiative mechanic, build a small test with a few units and run through the activation sequence. Does the initiative check produce the tempo you want? Does the non-initiative player have enough to do, or are they sitting idle while you resolve activations? If you are designing a supply system, trace a few supply lines on the map and check whether the rules produce realistic constraints. Can units operate too far from their depots? Do the supply costs create meaningful decisions, or are they just bookkeeping?

Testing in isolation will not catch interaction problems between systems. It will not tell you whether the supply rules break when combined with the movement rules under specific terrain conditions. But it will tell you whether each individual system is viable before you invest the time to integrate everything. It is a filter. It catches the worst problems cheaply.

Finding Playtesters

I find playtesters through Facebook and through my mailing list for Conflict Simulations LLC. This is not a great approach, and I am aware of that. My social media presence is minimal, and my outreach methods are about a generation behind where they should be. But I have been designing games long enough that I have accumulated a network of wargaming friends, many of them through Facebook, and I can usually find people willing to test a new design.

The pool of potential playtesters matters less than the quality of the feedback they provide. You want playtesters who will stress-test your game and tell you things you do not want to hear. Positive feedback feels good. A playtester who tells you the game is great and they had a wonderful time is not helping you improve the design. You need the playtester who tells you the combat system produced a result that made no historical sense on turn three. The one who tells you the setup took forty-five minutes because the instructions were ambiguous. The one who says the supply rules are confusing and they had to invent a house rule to cover a gap.

When you recruit playtesters, make it clear that you want criticism. Tell them you are looking for problems, not praise. Tell them the most useful thing they can report is something that did not work: a rule that was unclear, a mechanic that felt wrong, a setup issue, a result that seemed broken. If they had to invent a house rule to get through the game, you need to know about it. If they hit a situation the rules did not cover and had to make something up, that is a rules gap and you need to close it. If they found a strategy that dominated the game with no counter, that is a balance problem and you need to fix it.

The worst playtest reports are the ones that say “we had fun” and nothing else. The best playtest reports describe specific problems with enough detail that you can reproduce them and take action.

Blind Playtesting

Blind playtesting is the moment someone plays your game with only the rulebook and no help from you. You are not in the room. You are not answering questions. The rulebook and the components are the only things standing between the playtester and a functioning game session.

This is where rules clarity is tested in a way that no amount of self-editing can replicate. You wrote the rules. You know what you meant. You will read your own rulebook and see the intended meaning because it is already in your head. A blind playtester reads what is on the page, nothing more. Every ambiguity they find is a rules problem. Every question they have to ask is a sentence you need to rewrite. Every situation where they guess at the correct procedure is a gap in your rules.

Blind playtesting is uncomfortable because it exposes problems you thought you had solved. I have had playtesters interpret a rule in a way that was the exact opposite of what I intended, and when I reread the rule, I could see how they got there. The words on the page supported their reading. My intent was irrelevant because my intent was not in the rulebook. This is the most valuable category of feedback you will receive, and you cannot get it any other way.

Blind playtesting happens late in development, after you have done enough internal testing to believe the systems work. Sending out a blind playtest kit before the game is ready wastes your playtesters' time and their goodwill. They will hit mechanical problems that you already know about, and they will lose faith in the project. Wait until the game plays well in your hands before asking someone else to learn it from scratch.

VASSAL and Digital Playtesting

VASSAL, Tabletop Simulator, and other digital platforms expanded the playtesting pool for wargame designers. Before these tools, playtesting required physical proximity. After the shift to online play, accelerated by COVID, most playtesting networks became geographically dispersed. Digital playtesting is now the default for many designers.

A word of caution before you invest in a VASSAL module or a Tabletop Simulator setup: make sure you have playtesters who will use it. I have paid to have VASSAL modules built for my games, and I have ended up with modules that nobody used. Somewhere between three and eight times this has happened, and I have blocked out the exact count for my own sanity. A module costs money if you are paying someone to build it, and it costs significant time if you are building it yourself. Neither investment is worthwhile if your playtesters do not use the platform.

Before commissioning or building a module, ask your playtesters what tools they use. If your testing group plays on VASSAL, build a VASSAL module. If they use Tabletop Simulator, build for that. If they prefer print and play, send them print and play files. The platform should match your testers, not the other way around.

When digital playtesting works, it is a genuine accelerator. You can schedule sessions across time zones, iterate on the module between sessions without reprinting components, and log game states for later analysis. The speed of iteration is faster because updating a digital module takes less time than reprinting counters and maps. But the tool is only as useful as the people using it.

Processing Feedback

Not all feedback is equal, and learning to distinguish useful feedback from noise is a skill that develops over time.

The most useful feedback describes a problem. "The combat results on turn four made no sense because a 3:1 attack against a unit in a city produced a defender eliminated result" is actionable. You can go back to the CRT, check the odds, check the modifiers, and determine whether the result is within the intended range or whether something is miscalibrated. "The combat system feels off" is less useful, but it tells you something is wrong even if it does not tell you what. "Change the 3:1 column to produce more retreats" is the least useful type of feedback because it is a proposed solution rather than a problem description. The playtester may be right about the solution, but focus on understanding the problem first. The playtester experienced something that felt wrong. Your job is to figure out what actually caused that feeling and whether the fix they suggest addresses the root cause or just masks it.

Listen for blockers. A blocker is anything that stops a playtest session from continuing: an ambiguous rule that the players cannot resolve, a setup error that puts units in impossible positions, a game state that the rules do not cover. Blockers are the highest priority feedback because they prevent people from playing your game at all. House rules that playtesters had to invent are indirect evidence of blockers. If two different playtest groups invented the same house rule, you have found a gap in your rules that everyone trips over.

Listen for patterns. A single playtester saying the game is too long might reflect their preferences. Three playtesters saying the same thing, without talking to each other, tells you something structural about pacing. One playtester saying the attacker is too strong might be a player who lost. Four playtesters saying the attacker is too strong is a balance problem.

Do not argue with playtesters about their experience. If they tell you something did not work, it did not work for them. You can disagree about the cause. You can disagree about the fix. But their experience at the table is data, and dismissing it because it does not match your expectations is how designers ship broken games.

Playtesting and Novelty

Testing becomes more critical the more your design relies on original systems. Tried and true wargaming conventions, a standard odds-based CRT, hex movement with terrain costs, IGO-UGO turn structure, carry decades of accumulated testing built into them. Thousands of games have used these mechanics. The failure modes are well understood. If you build your game on established conventions, you can get away with less testing because the individual systems have proven track records.

When you introduce original mechanics, you lose that safety net. A new activation system, a novel combat resolution method, an unconventional approach to supply or command. These systems have no track record. Their failure modes are unknown. They will interact with other systems in ways you have not predicted, because nobody has built this combination before. The only way to discover how they break is to play them until they break.

This is not an argument against innovation. Original mechanics are what push the hobby forward. But they demand more testing than established ones, and you should plan your development timeline accordingly. If your game includes one or two original systems surrounded by conventional ones, the conventional mechanics will not surprise you. The original ones will.

Case Study: *With The Hammer*

With The Hammer is the game I am most proud of as a designer, and it is also the game where collaboration during testing changed the design the most.

The game covers the German Peasant War of 1525, a subject I became fascinated with after reading Andrew Drummond's *The Dreadful History and Judgement of God on Thomas Müntzer*. The conflict is the largest organized popular uprising in European history before the French Revolution, a mass movement of peasants who had decided they had tolerated enough exploitation from their noble landlords. Thomas Müntzer, a radical preacher, became the figurehead of this rebellion, signing his letters "Thomas Müntzer with the Hammer."

The design went through multiple failed prototypes before I found the right framework. I had started by trying to build a Eurogame, inspired by *Paladins of the West Kingdom* and other designs I admired. But my wargamer instincts kept pushing the prototypes toward literal simulation, and the Eurogame shells felt like reskins of existing games. Two or three prototypes went nowhere. The breakthrough came when I played Plantagenet in GMT's *Levy and Campaign* series. That game provided a model of medieval logistics that made sense to me, and I could see how to deconstruct and adapt its DNA into something more approachable for nonwargamers.

The core asymmetry of the design, Peasant leaders preaching and inspiring bands to join the rebellion while Noble armies muster forces and attempt to suppress the uprising through a combination of military force and negotiation, was clear from the start. But translating that asymmetry into a balanced game required extensive iteration during testing.

I brought the design to Fred Serval, who agreed to develop it. Fred's contributions were significant. Peasant leaders gained unique historical abilities: Thomas Müntzer received "Sermon to the Princes," which allows him to attempt to immobilize Noble armies through negotiation. Fred added event cards that alter the rules for a turn or grant abilities to either side. The action allowance system, which determines how many things each side can do per turn, went through multiple revisions to balance the asymmetry between the fast-moving but fragile Peasant leaders and the slower but more powerful Noble armies.

The most instructive moment came during a live discussion on Fred Serval's YouTube channel with Professor Doug Miller, a historian who had been generous in answering my research questions throughout the project. During that conversation, Miller suggested that propaganda markers should have a more meaningful mechanical role. Fred took that suggestion and developed it into a rule that Peasant leaders can only end their movement in locales where propaganda markers are present. That single constraint transformed the game. It gave the Peasant player a logistical puzzle that mirrored the historical challenge of maintaining a rebellion across a dispersed geographic area. You cannot march your leaders wherever you want. You have to preach first, spreading your influence into new areas before your leaders can operate there. The Nobles, in turn, can preach to remove propaganda markers, shrinking the Peasant player's operational space.

This rule did not come from my solo design work. It came from a collaborative discussion during testing, suggested by a historian, refined by a developer, and integrated into a game that was improved by it. The victory point thresholds, the action allowances, the balance between Peasant mobility and Noble military strength, all of these required iteration through testing. The strong core idea was there from the start, but the game that shipped is better than what I would have produced alone.

The lesson is not that you need a developer and a historian on call. The lesson is that playtesting, especially with people who bring different perspectives, surfaces ideas and solutions that a solo designer cannot reach. You see your game from one angle. Playtesters see it from others. A suggestion that sounds minor in conversation, "what if propaganda markers restricted Peasant movement," can solve a design problem you had been struggling with for weeks.

With The Hammer made the front page of BoardGameGeek with a design diary, and it is the design I point to when someone asks what I am most proud of. It is also the design where I was most willing to accept outside input and let the testing process reshape my original vision. Those two facts are not a coincidence.

Rules Writing

I have written close to thirty rulebooks, and the early ones were bad. Players had to guess what I meant in places where I thought I had been clear. Rules I considered unambiguous produced questions on BoardGameGeek. Mechanics I thought I had explained generated emails asking for clarification. I knew what the game was supposed to do. I had not learned how to get that knowledge out of my head and onto the page in a form that someone else could follow without help.

A wargame is a miniature algorithm. It has inputs, procedures, conditions, exceptions, and outputs. Your rulebook documents that algorithm for an operator, and if the documentation fails, the operator runs the algorithm wrong. The player misapplies a modifier, resolves combat on the wrong column, stacks units in a configuration you never intended, and produces a game state that has nothing to do with the simulation you designed.

You have experienced this as a player. You sit down with a new game, reach a situation the rules do not address, and spend fifteen minutes arguing with your opponent about what the designer meant. You send the designer an email, or post on BGG, or invent a house rule and move on. The designer had an answer when they wrote the rules. They did not get it onto the page.

Rules-writing improves through practice, feedback, and iteration. The skill overlaps with game design but the two are separate disciplines. I know brilliant designers who write terrible rulebooks, and clear rules writers whose games are mediocre. You need both skills, and this chapter is about developing the one that gets less attention.

The Case System

The SPI case system is the gold standard for wargame rules, and variations of it appear in games from GMT, MMP, Compass, Decision, and every small publisher I am aware of, including my own. If you have played a wargame published in the last forty years, you have read rules organized this way.

The system uses numbered sections and subsections arranged in a hierarchy. Major systems get top-level numbers: 1.0 Introduction, 2.0 Scale, 3.0 Important Concepts, 4.0 Sequence of Play, and so on. Subsections nest beneath them (3.1, 3.2, 3.3) and sub-subsections go deeper (3.1.1, 3.1.2). The numbering gives a permanent address to every rule in the game. A player aid card can say "see 6.1.4" and the player finds that rule without hunting through paragraphs. An errata sheet can say "replace 10.3 with the following" and nobody wonders what is being corrected.

The case system mirrors the structure of the game itself. A wargame turn is a sequence of phases. Each phase contains specific actions governed by specific rules. The rulebook follows that sequence, grouping rules under the phase where they apply. A player working through their first turn reads the rulebook in order and encounters rules in the same sequence they need them. A veteran on their tenth game jumps to a case number and resolves a question in seconds.

The sequence of play appears early in the rulebook because it is the skeleton that the rest of the rules hang on. Once the player knows the order of phases, each subsequent section fills in the details. The SOP tells you what happens and when. The case sections tell you how.

This structure makes sense beyond wargames. *Monopoly* has a sequence of play: roll dice, move, resolve the space, check for special conditions, pass the turn. So does every board game. Wargames formalize the structure more than most because they carry more moving parts, more exceptions, more situations where the correct procedure is not intuitive. The case system responds to that complexity by imposing order on the rules so the player does not have to impose order themselves.

Rules-Writing as Technical Writing

A finished rulebook is an exercise in technical and instructional writing. You need different muscles for this than for designing the game, and different muscles than for creative or academic prose.

Technical writing demands precision. A sentence should say one thing, say it clearly, and leave no room for a second reading. When you write "units may not enter enemy-occupied hexes," that statement needs to hold in all cases, or you need to list the exceptions right there. If engineers can enter enemy-occupied hexes under specific conditions, that exception belongs with the movement rule, not four pages later in the engineer section where a player resolving movement will never find it.

Instructional writing demands sequence. The reader needs to encounter information in the order they need it. If a rule depends on a concept defined elsewhere, you either define the concept first or provide a cross-reference. Do not assume the reader has memorized the entire rulebook before applying any individual rule. They are learning your game while they play it for the first time.

The hard part is that you already know how your game works. You designed it. You have played it dozens of times. When you read your own rules, you fill in gaps with knowledge that lives in your head but not on the page. A sentence that reads as clear to you is ambiguous to someone who does not share your mental model. Blind playtesting, discussed in the previous chapter, is the best test of rules clarity. A blind playtester reads only what is written.

A wargame rulebook also carries some academic weight. Your introduction contextualizes the conflict. Your scale section explains why you chose these unit sizes and time increments. Your design notes, if you include them, explain the historical reasoning behind specific mechanical choices. All of this needs to serve the primary job: getting the player from the box to a functioning game session.

When to Write the Rules

Some designers write comprehensive rules before building a prototype, using the rulebook as a design document that forces them to think through procedures in advance. Others write rules last, once the game is internalized through months of playtesting.

I write rules last. By the time I sit down to write a complete rulebook, I have played the game enough that every system, every interaction, every edge case is something I have encountered and resolved. I keep a bare-bones copy during testing, enough to keep the procedures straight, but the real rulebook comes after the design is finished. You do not waste time writing detailed rules for systems that get cut or reworked during development. You write them once, for the game that exists.

The danger of writing last is that you are writing when you are most blind to your own assumptions. After months inside the design, procedures that are second nature to you may not be obvious to a first-time player. You skip steps because you have internalized them. You use terminology without defining it because the definitions feel so obvious you forget they need stating. I have fallen into this trap more times than I want to count.

Whatever your timing, write as if the reader knows nothing about your game except what you put on the page. Assume they understand wargaming conventions in a general sense. Most wargamers know what a ZOC is, what a CRT does, what stacking means. But they do not know how your ZOC works, what your CRT produces, or what your stacking limits are. The player who has played a hundred wargames still needs you to tell them whether entering your ZOC costs all remaining movement points or just one extra, whether your ZOC prevents retreat or costs a step loss on retreat, whether friendly units negate your ZOC. These implementation details vary from game to game, and you need to spell out each one.

Calibrating Complexity

Match your rulebook to your game. A simple game does not need to read like the ASL manual. A complex simulation cannot afford to leave out procedures players need.

Approach your player assuming they will not know anything you do not explain. Experienced wargamers still approach your specific game as beginners. The mechanisms may be familiar in concept, but your implementation, your values, your interactions between systems are all new. If your movement system uses terrain costs, list every terrain type and its cost. If your combat system uses modifiers, list every modifier and when it applies. Do not rely on the player deducing anything from general wargaming experience.

At the same time, know your audience. A game aimed at experienced wargamers does not need to explain what a hexagon is. An introductory game might. The tone and detail level should match the complexity of the game and the experience of the intended player.

Separating basic and advanced rules is one of the better techniques for managing complexity in a single rulebook. Core mechanics go in the main body. Advanced, optional, and additional complexity goes in a marked section that players can ignore until they are comfortable with the base game. *Sedan 1940* uses this approach. The basic rules cover everything needed for a complete game. The advanced rules add command and control, headquarters, hidden units, and variable artillery results in a self-contained section that tells the player what changes, what new procedures it introduces, and how it modifies existing rules. Players can learn the game in stages rather than absorbing the full system at once.

Examples of Play

An example of play takes an abstract rule and shows it in action. Specific units, specific terrain, a specific die roll, a specific outcome. The rule tells the player what to do. The example shows them what it looks like. For any mechanic that involves multiple steps, modifiers, or conditional outcomes, one example teaches more than another paragraph of explanation.

Combat resolution is where examples matter most. Walk through selecting the attacking and defending units, calculating the odds, applying terrain modifiers, shifting the column, rolling the die, interpreting the result, applying losses. A player who follows that walkthrough can verify their understanding at each step and then apply the same procedure to their own combats with confidence. Without the example, they are guessing whether they assembled the steps correctly.

Place examples right after the rule they illustrate. Not in an appendix, not at the end of the chapter. The player reads the rule, reads the example, and moves on. Separating the two forces cross-referencing between locations, which is the kind of friction a rulebook should eliminate.

Include as many examples, diagrams, images, and tables as you can produce. A map excerpt showing a combat situation with units, terrain, and modifiers labeled communicates faster than a written description of the same situation. This is an area where I have always been weak. My own manuals lack diagrams and visual aids, a product of limited graphic design ability on my part. If you have the skills or the budget to include visual examples, use them. A labeled diagram of a combat situation will do more for comprehension than the most careful prose you can write.

Player Aids

A player aid card sits on the table next to the game and holds the information players reach for most often during play: the CRT, the terrain effects chart, the sequence of play, combat modifiers, movement costs. If a player would otherwise flip through the rulebook to find it mid-turn, it belongs on the aid.

A good player aid eliminates rulebook lookups. The player glances at the card, finds the modifier or terrain cost, resolves the action, and keeps moving. A bad one, missing a critical table or carrying wrong values, sends the player back into the rulebook and breaks the flow of the game.

Player aids also matter during playtesting. A playtester flipping through your rulebook to find a modifier during combat is testing your rulebook's navigation, not your game. Give them the reference material on a card and they will spend their time finding problems with the design instead of problems with the index.

Build your player aids early. They will change as the game changes. But having reference sheets from the first playtest forward makes every session more productive.

Managing Ambiguity

Write your rulebook to be as unambiguous as you can, within reason. You will not anticipate every edge case. You can minimize the common sources of confusion and address the situations most likely to produce arguments at the table.

Scope is the most frequent culprit. A rule says "units may not move through enemy units." Does that apply during retreat? During advance after combat? During strategic movement? If yes in all cases, say so. If exceptions exist, list them with the rule. Every general statement in your rules needs this check: are there situations where the general statement breaks? Address those situations in the text.

Inconsistent terminology causes problems that look like ambiguity but are really carelessness. If you call it "Assault Combat" in one section and "Close Combat" in another, the player cannot tell whether these are two different mechanics or two names for the same one. Pick your terms early, define them in an abbreviations or concepts section, and use them with discipline throughout. The abbreviation table that opens the Important Concepts section of a rulebook is a contract with the reader: these are the terms I will use, and they will mean the same thing every time.

Cross-references tie the rulebook together. When a rule in the movement section interacts with a rule in the combat section, a parenthetical case number lets the player find the related rule without searching. “Units in an EZOC may not use Strategic Movement (see 13.1)” tells the player where to look. Without the reference, they have to scan the entire rulebook for the Strategic Movement rules.

You will not achieve perfect clarity. Every rulebook ships with ambiguities that surface only when thousands of players encounter situations the designer and playtesters never hit. Errata and FAQ documents exist for this reason. But every ambiguity you catch before publication is a question that hundreds of players will not have to ask.

Other Designers’ Rulebooks as a Resource

Other designers’ rulebooks are one of the richest and cheapest resources available to you. Every major and minor wargaming publisher offers game rules for free, as downloads on their websites or bundled with VASSAL modules and print-and-play kits. It costs nothing to read them.

Read rulebooks for structure and formatting, not just mechanics. How does a publisher you admire organize their sequence of play section? How do they handle cross-references? How do they format examples of play? How do they present terrain effects? What goes on their player aids? You will find approaches you like and approaches you do not. Both teach you something. The good ones give you models to adapt. The bad ones show you what to avoid.

Pay attention to how different publishers balance completeness against readability. Some rulebooks are exhaustive, covering every edge case at the cost of length and density. Others are concise, covering core procedures cleanly but leaving some situations to player interpretation. The right choice depends on your game, your audience, and how much mechanical complexity you are asking the player to manage.

Reading other rulebooks also exposes you to structural ideas you may not have considered. Layered rules with basic and advanced tiers. Quick-start guides that get players playing before they finish the full rulebook. Extended play examples that walk through an entire game turn. Annotated designer notes that explain why a rule exists. All of these are techniques you can steal. Rules writing does not reward originality of format. If another designer solved a structural problem you are facing, use their solution.

Case Study: From *1916* to *Sedan 1940*

My own rulebooks across a decade of publishing show how rules-writing improves through experience, and how the case system supports that improvement when you lean into it.

1916, my game on Verdun, was one of my earliest published rulebooks. It covers a ten-turn operational game at the division level, centered on Administrative Points as the core resource. The case system is present. Sections are numbered. Important Concepts appears early. The sequence of play is laid out. A wargamer can learn the game from it.

But it carries the problems of an early rulebook. The unit descriptions section opens by listing five unit types, then describes seven. The gas rules explain that a roll of 1-3 shifts the CRT one column left (gas blows back at the Germans) and a roll of 4-6 shifts the CRT one column left (gas blows into the French). Read that again. Both results say "left." The intended asymmetry is clear if you already understand the game: 1-3 hurts the attacker, 4-6 hurts the defender. But the text as written says the same thing happens on both results. A reader encountering this for the first time cannot resolve it without guessing or emailing me.

The stacking rules mix different constraints in a single block (same corps, different corps, infantry, non-infantry) that requires careful parsing to untangle. The Administrative Points section does not explain the conceptual framework of the system until much later, in the design notes, where I write that units are "buckets for Administrative Points." That sentence is the conceptual key to the entire game. It belongs in the rules section, not the design notes. A player reading the rules in order does not encounter it until after they have already struggled to understand AP in the abstract.

I knew what I meant. I did not get what I meant onto the page.

Sedan 1940, published several years later, shows what practice does. The abbreviation table at the top of Important Concepts is clean, comprehensive, and lists every abbreviation the player will encounter. Each unit type gets its own numbered subsection with consistent formatting: what the unit is, what values it has, what special rules apply, and a cross-reference to any relevant case number. The sequence of play is laid out and each phase gets its own section, in order, with relevant rules grouped under the phase where they apply.

The basic/advanced split is one of the strongest structural choices in the Sedan rulebook. The advanced rules section opens by telling the player what it adds (Command, Headquarters, Hidden Units, Variable Artillery Results), what changes from the basic game, and states that where the advanced rules conflict with the basic rules, the advanced rules take precedence. That last sentence is small and important. It tells the player which version of a rule governs without ambiguity.

Sedan's combat system is more complex than anything in *1916*. It runs dual combat types (Fire Combat and Assault Combat), each with its own table and modifiers. The Assault Combat Table produces compound results: attacker disruption, defender disruption, exploitation allowance, low fuel, low ammo, sometimes combined in a single result. The rules handle this by giving each result type its own numbered subsection under the ACT rules. Attacker disruption is 6.1.4.3.3. Exploitation allowance is 6.1.4.3.4. Low fuel and low ammo is 6.1.4.3.5. A player who rolls a compound result and forgets how exploitation works goes to 6.1.4.3.4 and finds the answer. They do not need to reread the entire combat section. The case system is doing its job.

The design notes in Sedan are tighter than *1916*'s. Instead of burying the conceptual framework in the back of the book, the Sedan notes focus on specific design choices: why machine gun companies are abstracted into unit ratings, why non-infantry units require Overwatch to use Opportunity Fire, how the German concept of Auftragstaktik influenced the activation structure. These notes enrich the player's understanding of the game. They do not compensate for gaps in the rules.

I did not become a better writer in some general sense between these two games. I learned, through feedback and through hearing players tell me they could not understand rules I thought were clear, that rules-writing takes deliberate effort separate from game design. You test every sentence against one question: can a reader who has never played this game understand what I am telling them to do? If the answer is no, you rewrite the sentence. You do this hundreds of times across the length of a rulebook, and each time you get a little better at writing sentences that pass on the first try.

My early rulebooks generated more errata. My later ones generated fewer. The case system was there from the start. What improved was my ability to use it: putting the right information in the right place, anticipating the reader's questions, writing with the discipline that technical writing demands. That improvement came from writing rulebooks, shipping them, hearing what confused people, and doing better the next time. You learn rules-writing by writing rules.

Organization and Version Control

I am an organizational mess. I have been an organizational mess for the entire time I have been designing wargames, and I am aware that this is a problem, and I have not fixed it. What I have done is find a tool that compensates for it well enough that my projects ship on time more often than they do not. That tool is Basecamp, and I will talk about it and its alternatives in this chapter. But the tool is secondary to the principle, which is this: you need a system for tracking your files, your versions, and your communications, and you need the discipline to use it. The specific system matters less than having one at all.

I am telling you this because every one of those failures has happened to me, and the case study later in this chapter describes the worst instance in detail.

The Problem

A wargame in development generates a lot of files. Map drafts in multiple versions as you revise terrain and hex layouts. Counter sheets revised with every unit rating change. An order of battle (OOB) spreadsheet that feeds those counter sheets. A rulebook that goes through dozens of revisions between first draft and publication. Charts and player aids that need to match the current rules. Playtest notes from multiple testers. Correspondence with your developer, your artist, your printer. VASSAL module files if you are building one. All of this accumulates.

Each file changes over the life of the project. The map you sent your artist in January is not the map you playtested in March, because you moved a victory hex and added a river crossing. The OOB you finalized in February lost three units in April after playtesting showed they unbalanced the game. The rulebook exists in at least four versions: the one you sent to playtesters, the revision after their feedback, the one you sent to your developer, and the one your developer sent back with changes.

Without a system for tracking which version is current, you will send the wrong file to someone who matters. Your artist produces counters from an outdated OOB. A playtester spends an evening with a superseded rulebook. A printer manufactures maps with wrong terrain. Each of these costs time, money, goodwill, or all three.

What a System Needs to Do

Your organizational system needs to handle four things.

File storage with version history. You need to find the current version of any file in your project and look at previous versions when something goes wrong. Cloud tools like Google Drive, Dropbox, and Basecamp keep revision histories automatically. If you work locally, you need a naming convention that makes the current version obvious: "Sedan1940_Rulebook_v3.2.pdf" where the version number increments with each revision. Whatever convention you pick, stick to it across the entire project.

Collaboration and sharing. Your developer, artist, playtesters, and printer all need access to the right files. Your artist needs the current OOB spreadsheet and counter layout specifications. Your playtesters need the current rulebook, charts, and map. Your printer needs final production files. A shared workspace where you control access prevents the wrong person from grabbing the wrong file.

Task tracking and communication. When your developer sends back a revised rulebook, you need to know what they changed and why. When a playtester reports a problem, you need to track whether you addressed it. Email works when two people are involved. It falls apart at five or six, because conversations fragment across threads and nobody can reconstruct who said what.

Backup. Hard drives fail. Cloud services have outages. Your project files need to exist in more than one place. Cloud storage handles this by default. If you work locally, back up to a second drive or cloud service periodically. Losing a month of work on a counter sheet to a dead hard drive is preventable.

Tools

I use Basecamp. It costs $15 a month, which I justify because it makes all four of the above requirements easy without me having to think about any of them. I organize each game as its own project in Basecamp. Files go in the file storage. Conversations happen in message boards. Task lists track what needs doing and who is doing it. My developer, my artist, and my playtesters all have access to the project and can find what they need without me acting as a file-distribution service.

Basecamp is not the only option and may not be the best option for your situation. The free alternatives cover the same ground with different trade-offs.

Google Drive gives you file storage with version history, sharing with permissions, and the ability to collaborate on documents in real time. Google Sheets is where most of my OOB work happens regardless of what project management tool I use. The revision history in Google Docs and Sheets lets you roll back changes and see who changed what. For a solo designer or a designer working with one or two collaborators, Google Drive may be all you need.

Dropbox handles file storage and sharing well. Its version history lets you recover previous versions of files. It lacks built-in task tracking or discussion tools, so you would need to supplement it with email or a messaging platform for communication.

Trello, **Notion**, and similar tools provide task tracking and project organization. They are free or cheap at the scale a wargame designer needs. They can store files but are not built primarily as file storage systems, so you would pair them with Google Drive or Dropbox for the files and use the tool for task management and communication.

Git and GitHub handle version control with a precision that none of the above tools can match. If you are comfortable with command-line tools or learn to use a graphical Git client, Git tracks every change to every file in your project, lets you compare any two versions, and makes it straightforward to collaborate with others. The learning curve is steep for someone who has never used version control software. I would not recommend it as your first organizational tool unless you are already technical. But if you are comfortable with it, it is the most robust option for tracking file versions.

Do not keep your rulebook in Google Drive, your OOB in Dropbox, your playtest notes in email, and your task list on a sticky note. Put it all in one place. When everything lives in a single system, you know where to look. When it is scattered, you spend time searching instead of designing.

Naming Conventions and Folder Structure

If your organizational tool does not handle version tracking for you, you need naming conventions that make the current version of any file obvious at a glance.

Include the project name, the file type, and a version number in every filename. "1864_Rulebook_v12.1_Final.pdf" tells you what game, what file, and what version. "Rulebook_new_FINAL_v2.pdf" tells you nothing useful except that someone has lost control of their naming scheme.

Be honest about the word "Final." A file named "Final" that gets revised becomes "Final_v2" which becomes "Final_v2_ACTUAL" which becomes a naming disaster you have seen on your own hard drive and someone else's. Use version numbers. They do not drift.

Organize your project folder by file type: a folder for maps, a folder for counters, a folder for rules, a folder for playtest feedback, a folder for correspondence. Within each folder, the naming convention keeps versions straight. When you need the current counter sheet, you go to the counters folder and grab the highest version number. When your artist needs the current OOB, you point them to the OOB folder and tell them the version number.

Date-modified sorting is a fallback, and I use it more than I should. Sorting files by date and grabbing the most recent one works until you have two files modified on the same day, or until you open an old file to check something and its modification date updates. Explicit version numbers are more reliable. I know this and I still default to date sorting half the time, which is why I pay for Basecamp to handle it for me.

The Production Pipeline

Organization becomes critical when your game moves from development into production. During development, version errors are recoverable. You send a playtester the wrong rulebook, they tell you something seems off, you send the right one.

During production, version errors cost real money. Your counter artist works from the OOB spreadsheet. If they use an outdated version, the counters ship with wrong values and you do not catch it until the proof arrives. Your map artist works from your latest draft. If that draft predates your last terrain change, the published map has incorrect terrain. Your printer manufactures whatever files you send them. Wrong files in, wrong components out.

Each handoff in the pipeline is a point where a version error can enter the process. Counter manifests need to match OOBs. Map files need to match the latest terrain rulings. Rulebook versions need to incorporate all playtest changes. Player aids need to match the current rules.

A checklist before each handoff catches these. Does the counter sheet match the OOB? Does the player aid match the rulebook? Does the map reflect the latest playtest revisions? Fifteen minutes of verification saves weeks of rework.

Case Study: *1864: On to Jutland!*

1864: On to Jutland!, my game on the Second Schleswig War, is the clearest example in my catalog of what happens when the production pipeline breaks down.

The game itself came together well. It started as one of the more complicated designs I had made, using an in-depth system of operational movement followed by tactical combat inspired by Frank Chadwick's GDW Crimea game. That system would have added hours to playtime, which was the opposite of what the 2140 series is trying to do, so I scrapped it and rebuilt the game around a simpler framework. The perpetual combat mechanic, where players can keep attacking as long as they have units adjacent to the enemy but disrupt themselves after the second assault, was one of the ideas I salvaged from that earlier version. The final game plays quickly and the system works.

The production side did not go as smoothly. 1864 shipped to customers with a playtest map and half-finished counters. Not a pre-production proof. Not an early review copy. The actual retail product that people paid for arrived with components that were not final.

I had to replace components for the customers who received those copies. For a small publisher like Conflict Simulations LLC, replacement costs come out of the margin that was supposed to make the game worth producing. It also costs credibility. A customer who opens a new game and finds playtest-quality components does not think "organizational mistake." They think "this publisher does not care about quality." You do not get to explain the circumstances.

The wrong files went to the printer. Somewhere between the finished production files and the print order, an older map and an incomplete counter sheet were substituted for the final versions. Whether that happened on my end or the printer's matters less than the fact that no verification step caught it before the games were manufactured and shipped.

A checklist would have prevented this. Open the map file, confirm it matches the rulebook's terrain descriptions. Open the counter sheet, cross-reference it against the final OOB. Check the player aids against the current rules. Fifteen minutes. Skipping those fifteen minutes cost me money, customer goodwill, and weeks spent sorting out replacements.

I learned from this. My subsequent games have not had the same problem, in part because I got better at verifying files before sending them, and in part because Basecamp gives me a single location where the current version of every file is clearly identified. When I send files to a printer now, I pull them from Basecamp rather than hunting through my local hard drive for the most recent version. The tool compensates for my weakness without fixing it. I still default to sloppy habits when I am not paying attention. But a system that makes the right behavior easier than the wrong behavior is often enough.

Discipline Over Tools

Tools are concrete and easy to recommend. Using them consistently is the hard part. Every organizational system works when you set it up. The question is whether you maintain it under pressure, when you are focused on fixing a broken CRT and updating a version number feels like busywork.

I am not good at this. My natural tendency is to save files with vague names, forget to update version numbers, and rely on date sorting to find the current file. Basecamp compensates, but only because I have made it the default place where project files live. If I kept some files in Basecamp and some on my desktop, the system would collapse.

Pick a system. Use it for everything. When you feel the pull to save a quick revision to your desktop instead of the shared workspace, resist. When you want to name a file “rules_new.pdf” instead of “1864_Rulebook_v12.1.pdf,” take the extra ten seconds. The discipline is harder than choosing the tool, and it matters more. Nobody who buys your game will see your design process. They will see the box they opened, and the components inside need to be the ones you intended to ship.

Development

For all but a few of my more recent games, Matt Ward developed every title I published through Conflict Simulations LLC. His name appears in the credits of *Sedan 1940*, *1916*, *1864: On to Jutland*, and most of the rest of the catalog. Matt brought exhaustive expertise from his work as a developer on the *Panzer Grenadier* series for Avalanche Press, and he improved my games tenfold from what they would have been without him. I am stating this plainly because this chapter is about why you need a developer, and the strongest argument I can make is my own experience.

Development is the process that happens between "the designer thinks the game is done" and "the game is ready for players." Look at the credits of any game published by GMT, MMP, Compass, or most other publishers, and you will see a developer listed separately from the designer. That credit exists because the work is real and the skills are distinct.

What a Developer Does

A developer's job is to make sure the game accomplishes the goals it sets out to do while remaining fun and playable as a commercial product. This requires a distinct skillset from design. The designer creates the game. The developer pressure-tests it from the outside, with fresh eyes and no emotional attachment to any particular mechanic.

A developer reads your rules and finds the ambiguities you cannot see because you wrote them. They play your game and find the strategies that break it, the units that are too strong or too weak, the situations where the rules do not cover what happens. They look at your CRT and ask whether the results produce outcomes that match the historical conflict. They look at your order of battle (OOB) and ask whether the unit ratings create the force balance the game needs. They look at your sequence of play and ask whether the phase order creates the right rhythm.

A developer also handles practical concerns that designers tend to neglect. Does the counter manifest match the OOB? Are the setup instructions unambiguous? Does the player aid contain everything the player needs? Are the victory conditions clear and achievable? These are the details that separate a game that plays well on the designer's table from a game that plays well on a stranger's table.

The developer is the first reader who does not share your mental model of the game. As I discussed in Chapter 18, rules-writing requires stepping outside your own understanding. A developer forces that step. They read what you wrote, not what you meant, and they tell you where the two diverge.

The Difference Between Development and Playtesting

Development and playtesting overlap but serve different functions. A playtester reports that a 3:1 attack against a unit in a city produced a defender eliminated result that felt wrong. A developer looks at the CRT, checks the modifiers, determines whether the result is a calibration error, and recommends a specific fix. The playtester provides data. The developer interprets the data and acts on it.

This requires experience with game systems, familiarity with how different mechanical approaches perform in practice, and enough design literacy to propose solutions that address root causes rather than symptoms. Most playtesters tell you something feels wrong. A developer tells you why it feels wrong and how to fix it.

Self-Development

Self-development is tricky. It may feel productive but it accomplishes little unless you can genuinely detach yourself from your own design. After spending months building a game, your attachment to specific mechanics, specific unit ratings, specific rules structures makes it difficult to evaluate them objectively. You know why you made each decision. A developer does not carry that history and can look at the decision on its own merits.

I manage to self-develop my more recent games by tightly scoping designs and being extra critical, a skill that comes from experience and from years of watching Matt Ward find problems I had missed. But self-development is a compromise, not a replacement for having another informed set of eyes on your game. If you can find an experienced developer willing to work on your project, take that opportunity. The game will be better for it.

The difficulty is finding developers. The wargaming hobby does not have a large pool of experienced developers, and the ones who exist are busy. If you are self-publishing, you may need to develop your own game out of necessity. In that case, the best advice I can offer is to create as much distance as possible between yourself and the design before attempting to develop it. Set the game aside for weeks. Come back to it with as close to fresh eyes as you can manage. Read your own rules as if you have never seen the game before. Play it as if you are a stranger encountering it for the first time. You will not fully succeed at this. You will still fill in gaps from memory and overlook ambiguities that are invisible to you because you know what you intended. But the effort produces better results than developing the game while you are still in the middle of designing it.

Working with a Developer

When you work with a developer, the dynamic is collaborative but the roles are distinct. You are the designer. You created the game. You have the vision for what it should be. The developer's job is to help you realize that vision in a form that works for players who do not share it.

A good developer will challenge your decisions. They will tell you that a mechanic you love does not work, that a subsystem you spent weeks building should be cut, that your unit ratings are wrong. This is uncomfortable, and you need to accept it as part of the process. The developer is not attacking your game. They are improving it. The instinct to defend every design choice is natural and counterproductive. Listen to the problem before dismissing the proposed solution. If you disagree with the fix, engage with the problem. Maybe a different fix exists. But the problem itself is data, and ignoring it because you do not like the diagnosis is how games ship with known issues.

A bad developer will try to redesign your game. Development is refinement. If a developer wants to overhaul your core systems, they are designing a different game rather than developing yours. A developer tightens your rules, balances your systems, polishes your game. They do not change what the game is about. If the development process is pulling the game in a direction you did not intend, push back. The game is yours.

The practical workflow varies by publisher. When I worked with MMP on *Rostov '41*, the process was my first glimpse at a professional development workflow. MMP used Basecamp, the same project management tool I now use for my own company. I would send my ideas and notes as the designer. The development team, led by Lee Forester with contributions from Carl Fung and others, would reply with what worked and what would not, then coordinate playtesting based on those assessments. Design ideas flowed in one direction, development feedback flowed back, and the playtesting schedule was built around the current state of the game rather than testing whatever happened to be ready.

Watching that process taught me how a professional company does it correctly. The designer proposes. The development team evaluates. Playtesting generates data. The developer interprets the data and recommends changes. The designer approves or negotiates. The cycle repeats until the game is ready for production. Each role has a defined function and the workflow prevents any single person from operating in isolation.

Compare that to self-publishing through CSL, where I was the designer, Matt Ward was the developer, and the communication was less formal but the roles were the same. Matt would go through the game systematically, identify problems, propose fixes, and send me a revised rulebook or a list of recommended changes. I would review his work, accept most of it, push back on the occasional change I disagreed with, and we would iterate until we were both satisfied. The process was smaller in scale but the principle was identical: the designer creates, the developer refines, and neither role replaces the other.

The Final Checklist

Before a game goes to production, someone needs to verify that every piece fits together. This is development's last function, and skipping it leads to the kind of production failures discussed in Chapter 18.

Has every rule been tested? Not in isolation, but in the context of a full game. Rules that work in theory can break when they interact with other rules under conditions that only arise during actual play. If a rule has not been tested in a full game session, it has not been tested.

Does the OOB match the counter manifest? Every unit in the OOB should have a corresponding counter, and every counter should appear in the OOB. Mismatches here produce games where players cannot find a counter the setup requires, or where extra counters exist that the rules never reference.

Are the setup instructions unambiguous? Can a player who has never seen the game before place every unit on the map without guessing? Setup instructions that reference "the area around Rostov" are ambiguous. Instructions that say "hex 46.09" are not.

Does the CRT produce historical results across a reasonable number of plays? Not every game will produce the historical outcome. That is the point of playing. But the range of plausible outcomes should include the historical result, and the CRT should not consistently produce results that are wildly ahistorical. If your game about Sedan in 1940 consistently produces French victories with the historical setup, something is miscalibrated.

Do the player aids match the current rules? Every modifier, every terrain cost, every CRT column on the player aid should match what the rulebook says. Any discrepancy between the two produces confusion during play and errata after publication.

Is the rulebook internally consistent? Does it use the same terminology throughout? Do cross-references point to the correct case numbers? Does the sequence of play in Section 4.0 match the phase order used in the rest of the rules?

This checklist is not exhaustive, but it covers the categories where errors most commonly survive into published games. A developer working through this list catches problems that save time, money, and the publisher’s reputation. A designer working through it alone catches some of them. Either way, the list needs to be worked through. Skipping it is how games ship with playtest maps.

Case Study: *Rostov ’41* and MMP

Rostov ’41, my Standard Combat Series game on the German drive toward Rostov-on-Don in late 1941, was published by Multi-Man Publishing. It was my first game with a major publisher, and the development experience shaped how I think about the entire process.

The SCS is Dean Essig’s system. Designing within it meant working inside an established rules framework where the core mechanics (movement, combat, supply, exploitation) were already defined. My job as the designer was to build the game-specific rules on top of that framework: the scenarios, the OOB, the special rules that capture what makes the Rostov campaign distinct from other SCS titles. The weather system, which transitions from clear to mud to freeze across the game’s timeline, was one of those specific elements. The mud rules restrict exploitation, overruns, air strikes, and artillery, simulating how the rasputitsa bogged down operations. The deep freeze opens major rivers to crossing but cripples German armor, reflecting the mechanical failures that plagued Panzer formations in the Russian winter.

Lee Forester led the development. The MMP team went through the game with a rigor I had not experienced in self-publishing. They tested scenarios systematically, tracking whether the victory conditions produced competitive games across multiple plays. They checked the OOB against historical sources and flagged discrepancies. They tested the weather transition rules to make sure the mud slowed both sides enough to matter without grinding the game to a halt, and the freeze shifted the balance toward the Soviets without making the German position hopeless.

Carl Fung contributed to playtesting, research, and proofreading. The MMP process involved multiple people checking each other's work, which catches errors that any single person would miss. A designer checks their own OOB and finds it correct because they built it. A second person checks the same OOB against the counter sheet and finds a unit that does not match. A third person reads the setup instructions and finds a hex reference that points to the wrong location. Each additional set of eyes catches a different category of error.

The published game benefited from this process in ways that are difficult to quantify but easy to feel when you play it. The rules are tighter than they would have been without development. The scenarios are more balanced. The special rules interact cleanly with the SCS framework. The credits list reflects the reality: series design by Dean Essig, game design by me, development by Lee Forester, with a team of playtesters and proofreaders who each contributed to making the finished product better than what any one person could have produced alone.

Working with MMP taught me that development is not an optional step for designers who lack confidence. It is a standard part of professional game production, performed by people with specific skills, producing results the designer alone cannot replicate. Major publishers budget time and resources for development because the games that skip it show the gaps.

Publishing Your Game

You have a game. It works. It has been developed, playtested, and revised. The rules are written, the counters are designed, the map is drawn. Now you need to get it into the hands of players. This chapter covers the three paths available to you: traditional publishing, self-publishing, and digital-first distribution. Each has tradeoffs. None of them will make you rich. I know this because I have tried all three, and this chapter is as much a cautionary tale as it is a guide.

Wargame publishing is not a viable career path for most people. The market is small. The margins are thin. The audience, while passionate, is not large enough to support a full-time income for anyone but a handful of established publishers, and even they operate on margins that would horrify someone used to mainstream business. If you are designing wargames because you love it and want to see your work in print, this chapter will help you do that. If you are designing wargames because you want to make a living, I would encourage you to reconsider that plan. I say this as someone who tried it and learned the hard way that creative ability and business ability are separate skills, and possessing one does not guarantee the other.

The Traditional Publisher Path

Traditional publishing means submitting your game to an established publisher (GMT, MMP, Compass Games, Decision Games, Revolution Games, and others) who handles production, distribution, marketing, and sales. You design the game. They do everything else.

This is the most stable path and the one most likely to produce a game you are proud of. A good publisher has professional graphic designers, experienced developers, established printing relationships, and distribution channels that reach retailers and customers worldwide. Your game will look better than anything you could produce yourself, and it will reach more people. The publisher's reputation lends credibility to your design. A first-time designer published by GMT or MMP enters the market with an implicit endorsement that no amount of self-promotion can replicate.

The tradeoff is time and money. Publisher timelines are long. Submitting a game, getting it reviewed, entering the development queue, reaching production, and arriving in players' hands can take years. GMT's P500 system, where a game goes on a pre-order list and enters production once it reaches 500 orders, is the industry standard for managing financial risk. It works for the publisher and for the customer. For the designer, it means waiting. Your game sits on a list. Orders trickle in. You check the P500 page to see whether the number has moved. It hits the threshold, enters production, and ships. Or it does not hit the threshold, sits on the list, and you wait longer.

The financial return is modest. Royalty structures vary by publisher, but a wargame designer can expect somewhere between a few hundred and a few thousand dollars for a typical print run. Larger publishers with bigger print runs and higher price points pay more, but the per-game income supplements your income. It does not replace it. You are not going to quit your day job on wargame royalties.

You get a product done right in exchange. Professional production values. Wide distribution. A game that sits on retail shelves and appears in convention dealer halls. Your name in the credits of something that looks and plays like a professional game, because professionals made it. For most designers, and all first-time designers, this is the right path. Submit your game. Be patient. Let the publisher do what publishers do.

I was not patient enough for this. That is a personality flaw, not a criticism of the model.

Self-Publishing

Self-publishing means you handle production, distribution, and sales yourself, either through a business entity or your own personal operation. You find printers. You source components. You manage inventory. You ship boxes. You handle customer service. You are no longer a designer. You are a business owner, and those two jobs have almost nothing in common.

Self-publishing is more accessible now than it has ever been. Print-on-demand services can produce small runs of maps and rulebooks. Counter sheets can be printed by specialty shops. Kickstarter and other crowdfunding platforms give you access to upfront capital and a built-in marketing channel. Wargame Vault and similar platforms handle digital distribution with no inventory risk.

I have opinions about crowdfunding that I will keep brief. Kickstarter campaigns work for some designers and some products. They provide capital before production, which solves the cash flow problem that kills most small publishers. They also create obligations. You are taking money from people in exchange for a promise to deliver a product. If you cannot deliver, because production costs more than expected, because your timeline slips, because life intervenes, you have people's money and they do not have a game. If you use crowdfunding, scope your project conservatively, build in a margin for things going wrong, and deliver what you promised. The community remembers who follows through and who does not.

Do not start your own publishing company unless you have significant capital, a realistic business plan, and the discipline to execute it. Running a business requires skills that have nothing to do with game design: financial management, vendor relationships, inventory control, customer service, marketing, accounting. You need to make promises you can keep and keep the promises you make. I can design a wargame. I could not, as it turned out, run a publishing company with the consistency and professionalism the job requires.

The financial discipline alone is a serious barrier. You need to pay printers on time. You need to pay artists on time. You need to pay developers, graphic designers, and anyone else whose work your product depends on. If cash flow problems mean vendors wait months for payment, if production delays mean customers wait years for products, you damage relationships that you cannot repair. The wargaming industry is small. Your reputation as a business partner follows you in ways that your reputation as a designer does not.

Digital-First Publishing

Digital-first means releasing your game as a print-and-play file, a VASSAL module, a Tabletop Simulator implementation, or a combination of these, before or instead of pursuing physical production.

This path has the lowest barrier to entry and the lowest financial risk. You create the game files. You upload them to a platform. Players download them and print or play them digitally. Your costs are your time and whatever software you used to create the files. You carry no inventory, deal with no printers, ship no boxes.

Wargame Vault is the primary commercial platform for digital wargame distribution. It handles payment processing and file delivery. You set the price. They take a percentage. Players download the files. The per-sale revenue is small, most print-and-play wargames sell for modest prices, but the per-sale cost is near zero, so the margin is high in percentage terms even if the absolute numbers stay low.

Digital-first publishing also serves as a proving ground. A game that generates interest as a print-and-play can attract the attention of traditional publishers. Several games that started as free or low-cost digital releases have been picked up by publishers for physical production. If your goal is eventually to see your game in a box on a shelf, releasing it digitally first lets you build an audience, gather feedback, and demonstrate market interest without financial risk.

The limitation is reach. Digital wargames sell to a fraction of the audience that physical games reach. Many wargamers prefer physical components. A print-and-play game requires the player to invest time and materials in producing their own copy, and many players will not do that for a game they have not already decided they want. Digital-first works well as a starting point or a supplement. It works less well as a primary publishing strategy if your goal is to reach the broadest possible audience.

Marketing a Wargame

Marketing in this hobby is almost entirely word of mouth. The community is small enough that a good game finds its audience through forum discussions, BGG ratings, convention demos, and review copies sent to the handful of people whose opinions carry weight.

Your BoardGameGeek page is your storefront. Most wargamers will look there before buying. Good photographs of the components, a clear description of what the game covers and how it plays, and a responsive designer who answers questions in the forums all contribute to sales. BGG is also where your reputation lives. Positive ratings accumulate slowly. Negative experiences spread quickly.

Convention presence matters if you can manage it. A demo at ConsimWorld Expo, WBC, or a regional convention puts your game in front of players who can try it before buying. In-person demos convert browsers into buyers more effectively than any online marketing.

Review copies sent to respected reviewers and content creators generate visibility. A thoughtful review from someone the community trusts is worth more than any advertising you could buy. Advertising in this market does not work. The audience is too small and too savvy for it.

Social media will make or break you, and I say this from experience. As a solo publisher, there is no separation between you and your business. Your personal social media behavior reflects on your company. Your arguments, your tone, your interactions with other members of the community all shape how people perceive your business. If you are the kind of person who gets into arguments online, as I am, understand that every argument costs you potential customers. People do not want to give money to someone they find unpleasant, even when that person is correct about the substance of the argument. Being right and being combative about it produces enemies who remember the combativeness long after they forget what the fight was about.

Be thoughtful about your online presence. Be professional even when others are not. Respond to criticism of your games with grace and to personal attacks with silence. Your reputation is half your business in a community this small. Protect it. I did not, and the cost of that carelessness compounded over years.

Pricing Your Game

The relationship between component quality, print run size, and retail price is the central economic reality of wargame publishing.

A small print run costs more per unit. Printers charge setup fees that spread across the total quantity, so five hundred copies of a game cost much more per copy than two thousand. Component quality adds cost at every level: mounted maps cost more than paper maps, die-cut counters cost more than uncut sheets, linen-finish boxes cost more than plain cardboard. Every upgrade improves the product and increases the price.

A ziplock game with a paper map and a single counter sheet can retail for thirty to forty dollars with reasonable margins. A boxed game with mounted maps, multiple counter sheets, and professional packaging might retail for eighty to a hundred and twenty dollars. Margins on both can land in a similar range, because the higher-priced game also costs more to produce.

Pricing too low is a common mistake for self-publishers. The instinct to keep prices accessible is understandable but the math does not work if your price fails to cover production, shipping, and platform fees while leaving enough margin to fund your next project. Price your game based on what it costs to produce and what the market will bear. Look at comparable games from established publishers and price in the same range. Wargamers expect to pay wargame prices. Underpricing your game does not attract more buyers. It signals that the product is lower quality.

What I Would Do Differently

I started Conflict Simulations as a publishing company because I was impatient and because early results were encouraging. Games sold. Money came in. The operation felt viable. I lacked the financial discipline and business experience to keep it that way.

I made decisions that seemed reasonable at the time and proved costly in hindsight. The most damaging was offering pre-orders on games that were not yet finished, collecting payment upfront for products that did not exist in final form. Pre-order revenue was supposed to fund production. Instead, I was taking on obligations I could not fulfill on the timeline I promised. Some games took far longer to complete than projected. Some are still unfinished. The people who paid for those games trusted me with their money, and where I failed to deliver on time, I damaged that trust.

I have worked to make good on those commitments, fulfilling orders where possible, processing refunds when asked. But the damage to my reputation was done, and reputation in this community does not rebuild fast. Pre-orders are not wrong in themselves. Publishers use them responsibly all the time. Mine were problematic because I did not have the production infrastructure or financial discipline to back the promises I was making. I was a designer playing at being a publisher, and the two jobs require different skills.

I was too ambitious in scope. I published close to thirty games in less than a decade. That pace required cutting corners I should not have cut, shipping products before they were ready, and spreading my attention across too many projects at once. Quality suffered where it should not have. I commissioned art for games that remain unfinished, money that made sense in the context of an ambitious production schedule and makes no sense in retrospect.

The burnout came around 2020. My personal life was falling apart, I was moving constantly, and the business was the last thing I could focus on. Production slowed. Communication with customers and partners suffered. Problems compounded. The operation that had felt manageable when energy and optimism were high became unmanageable when both ran out.

The company still exists. A handful of retailers stock my games. Revenue from those sales and from my full-time job covers operating costs. But the operation I envisioned when I started, a functional small publisher producing a steady stream of quality wargames, did not survive contact with my own limitations as a businessman.

I share this to give you information I wish someone had given me. You can learn to design games, write rules, develop scenarios, create maps. The business work, managing cash flow, maintaining vendor relationships, fulfilling obligations on time, keeping your public persona professional, knowing when to say no to a project you cannot complete, that is the part that determines whether your publishing venture survives.

Be honest with yourself about which skills you have and which you lack. If you can design games and run a business, self-publishing can work. If you can design games but not run a business, find a publisher. The mistake is taking on responsibilities you cannot meet and letting other people bear the consequences.

Case Study: The Economics of Small-Press Wargame Publishing

Most designers have no idea what the numbers look like until they are already committed. Here they are.

A typical small-press wargame (paper map, one or two counter sheets, rulebook, ziplock bag) costs eight to fifteen dollars per unit to produce at a run of three to five hundred copies, depending on your printer, your component choices, and where you print. Setup fees, shipping from the printer to your location, and packaging materials add to that baseline.

Retail price for a ziplock game in this market ranges from roughly thirty to fifty dollars. If you sell direct through your own website or at conventions, you keep the full margin. If you sell through distributors, you sell at roughly forty to fifty percent of retail. If you sell through a platform like Wargame Vault or Noble Knight, the platform takes a percentage.

At a five hundred copy print run, selling through a mix of direct and distribution, a successful ziplock game might gross ten to fifteen thousand dollars over its lifetime. Subtract production costs, shipping, platform fees, art commissions, and replacement components, and the net shrinks fast. A game that sells out its print run at these numbers might net a few thousand dollars over two to three years. That does not account for your time, which in a self-publishing operation is substantial.

These numbers improve with scale. A publisher producing multiple titles per year amortizes fixed costs across a larger catalog.

Backlist sales provide ongoing revenue. A reputation for quality brings repeat customers. But reaching that scale requires capital, and the early years of any small publishing operation are the most capital-intensive and least profitable.

The math works if you treat it as a hobby that partially pays for itself. The math does not work if you need it to pay rent. If someone tells you otherwise, they are either working at a scale you do not have access to yet, or they are not telling you the full story.

I tried to make the math work as a primary income source. It did not, and the attempt cost me more than money. Learn from that. Design your games because you love the work. Publish them because you want to share the work. Build the business side with eyes open, expectations realistic, and a day job you can fall back on. The wargaming world needs more designers. It does not need more designers who burned out trying to turn a passion into a paycheck.

Beyond Hex and Counter

I prefer simple hex-and-counter games. I have for as long as I have been designing them. I came to wargaming in my mid-twenties through John Tiller's PC games, not through a childhood spent punching counters out of SPI flats, and by the time I discovered the physical hobby I had already internalized the grammar of hex-based operational gaming through hundreds of hours on a screen. Breaking out a copy of *Soldiers* or *Battle for Germany* will always be more enjoyable to me than spending hours learning a new system, no matter how good the new system is. This is stubbornness on my part, and I am comfortable admitting it. The ritual of the hex grid, the CRT, the counter tray, the movement phase, these are the mechanics I adopted and the ones I return to when I sit down to play or design.

But the hobby has moved past me in certain respects, and I have come to respect that movement even if I do not always participate in it. Wargame design is pushing harder toward new frontiers than at any point in its history. Designers like Fred Serval and Mark Herman seem to produce new ideas on a monthly basis. Amabel Holland and Volko Ruhnke have built entire design philosophies that challenge the conventions I learned on. New designers arrive with fresh systems at a rate that was not possible before the internet collapsed the distance between an idea and a prototype.

The tools are better. The audience is more open. The space for experimentation has never been wider.

Traditional hex-and-counter wargames will retain their audience. The success of publishers like VUCA Simulations proves this. Their production quality is exceptional, their game designs are sharp, and they demonstrate that the traditional format can be executed at a level that competes with anything on the market. I have never owned one of their games, but I have seen enough of them to know that the craft is beyond well done. The format is not dying. It is being joined by alternatives that expand what a wargame can be without replacing what it already is.

New Formats and Hybrid Designs

Block games, dice games, trick-taking wargames, card-driven games that borrow from euro design, resource management systems layered onto historical simulations. A decade ago I would have been skeptical that any of these could produce a satisfying wargame. I have played enough of them now to know I was wrong. The combination of euro-style mechanical elegance with wargame-style historical grounding has produced games that I would not have thought possible, and designers who are comfortable working in both traditions are producing work that neither tradition could have generated alone.

I do not design in these hybrid spaces often. My instincts pull me back to hex-and-counter, and my design language is rooted in SPI's grammar. But I respect and accept that there are more ideas to be had in combining systems with wargames than I previously assumed. The designers doing this work are expanding the vocabulary of the hobby, and the games they produce reach players who would never have picked up a traditional hex-and-counter title.

Digital Tools and Prototyping

VASSAL is an essential tool for prototyping in this industry. It allows designers to build functional digital prototypes, test remotely with playtesters across the world, and iterate on designs without printing a single component. Most published wargames have VASSAL modules, and many designers use VASSAL as their primary development environment before physical components exist.

VASSAL has a learning curve, though, and there is room for a determined developer to build something better. Rally the Troops proves it is possible. A cleaner, more intuitive prototyping tool aimed at wargame designers would fill a gap that VASSAL occupies by default rather than by excellence. The functionality is there. The user experience could be improved.

Tabletop Simulator serves a similar function with a different approach. It provides a 3D physics sandbox where you can place components, flip counters, and roll dice in a virtual space that mimics a physical table. For playtesting and remote play, it works fine. It lacks the automation that VASSAL provides (automatic rules enforcement, scripted procedures), but the trade-off is a more tactile experience that some players prefer.

Browser-based prototypes are another option, and one I have explored with my current design. A game built as a web application can be played by anyone with a browser, requires no installation, and can incorporate automation, visual feedback, and data logging that physical prototypes cannot. The development cost is higher, but for a designer comfortable with digital tools or willing to collaborate with a developer, the results can be impressive.

Wargames Beyond Entertainment

As discussed in Chapter 1, wargames have served military training since the Prussian Kriegsspiel of the early nineteenth century. That tradition continues today. The most instructive modern example might be Millennium Challenge 2002, a joint forces exercise run by the United States military to test new operational concepts. Lieutenant General Paul Van Riper commanded the opposing force, playing the role of an unnamed Middle Eastern military commander. Van Riper used unconventional tactics: motorcycle couriers instead of radio communications to avoid electronic surveillance, swarm attacks with small boats against naval vessels, cruise missiles launched from civilian aircraft. In the simulation, his forces sank a carrier battle group and inflicted casualties that would have killed approximately twenty thousand service members in a real engagement. The exercise controllers reset the game, restricted Van Riper's tactics, and scripted the outcome to ensure the US force succeeded. Van Riper stepped down as commander of the opposing force in protest.

The episode captures wargaming in professional contexts at its best and worst. A wargame can expose assumptions, reveal vulnerabilities, and generate outcomes that the participants did not anticipate. Van Riper's swarm tactics exposed a genuine vulnerability in naval force protection doctrine. The danger is that the people commissioning the wargame may not want their assumptions exposed, and the institutional pressure to produce a predetermined outcome can override the exercise's analytical purpose. A wargame that cannot produce an uncomfortable result is not a wargame. It is a briefing with dice.

Beyond military applications, wargames serve educational institutions, corporations, and government agencies as tools for teaching decision-making under constraint. Matrix games are a good example of how wargames can be purpose-built for organizational use. A matrix game requires little more than a scenario, a set of structured arguments, and an umpire to adjudicate results, much like a classic Kriegsspiel. Players propose actions, justify them with arguments, and the umpire assesses the likelihood of success. The format is flexible enough to model anything from a military campaign to a corporate crisis to a diplomatic negotiation.

The critical thinking that wargames develop is context-driven. You are not solving a puzzle with a single correct answer. You are making decisions with incomplete information, managing competing priorities, accepting risk, and living with consequences. The freedom a wargame gives you is the freedom to try anything within the rules of the simulation, but that freedom is also the freedom to fail. You learn from the failure because the simulation shows you why your decisions produced the outcome they did. A lecture cannot produce that feedback loop. A game where your choices had consequences you can trace does it in two hours.

The best way to learn about the future is to study the past, and simulations like wargames serve that purpose for organizations ranging from defense ministries to business schools. The growing interest from defense institutions in commercial wargame design reflects a recognition that hobbyist designers, working with historical data and iterating through playtesting, sometimes produce simulations that are more rigorous and more honest than purpose-built institutional exercises. A game designed to sell to an audience of informed hobbyists cannot afford to produce ahistorical results. A game designed to validate an institutional conclusion can.

Case Study: *A Mighty Fortress Is Our God*

Most of my published games are hex-and-counter operational simulations. *Sedan 1940*, *1916*, *Rostov '41*, the catalog runs on hexes, CRTs, and movement points. *A Mighty Fortress Is Our God* is not that kind of game.

AMFIOG is a solitaire game about the Anabaptist commune of Munster, 1534 to 1535. You play as the commune's leadership, managing resources, making decisions about how to interact with the besieging forces outside the walls, and trying to survive long enough to achieve your goals. There is no hex map. There are no counters in the traditional sense. The game runs on a dice pool, a resource management system, a queue of outsider cards representing the forces and factions surrounding the city, and a fate deck that drives narrative events.

Each turn, you draw dice from a bag. The dice come in different colors representing different capabilities: white for general actions, yellow for piety, red for military, blue for influence, purple for special. You allocate these dice to actions: negotiating with outsiders, raiding besieging forces, issuing decrees, managing internal unrest, fortifying defenses. The dice you spend are exhausted. The dice you lose to casualties or events leave the bag permanently. Your resource pool shrinks as the siege progresses. Every decision about how to spend a die is a decision about what you cannot do with that die later.

The outsider queue presents six cards at a time, each representing a faction or group outside the walls: nuns, nobles, landsknechts, merchants, peasant levies. You can negotiate with them (spend dice to meet their resistance value, gain their cooperation and ongoing passive abilities) or raid them (spend military dice, gain resources, risk casualties). Each card has different resistance values, different rewards for negotiation versus raiding, and different passive effects once converted to your cause. Graf Dhaun, a noble commander with a resistance of 4 and a siege value of 2, is a different proposition than a group of nuns with a resistance of 1 and no siege contribution.

Fate cards fire each turn and introduce narrative events drawn from the historical record: The Secret Passage Revealed, Sudden Assault, Plague in the Bishop's Camp, The Girl Who Would Be Judith, Luther Denounces Munster. Each card carries mechanical effects that alter the game state. Some are catastrophic. The Secret Passage and Sudden Assault events can breach the walls and end the game if your defenses are inadequate. Others provide opportunities or force difficult choices.

The game tracks siege strength, defense, entrenchment, bombardment, consumption, unrest, cohesion, piety, influence, traitor pressure. Each track interacts with others. High unrest can trigger a breach. Low food increases consumption, which increases unrest. Piety and influence affect your ability to negotiate with outsiders and issue decrees. The system creates a web of interdependent pressures where improving one track often worsens another, and you are constantly managing trade-offs with a shrinking pool of resources.

None of this resembles the hex-and-counter games that make up the rest of my catalog. AMFIOG uses no hexes, no CRT, no movement points, no stacking limits. The design language is closer to a euro-style resource management game than to anything SPI published. The historical argument is delivered through the fate deck and the outsider cards rather than through unit ratings and terrain modifiers. The simulation runs on resource pressure and probabilistic events rather than on spatial positioning and combat odds.

I designed it this way because the subject demanded it. The siege of Munster was not an operational problem. It was a resource problem, a political problem, a religious problem, and a survival problem. The people inside the walls were not maneuvering armies on a battlefield. They were managing food, maintaining morale, negotiating with enemies, and making desperate choices under mounting pressure. A hex map would have been the wrong tool. The dice pool and resource tracks capture the texture of the siege in ways that a traditional wargame format could not.

Players responded to the game differently than I expected. I anticipated that the wargaming audience would find the format alienating, too far from their expectations of what a wargame looks like. Some did. Others engaged with it because it did not look like what they were used to. The mechanical novelty created curiosity that a thirty-first hex-and-counter game from a small publisher would not have generated. The solitaire format helped. A player trying a strange new system alone at their table is more willing to experiment than a player trying to convince an opponent to learn unfamiliar rules.

AMFIOG is still in development and may not be my best game. It represents the kind of design thinking this chapter is about. The format should serve the subject. If your subject fits a hex map and a CRT, use a hex map and a CRT. If your subject demands something else, build something else. The hex grid is a tool, not a constraint. When it fits, it is the best tool available. When it does not, put it aside and find the tool that does.

AI and the Designer's Toolkit

This is a chapter I need to write carefully, because the subject generates more heat than light in the wargaming community and in creative communities more broadly. I have strong opinions about AI tools. I use them daily. I have built tools that use them. I also understand why some designers want nothing to do with them, and I respect that position even where I disagree with it. If you are reading this chapter and you have already decided that AI has no place in your design process, skip to the next chapter. Nothing I say here will change your mind, and I am not trying to. This chapter is for designers who are curious about what these tools can do and want an honest account from someone who has used them, made mistakes with them, and figured out how to make them useful.

What AI Does Poorly

AI is bad at taste. It cannot tell you whether your game idea is good. It cannot tell you whether a mechanic is elegant or clumsy. It cannot look at your combat results table and feel whether the outcomes produce the right tension. It does not know what "right" means in the context of your specific design, because "right" is a judgment that requires historical knowledge, design experience, and an understanding of what you are trying to say with this game. AI does not have an argument. You do.

If you ask AI to generate a game concept from scratch, you will get something generic. If you ask it to design your mechanics, you will get something that functions but lacks personality. If you ask it to write your rules without heavy guidance, you will get prose that reads like it was written by a committee that has never played a wargame. The output looks competent at a glance and falls apart under scrutiny. This is slop, and the wargaming community can spot it faster than most audiences because wargamers are a well-read, detail-oriented group who have spent decades developing strong opinions about what good design looks like.

AI is also bad at knowing when it is wrong. It will present incorrect information with the same confidence as correct information. It will fabricate order of battle (OOB) data, invent historical events, and cite sources that do not exist. If you are not knowledgeable enough about your subject to catch these errors, AI will make your game worse, not better. The designer's historical knowledge is the quality filter. Without it, you are building on sand.

What AI Does Well

AI is good at tasks where the designer provides the knowledge and the AI provides the labor. Research acceleration is the clearest example. If you are designing a game about a specific battle and you need to cross-reference OOB data across multiple sources, AI can help you organize and compare that data faster than you can do it by hand. If you have a stack of academic sources and need to find every reference to a specific unit, engagement, or logistical detail, AI can search and summarize faster than you can read.

The key word is "faster." AI does not do research you could not do yourself. It does research you do not have time to do yourself, which for a designer working a full-time job and designing games in whatever hours remain, is the difference between a project that moves forward and one that stalls.

I built a tool called Bookfinder General that searches for, downloads, and extracts text from research books, then lets AI read and summarize the content. The tool exists because I needed a way to get through large volumes of historical material quickly without skipping the research. The AI reads the book. I read the AI's summary. I go back to the original source for anything that matters. The AI is doing the first pass, the equivalent of skimming a book to find the relevant chapters. The historical judgment about what to do with that information remains mine.

AI is also good at generating structured output from unstructured input. If you have a pile of handwritten notes, scattered design ideas, and half-formed mechanics, AI can help you organize that material into something coherent. It will not tell you which ideas are good. It will help you see all your ideas in one place so you can make that judgment yourself.

And AI can generate code. If you need a Monte Carlo simulator to run five thousand games and tell you which fate cards are killing your players, AI can write that simulator in an afternoon. If you want a browser-playable prototype that renders your game state and lets a playtester try your design without printing a single component, AI can build that too. I have done both of these things for AMFIOG, and the tools they produced are things I could not have built on my own. The simulator runs five thousand games per strategy profile and generates statistical breakdowns: win rates, loss reasons, resource flows by turn, dice economy, danger scores for every fate card in the deck. The browser version renders the full game with dice allocation, outsider card interaction, and an event log. These are development tools that would have required hiring a programmer a few years ago. Now they require describing what you want to a competent AI and iterating until it works.

The Source of Truth Problem

The difference between AI that produces slop and AI that produces useful work is whether you give it a source of truth.

AI left to its own devices will generate plausible-sounding content from statistical patterns in its training data. It will average out the design space and give you something that resembles a wargame the way a composite sketch resembles a face. It is recognizable but has no identity. This is what happens when you ask AI to "design a wargame about Gettysburg" with no further guidance.

AI given a source of truth, your research, your design document, your OOB compiled from primary sources, produces different results. It is no longer generating from averages. It is working from your data, following your constraints, building within your framework. The output still requires your judgment to evaluate, but the starting point is your knowledge rather than a statistical composite of the internet.

I have generated OOBs and rulesets with AI assistance that I would not call slop. The reason they are not slop is that I forced the AI to work from historical sources rather than its training data. With Bookfinder General, the AI reads the actual book before summarizing it. With DevForge, my game development tool, the AI reads your game design document before writing a line of code. The GDD is the source of truth. The AI implements your vision rather than substituting its own. The pattern is the same in both cases: you provide the knowledge, you define the constraints, and the AI operates within them.

This does not make AI infallible within those constraints. It still makes mistakes. It still needs oversight. But the mistakes are different in kind from the slop that comes from unconstrained generation. They are implementation errors rather than conceptual errors, and implementation errors are easier to catch and fix.

The Ethical Question

Some designers reject AI tools on moral or philosophical grounds. They object to how the training data was collected, to the labor practices of the companies building these systems, to the environmental cost, or to the principle that creative work should be done by humans. I respect these positions. They are held by people I admire in this hobby, and I am not going to tell anyone they are wrong for holding them.

My own position: I do not believe there is such a thing as ethical consumption under capitalism, and I believe this more now than I did before I started working a minimum-wage job full time. The supply chains behind your computer, your software, your printer, your phone, the paper your rulebook is printed on, the shipping that delivers your game, none of it is clean. Refusing to use AI tools is a personal choice, and I have no quarrel with anyone who makes that choice. Framing it as a moral high ground while participating in every other exploitative system that modern commerce runs on is a distinction I do not find meaningful.

For those willing to use AI, the question is whether you use it well or poorly. Used well, with your knowledge as the foundation and your judgment as the filter, it is a force multiplier that lets a solo designer with limited time produce work that would otherwise require a team or remain unfinished. Used poorly, without knowledge or judgment, it produces work indistinguishable from not having done the work at all. The tool does not determine the quality. The designer does.

What AI Gave Me Back

The practical argument for AI tools in wargame design is about what becomes possible when time is no longer the bottleneck.

I burned out around 2020. The publishing company was struggling, my personal life was in disarray, and game design, the thing I had built my identity around, was the last thing I had energy for. For several years I did not design anything meaningful. The ideas were still there. The desire was still there. The time and energy were not.

AI tools changed that equation. I can do in an evening what used to take a week. Research that required days of reading can be surveyed in hours. A prototype that required learning a programming language or hiring a developer can be built in a conversation. A Monte Carlo simulation that would have been beyond my technical ability can be described, built, tested, and revised in an afternoon. The constraint that kept me from designing, not having enough hours in the day while working a full-time job, became manageable.

That matters more than the tools themselves. I am designing again. The creative part of my life is functioning again after years of dormancy. A designer who had run out of time found a way to get it back.

The bottleneck moved. For most of my design career, the gate between an idea and a working prototype was coding ability. A Monte Carlo simulator, a browser-playable rulebook test, an OSINT terminal that pulls live conflict data: I could imagine all of these. I could not build them, because building them required programming skills I did not have and could not justify spending years acquiring. AI removed that gate. Now the constraint is the one it was always supposed to be: design judgment. Knowing what is worth building and what to refuse.

That sounds like the obstacle got easier. It did not. Removing the coding gate does not automatically produce shipped work. I have started more projects than I have finished, and most of the unfinished ones stalled not because AI failed me but because I lacked the discipline to scope them correctly, to decide what to delegate and what to gate on my own judgment, to refuse the next shiny idea long enough to close the current one. The access is real. The discipline has to come from somewhere else.

What I found is that a wargame designer approaches this problem with a portable apparatus. Wargamers have spent decades thinking about how to model complex systems, how to separate the essential from the noise, how to build something that runs honestly rather than just plausibly. Those same habits of mind apply to operating AI. Two frameworks I developed out of my own frustration with AI's failure modes make the point. The Hammerstein framework takes a 1933 German general's typology of officer quality (clever-lazy, clever-industrious, stupid-lazy, stupid-industrious) and applies it to how a person operates an AI agent. The same officer who blindly executes bad orders without questioning them is the same AI user who accepts confident-but-wrong output without verification. Hammerstein's clever-lazy officer, the one who delegates execution while maintaining strategic judgment, is the model. GeneralStaff applies a different military concept, the staff structure that manages information flow and execution across a campaign, to orchestrating AI development agents across multiple projects. Both are wargame thinking applied to a new medium. That is not coincidence. It is what wargame designers do.

I am skeptical of AI. I use it daily and I built both of these frameworks out of that skepticism: I watched AI produce historically wrong OOBs with full confidence, watched it drift off the design document the moment a conversation grew too long, watched it optimize for sounding useful rather than being useful. The frameworks exist to catch those failure modes structurally. I run them myself. That skepticism is the credential, not the output.

Case Study: A Mighty Fortress Is Our God and the Long Road Through AI

AMFIOG had been stuck in my head for over a year before AI got involved. I had notes. I had research on the Munster commune, the siege, the Anabaptist movement, the political and religious dynamics. I had the core idea: a solitaire game about managing a besieged commune, resource management under pressure, the slow constriction of a siege. I knew what I wanted the game to feel like. I could not figure out how to make the pieces fit together mechanically.

At some point, exasperated, I dumped all my notes into a conversation with ChatGPT and asked it to help me find the connective tissue. I told it the game should work as a State of Siege and euro hybrid. It suggested structural approaches that I felt stupid for not thinking of myself: the dice pool as a shrinking resource, the outsider queue as a spatial-economic system, the fate deck as a narrative driver that interacts with the mechanical state. The AI did not design the game. It saw connections in my own notes that I could not see because I was too close to the material.

The experience was not smooth. I had no idea how to work with AI at that point. My chat history was stuck in a buggy ChatGPT window. Conversations grew too long and the AI started forgetting context, contradicting earlier decisions, losing track of the design. It became a nightmare of trying to extract useful information from a deteriorating conversation. I figured out how to pull what I needed out of the text and how to work with command-line tools, which led me to Claude Code, where I could work with my original notes cleanly, maintain context across sessions, and continue development without fighting the tool.

Trial and error. That is the honest summary. The AI was useful from the beginning, but I was bad at using it, and the early tools were bad at maintaining coherence over long conversations. Both of those problems improved with time and practice. The design moved from a pile of notes to a functional prototype with a Monte Carlo simulator, a browser-playable version, and a rulebook that is further along than anything I have produced in years.

Being no longer too burned out to create again matters more than any specific tool. But the tools removed enough friction that designing within the constraints of a full-time job, limited energy, and no budget became realistic again. For a designer in that situation, and there are more of us than the hobby likes to acknowledge, AI is worth learning to use well. The discipline that turned my own trial and error into a working method is what I wrote about in *Clever Lazy: The Hammerstein Method.*

Go Design Something

I am not widely regarded as a great or consistent designer. My BGG ratings would confirm that. I have shipped games before they were ready, written rulebooks with errors I should have caught, made business decisions that cost me relationships and money, and burned through goodwill I spent years building. I have laid all of that out across the previous chapters because I think the mistakes are more useful to you than the successes.

Three of my games were nominated for Charles S. Roberts Awards. I do not mention this to brag. I mention it because the same designer who shipped rough products and ran a publishing company into the ground also produced work that the hobby considered worth recognizing. Both things are true at the same time. You do not need to be perfect to make something good. You do not need to get everything right. You need to care enough about the work to keep doing it after you have gotten things wrong.

The Drive

I talked earlier in this book about the subconscious drive toward creation, the pull toward making something that did not exist before. I want to be more specific about what that looks like in practice, because it is not the romantic process people imagine.

I am manic about creative work. When an idea takes hold, it becomes all-consuming. I will spend days inside a subject, reading sources, sketching maps, building orders of battle, running numbers in spreadsheets, sleeping poorly because my brain will not stop solving design problems. The period of obsession produces real output. Games get built during those bursts. But the intensity is not sustainable, and it is not always healthy.

I do myself no favors with how hard I focus. The manic burst is validating. You feel productive, you feel like the work matters, you can see the game taking shape in front of you. Then the burst ends and you are exhausted and behind on everything else in your life. The body keeps a tab. The mind keeps a tab. I am approaching forty and I have become more familiar with those tabs than I would like.

I used to think that pressure made better art. Financial pressure, deadline pressure, the pressure of needing a project to succeed because the alternative was worse. I am no longer sure that is true. What pressure does is force honest outcomes. When you are under pressure, you do not have the luxury of overthinking. You make decisions and live with them. Sometimes those decisions are good because the pressure stripped away the hesitation that would have diluted them. Sometimes they are bad because you did not have the time or energy to think them through. Pressure does not improve art. It forces the designer to come to terms with the other factors at play: health, finances, relationships, mental state. You design the game with whatever version of yourself shows up that day.

My own experience has proven this. I have designed games during financial crises, through depression and anxiety disorders, through problems in my personal and professional life that I will not detail here because they are not the point. The drive was there through all of it. The drive does not care about your circumstances. It shows up whether you are ready for it or not, and it demands that you make something.

Recently I had two oral biopsies for suspected oral cancer. I am thirty-eight years old. I have smoked since I was thirteen, so the situation was not a surprise, but sitting in a waiting room for an hour past your appointment time knowing what the doctor is about to do recalibrates your priorities faster than any self-help book. As of this writing, the results have not come back yet. But the experience has already left a mark regardless of what they say. The urgency I feel about creative work, the sense that there is not enough time to make everything I want to make, got louder. That urgency is useful in small doses. It gets you out of bed and in front of the computer. In large doses it feeds the same manic cycle that burns you out.

I do not have a solution to this. I am not going to pretend that I have figured out how to balance creative obsession with physical and mental health, because I have not. What I can tell you is that the work survives the chaos. The games I designed during the worst periods of my life are published. People play them. Some of those games are among my best. Others show the cracks of the circumstances that produced them. Both kinds taught me something about design that I could not have learned any other way.

What the Work Gives Back

Writing this book forced me to look at my own career more honestly than I had before. Every chapter sent me back through decisions I made and outcomes I produced, and the picture that emerged is a messy one. I would do most of it differently with the brain I have now. But I would still do it. The games I designed, the ones that work, give players an experience I could not have given them any other way. A simulation of a battle they read about. A decision space that puts them inside a commander's problem. A system that makes them feel the weight of choices that real people made under real constraints. That is worth doing, even when the person doing it is imperfect.

There is something about wargame design specifically that rewards this kind of intensity. The historical research pulls you into a subject until you understand it at a level most people never reach. The mechanical design forces you to translate that understanding into a system that another person can interact with. The playtesting shows you where your understanding was wrong. The whole process is a cycle of learning, building, and correcting that mirrors the scientific method more closely than most creative disciplines. Pressure, for all its costs, tends to force honest outcomes from that cycle. A wargame built under pressure will show you truths about its subject that a more leisurely design process might have papered over. The same dynamic explains why wargames built to simulate historical conflicts can reveal patterns that modern organizations recognize hundreds of years later. The simulation does not care about your comfort. It cares about whether the model is honest.

To You

You have read twenty-three chapters about how to do this work. You have more information now than I had when I started. You have better tools, better reference material, and a community of designers more willing to share what they know than at any point in the hobby's history.

Pick a conflict that interests you enough to spend months inside it. Read about it until you understand what happened and why. Ask yourself what question your game will answer, what argument it will make, what experience it will create for the player. Then start building. Your map will be rough. Your first CRT will produce strange results. Your rules will have gaps you cannot see. Your playtesters will find problems you missed. All of this is normal. All of this is the process.

Scope small. A single map and a hundred counters will teach you the same design skills as a monster game, and you can finish it, test it, and learn from it in a fraction of the time. Finish the game. Then design another one.

The hobby is small, but it is persistent. People have been designing and playing these games for over half a century, and the community has survived format changes, market contractions, the rise of digital gaming, and predictions of its own death that have been wrong for decades. It survived because the people in it care about the work. They care about history, about systems, about the challenge of modeling something complex and making it playable. They will care about your game too, if you give them something worth their time.

Go design something.

Appendix A: Recommended Reading

A list of the books, articles, and resources that shaped how I think about wargame design, military history, and the relationship between games and the conflicts they model. Some are easy to find. Some are out of print. One is a holy grail.

For research methodology and sources specific to wargame design, see also Chapter 3.

On Wargame Design

The Complete Wargames Handbook — James F. Dunnigan. The closest thing the hobby has to a foundational textbook. Revised and expanded as *Wargames Handbook* in later editions. Covers the design process from concept through production, written by the person who industrialized wargame design at SPI. Some of the specific advice is dated. The design philosophy is not.

SPI Staff Study #2: Wargame Design — SPI staff. If you can find a copy, this is the single most valuable document on wargame design methodology ever produced. A detailed internal study of how SPI designed games, covering research methods, system design, playtesting procedures, and production. Out of print, rare, and expensive when it surfaces. Worth the hunt.

Zones of Control: Perspectives on Wargaming — Pat Harrigan and Matthew Kirschenbaum, eds. MIT Press. A collection of essays from designers, historians, and academics on wargaming as a medium. Broad in scope, covering hobby games, professional wargaming, and digital simulations. Individual essays vary in quality but the collection as a whole is the best single-volume overview of the field.

Cardboard Ghosts: Using Physical Games to Model and Critique Systems — Amabel Holland. CRC Press. Holland's argument for games as an expressive medium, grounded in her own design work at Hollandspiele. Read this if you are interested in what games can say beyond simulating combat.

Mark Herman on Wargame Design — Mark Herman. Herman invented the card-driven game with *We the People* and has been producing influential designs for decades. This book covers his design philosophy and methodology.

OSG Design Dispatches and Newsletters — Operational Studies Group (Kevin Zucker). Design dispatches and newsletters covering operational game design philosophy, the Napoleonic series, and Zucker's approach to simulation. If you are designing at the operational scale, read these. Available through OSG's website and archives.

Ted Raicer's Design Writing — Raicer has written on wargame design across multiple publications. His perspective as the designer of *Paths of Glory* and the Dark series provides insight into both card-driven and traditional hex-and-counter design at the operational level.

Al Nofi's Historical and Design Writing — Nofi's contributions to establishing research standards in wargame design are discussed in Appendix E. His published works on military history reflect the same rigor he brought to game research at SPI.

On Military History

The books below are not design books. They are the kind of sources you will use when researching a game. I include them because they represent the standard of historical scholarship that good wargame design demands.

Imperial Bayonets: Tactics of the Napoleonic Battery, Battalion and Brigade as Found in Contemporary Regulations — George Nafziger. The reference for Napoleonic wargame designers. Nafziger compiled and translated tactical regulations from multiple armies, providing the primary-source data that informs unit ratings, movement values, and combat mechanics for the period.

Numbers, Predictions and War — Trevor N. Dupuy. Dupuy developed quantitative methods for analyzing combat effectiveness across historical periods. His Tactical Numerical Deterministic Model (TNDM) attempted to reduce combat outcomes to measurable variables. Whether you agree with his methodology or not, the data tables on historical advance rates, casualty ratios, and force multipliers are reference material for any designer calibrating movement values or CRT results. Appendix D of this book draws on his work.

Supplying War: Logistics from Wallenstein to Patton — Martin van Creveld. Cambridge University Press. The single best book on military logistics as a historical force. Van Creveld argues that logistics, not strategy or tactics, determined the outcome of most campaigns. If you are designing a game where supply matters (and Chapter 11 argues it should), read this first.

Alexander the Great and the Logistics of the Macedonian Army — Donald W. Engels. University of California Press. Engels reconstructed the logistical requirements of Alexander's army using ancient sources and modern calculations of food, water, and fodder consumption. His methodology for converting historical data into usable numbers is directly applicable to wargame design. Appendix D references his work on ancient march rates.

The Franco-Prussian War: The German Invasion of France 1870-1871 — Michael Howard. Routledge. The standard English-language account of the war. Howard's treatment of Prussian staff planning, French command dysfunction, and the operational dynamics of the campaign is essential reading for anyone designing a game on this period. His analysis of why the French lost goes far beyond "the Prussians had better guns."

The Campaigns of Napoleon — David G. Chandler. Macmillan. The comprehensive operational history of Napoleon's campaigns. Over a thousand pages covering every major campaign from Toulon to Waterloo. Chandler's maps and his analysis of Napoleon's operational methods are reference material for Napoleonic game designers. Kevin Zucker's *Library of Napoleonic Battles* draws on Chandler extensively.

David Glantz's Eastern Front Library — Glantz is the most important English-language historian of the Soviet-German war. *When Titans Clashed* (with Jonathan House) is the single-volume overview. *Colossus Reborn* covers Soviet force reconstitution. *Stumbling Colossus* examines the Red Army in 1941. *To the Gates of Stalingrad*, *Armageddon in Stalingrad*, and *Endgame at Stalingrad* form the definitive trilogy on that battle. If you are designing an Eastern Front game, Glantz is where you start and where you return when something does not add up.

On Games, Media, and Simulation

Society of the Spectacle — Guy Debord. Zone Books. Debord's critique of mediated experience has direct relevance to simulation design. His argument that representations of reality replace direct experience maps onto the relationship between a wargame and the conflict it models. Debord was also a wargame designer himself. His *Game of War* (Kriegspiel) is a two-player abstract strategy game modeling Napoleonic-era warfare, and his interest in the strategic and spatial dimensions of conflict informed both his theoretical work and his game design.

Finding Sources

Many of the best references for wargame design are out of print, published in limited runs, or scattered across magazine articles, convention presentations, and designer blogs. BGG designer diaries, InsideGMT articles, and C3i magazine contain design insights that no one has collected into book form. The wargaming community's design knowledge lives as much in forums and newsletters as in published books. Search for it. Chapter 3 discusses methodology for finding and evaluating historical sources. The same discipline applies to design sources.

Appendix B: Tools and Resources

Chapter 15 covers the core design tools (GIMP, VASSAL, Tabletop Simulator, word processors, pen and paper). This appendix expands the list to include production tools, research databases, communities, and resources that become relevant as your game moves from prototype toward publication.

Graphic Design and Production

GIMP — Free, open-source image editor. Covered in Chapter 15. Handles maps, counter sheets, charts, and player aids. Plugin support for hex grids. The learning curve pays for itself across the life of a project.

Inkscape — Free, open-source vector graphics editor. Better than GIMP for counter design because vector graphics scale without losing quality. Design a counter at any size and scale it to print resolution without pixelation. Also useful for clean chart and player aid layouts.

Scribus — Free, open-source desktop publishing. Handles multi-page rulebook layout with proper typesetting, columns, headers, and page numbering. The open-source alternative to Adobe In-Design. Use it when your rulebook outgrows a word processor.

LibreOffice Writer / Google Docs — Sufficient for rulebook drafting and early layout. Most of my rulebooks were written in a word processor before being formatted for print. Use styles and heading levels from the start so converting to a layout tool later is painless.

Counter and Component Design

NATO Joint Military Symbology (MIL-STD-2525) — The standard for military map symbols used across the hobby. Free reference documents available from the U.S. Department of Defense. If your game uses NATO-style unit symbols on counters, this is the source.

Counter templates — Several designers have shared counter templates for GIMP and Inkscape on BGG and ConsimWorld. Search for them before building your own from scratch. A good template with proper bleed, cut lines, and registration marks saves hours.

Print-on-demand counter sheets — Services like Wargame Print and The Game Crafter print die-cut counter sheets in small quantities. Quality varies. Order samples before committing to a full print run.

Map Resources

David Rumsey Map Collection (davidrumsey.com) — Over 100,000 digitized historical maps. Free to browse and download. A starting point for tracing historical maps into game maps.

Perry-Castañeda Library Map Collection (lib.utexas.edu/maps) — University of Texas collection of scanned maps, many from U.S. government sources and therefore in the public domain.

Old Maps Online (oldmapsonline.org) — Aggregates historical map collections from institutions worldwide. Search by location and date range.

Order of Battle Research

Nafziger Collection — George Nafziger compiled thousands of orders of battle (OOBs) across multiple eras. The collection was donated to the U.S. Army Combined Arms Research Library and is available online. The single largest OOB reference source in English.

Orbat.com — OOB data for twentieth-century conflicts, organized by campaign and date. Useful as a starting point, but cross-reference against primary sources.

National Archives (archives.gov) — U.S. military records, after-action reports, and unit histories. Available digitally for many WWII and later conflicts. The closest thing to primary OOB documentation you can access without visiting a physical archive.

Bundesarchiv (bundesarchiv.de) — German federal archives. Wehrmacht records, unit diaries, and organizational documents. Much of it is in German. If you are designing a game involving German forces, this is the primary source repository.

AI-assisted research tools — Tools like Bookfinder General (discussed in Chapter 22) can accelerate OOB research by searching for, downloading, and extracting text from staff histories and unit records, then letting you query the content for specific formations, strengths, and dates. I have found this approach useful for cross-referencing OOB data across multiple sources that would take days to read cover to cover.

Research and Note Organization

Obsidian — A free, offline-first note-taking tool that stores everything as plain Markdown files on your computer. Useful for organizing research across a design project: link notes to each other, tag sources by topic, build a web of connections between historical events, unit data, and design decisions. The learning curve is steeper than a simple notepad but the payoff is a research database you can search and cross-reference as your project grows.

Zotero — Free, open-source reference manager. Collects, organizes, and cites sources. If your game research involves academic papers, books, and archival documents, Zotero keeps them organized and generates citations. Overkill for a small project. Worth learning for a large one.

Prototyping and Playtesting

VASSAL — Covered in Chapters 15, 16, and 21. The standard for digital wargame prototyping and remote play. Free. Learn to build a VASSAL module.

Tabletop Simulator — Covered in Chapters 15, 16, and 21. Commercial product, requires purchase. 3D sandbox environment for remote playtesting.

Rally the Troops (rally-the-troops.com) — A browser-based platform for playing board wargames online. Cleaner interface than VASSAL for supported games. Demonstrates what a modern wargame prototyping platform could look like.

Publishing and Distribution

Amazon KDP — Print-on-demand for rulebooks and game books. No upfront cost. You upload a PDF, set a price, Amazon prints and ships on demand. Quality is acceptable for rulebooks and supplements. Not suitable for maps or counter sheets.

Wargame Vault (wargamevault.com) — Digital distribution for print-and-play wargames. Upload your files, set a price, the platform handles payment and delivery. The primary commercial platform for digital wargame sales.

The Game Crafter (thegamecrafter.com) — Print-on-demand for complete games: maps, counter sheets, cards, boxes, rulebooks. Higher per-unit cost than a traditional printer but no minimum order. Useful for prototypes and very small print runs.

Noble Knight Games (nobleknight.com) — Retailer specializing in wargames and hobby games. Accepts consignment from small publishers. A distribution channel for self-published physical games.

Communities

BoardGameGeek (boardgamegeek.com) — The central hub for the wargaming community. Game pages, forums, designer diaries, file uploads, ratings, and reviews. Your game's BGG page is its storefront. The wargaming subforum is active.

ConsimWorld (consimworld.com) — A forum community focused on conflict simulation. More wargame-specific than BGG. Active discussions on design, development, and playtesting. Many professional designers participate.

Discord servers — Multiple wargaming Discord communities exist for specific publishers, game series, and general wargame discussion. Search for them. They tend to be more informal and faster-moving than forums.

Conventions — ConsimWorld Expo, World Boardgaming Championships (WBC), Origins, and regional conventions. In-person playtesting and demos reach players that online channels do not. Chapter 20 discusses convention presence as a marketing tool.

Appendix C: Sample Design Exercise

You are going to design a wargame. A small one, on a single sheet of paper, with about fourteen counters. The historical research is provided. The design decisions are yours.

You will work through the full process: map, counters, rules, CRT, sequence of play, playtesting. The project is small enough to finish in a weekend. The skills transfer to anything you design after it.

The Situation

Tunisia, November 15 – December 10, 1942

Three days after the Operation Torch landings in Algeria and Morocco, the Allies pushed east toward Tunis. Get there before the Germans. They failed by a narrow margin.

Lieutenant-General Kenneth Anderson's British First Army advanced along two axes: a northern coastal route toward Bizerte and an inland route through the Medjerda River valley toward Tunis. His forces were light. Two infantry brigades, an improvised armored group, some paratroopers and commandos, and a

French garrison at Medjez el Bab with outdated equipment. Reinforcements trickled in over the following weeks, but Anderson never had enough of anything. His supply line stretched 560 miles back to Algiers along a single road and a single-track railway. The forward port at Bone could not handle the volume.

The Germans moved faster than anyone expected. Sicily to Tunis was 100 miles by sea. Within days of the Torch landings, Luftwaffe transports flew infantry battalions into Tunis and Bizerte. General Walther Nehring arrived on November 17 to take command and organized a defense from whatever he could find: replacement drafts formed into ad hoc battalions, paratroopers, Italian infantry, Luftwaffe ground personnel. The improvised force was thin, but it held the airfields at Tunis and Bizerte. That mattered more than anything else in the campaign. German fighters and Stukas operated five minutes from the front lines. The RAF flew from bases sixty miles to the rear.

By late November, Allied spearheads reached Djedeida, thirty kilometers from Tunis. That was as close as they got. The 10th Panzer Division arrived with Tiger tanks (among the first used in combat in North Africa) and counterattacked, driving the Allies back to Tebourba, then to Medjez el Bab. Winter rains turned roads to mud and grounded aircraft. A final Allied push on December 22–25 to take Longstop Hill, the high ground controlling the valley approach to Tunis, failed after three days. On Christmas Day, Eisenhower called off the advance.

The central question for your game: Can the Allied player reach Tunis before Axis reinforcements and logistics problems make it impossible?

The Research

In a real project, you would spend weeks assembling this from primary and secondary sources (Chapter 3). For this exercise, the research is provided.

Orders of Battle

Allied Forces — British First Army (Lt-Gen Anderson)

Formation	Type	Composition	Available
36th Infantry Brigade (78th Division)	Infantry	Three infantry battalions, artillery support	Start
11th Infantry Brigade (78th Division)	Infantry	Three infantry battalions, artillery support	Start
Blade Force (6th Armoured Division)	Armour	17th/21st Lancers (Crusader/Valentine tanks), armoured cars, motorized infantry, artillery	Start
French garrison (General Barré)	Infantry	Colonial troops, brigade strength, WWI-era weapons	Start, at Medjez el Bab
1st Guards Brigade (78th Division)	Infantry	Three Guards battalions	Turn 3–4
US Combat Command B (1st Armored Division)	Armour	Stuart light tanks, armored infantry	Turn 4–5

Axis Forces — XC Corps, later 5th Panzer Army

Formation	Type	Composition	Available
Division von Broich	Mixed	Ad hoc: Tunis Field Battalions, Italian Bersaglieri, improvised artillery, Luftwaffe ground troops	Start, at Tunis/Bizerte
Fallschirmjäger Group (Koch/Witzig)	Elite infantry	Two paratrooper battalions, lightly equipped but aggressive	Start
1st "Superga" Division (Italian)	Infantry	Standard Italian infantry division, southern sector	Turn 2–3
10th Panzer Division	Armour	Panzer regiment with Tiger elements, panzergrenadiers, divisional artillery	Turn 4–5

Axis Air Superiority: The Luftwaffe controlled the air throughout this campaign. German fighters and dive-bombers operated from airfields adjacent to Tunis with five-minute response times. The RAF flew from fields sixty miles behind the front. Your design needs to account for this. How you do it is up to you.

Geography

The operational area covers roughly 200 kilometers from Bone in the west to Tunis and Bizerte in the east. Two main routes:

1. **Northern route:** Tabarka — Sedjenane — Mateur — Bizerte. Rugged, wooded hill country. Roads poor.
2. **Medjerda valley route:** Souk el Arba — Béja — Medjez el Bab — Tebourba — Djedeida — Tunis. The main corridor, following the Medjerda River. The valley narrows toward Tunis, with

high ground (djebels) on both sides forming natural defensive positions.

Key locations:

- **Medjez el Bab:** Critical river crossing, roughly 60 km west of Tunis. Whoever holds this controls the main approach.
- **Tebourba:** 30 km west of Tunis. Scene of the heaviest fighting.
- **Djedeida:** Airfield complex, 30 km from Tunis. Farthest Allied penetration.
- **Longstop Hill (Djebel el Ahmera):** 900-foot hill dominating the valley east of Medjez. The battle that ended the race.
- **Mateur:** Road junction on the northern route.
- **Tunis:** The objective.

Terrain types you will need:

- Open/valley (the Medjerda corridor and coastal plain)
- Mountains/djebels (the Eastern Dorsal ridgeline, hills flanking the valley)
- Roads (few, poor quality, critical for supply and movement)
- Key towns (affect combat, supply, and victory)

Distances and Timing

- Bone to Tunis: ~200 km by road
- Medjez el Bab to Tunis: ~60 km
- Historical timeframe: November 15 – December 10 (25 days)
- At 6–8 turns, each turn represents roughly 3–4 days

The Weather Problem

Winter rains began in late November and worsened through December. Roads dissolved into mud. Vehicles bogged. Airfields flooded. Any movement in rain was slower and costlier. The rains favored the defenders.

Your Design Tasks

Work through these in order. Each corresponds to skills covered in the main chapters. Do not skip ahead. Decisions in early steps constrain later ones, and working within those constraints is part of what you are learning.

Task 1: Define Your Scales (Chapter 5)

Decide on your hex scale, turn scale, and unit scale. The research above gives you the raw numbers. You need to make them work together.

Questions to answer:

- How many hexes from Medjez el Bab to Tunis? (This determines hex scale.)
- How many turns in the game? (6–8, representing roughly 3–4 days each.)
- How far should a unit move in one turn? Infantry and armor move at different rates. Consult Appendix D for march rates and decide what fits your hex scale.

Constraint: The entire map must fit on a single A4 sheet of paper (210 × 297 mm). With hexes at roughly 15–20 mm across, that gives you a map of about 10–14 hexes wide and 15–20 hexes tall. Your design must work within that space.

A sketch on graph paper is fine. You are making a functional tool for testing your design, not art.

Task 2: Build Your Map (Chapter 5)

Draw a playtest map. It does not need to be pretty. It needs to be accurate enough to test your design.

Requirements:

- Two approach routes (northern and Medjerda valley) converging on Tunis
- Mountains channeling movement into corridors
- Key towns marked (Medjez el Bab, Tebourba, Djedeida, Mateur, Tunis, Bizerte)
- Terrain types clearly distinguishable (use colored pencils, hatching, or whatever works)
- A hex grid

Refer to: A real map of northern Tunisia. Free topographic maps are available online. Your map should reflect the actual terrain (mountain ridges, river valley, coastal plain) simplified to fit your hex scale.

Task 3: Design Your Counters (Chapter 6)

Create a counter manifest: the complete list of every counter in the game and what goes on each one.

Decisions to make:

- What information goes on each counter? (Chapter 6 covers this.) At minimum: unit name, type (infantry/armor), combat strength, movement allowance.
- How do you rate combat strength? Raw headcount? Combat effectiveness? A single number or attack/defense split?
- Do all infantry brigades rate the same, or do the Guards and the Fallschirmjäger get higher values? The French garrison with their outdated equipment? The Italian Superga division?

- How do you represent Blade Force and the 10th Panzer Division? Armor should feel different from infantry. How?

The order of battle (OOB) above gives you roughly five counters per side at game start, with two reinforcements per side arriving during play. That is fourteen total, a good number for this exercise. You may add markers for turn track, weather, or other game state as needed.

You do not need to print professional counters. Write the values on small squares of cardboard or paper. What matters is that every counter has clear, consistent information.

Task 4: Write Your Sequence of Play (Chapter 10)

Decide what happens each turn and in what order. A simple sequence for a game this size might be:

1. Weather Determination (if applicable)
2. Allied Player Turn
    a. Reinforcement Phase
    b. Movement Phase
    c. Combat Phase
3. Axis Player Turn
    a. Reinforcement Phase
    b. Movement Phase
    c. Combat Phase

But you need to decide: does the Allied player always go first? Does initiative shift? Is there a supply check? Where does air power fit?

Keep it simple. Every phase you add is a phase you have to write rules for, test, and balance. For a game on a single sheet of paper with fourteen counters, three to five phases per turn is probably right.

Task 5: Build Your Combat Results Table (Chapter 9)

The CRT determines whether your game works.

Decisions to make:

- Odds-based CRT or differential CRT? (Chapter 9 explains both.)
- What results are possible? Retreat, step loss, exchange, attacker back?
- How many columns? For a game this small, four to six.
- What modifiers shift the column or modify the die roll? Terrain? Unit type? Air superiority?

Test it on paper before you test it on the map. Set up a few hypothetical combats using your unit values and roll through the CRT. Does a 78th Division brigade attacking Division von Broich in mountains at Medjez produce a result that feels right? Does 10th Panzer counterattacking in open terrain feel devastating? If not, adjust the table.

Task 6: Write Your Rules (Chapter 17)

Write a complete rulebook for your game. For a game this size, aim for three to six pages. Two is ideal but tight once you add examples and a terrain effects chart. If you find yourself writing past six, that is a sign your design may be more complex than this exercise calls for. Use the case-numbering format described in Chapter 17.

Required sections:

1.0 Introduction (one paragraph: what the game is about) 2.0 Components (list of what is in the game) 3.0 Sequence of Play 4.0 Movement (including terrain effects) 5.0 Combat (including the CRT) 6.0 Reinforcements (when and where) 7.0 Special Rules (weather, air power, supply, whatever your design requires) 8.0 Victory Conditions

Victory conditions deserve thought. Chapter 12 covers this in depth, including scaled victory structures for conflicts where one side historically lost. The Allies lost at Tunis. If the Allied player almost always loses, your game is broken. If the Allied player almost always wins, your game is dishonest. You need conditions where both players feel they have a chance, and their decisions determine the outcome. One approach: scaled victory (decisive, marginal, draw) based on how close the Allies get and how many turns it takes.

Task 7: Your Signature Mechanic

A wargame worth playing has at least one mechanic that belongs to its specific conflict and no other. A generic system would miss it.

This campaign has several distinctive features worth modeling:

- **Axis air superiority** that degraded every Allied operation
- **Allied supply strain** that worsened the further east they pushed
- **The reinforcement race**, both sides feeding units in through different bottlenecks
- **Weather** that shut down operations without warning
- **The ad hoc nature of both armies**, neither side had a proper OOB and both were improvising

Pick one, or find your own, and design a mechanic that makes the player feel it. A modifier on a table does not count. A rule that says "subtract 1 in rain" does not count. You want something that changes how the game plays, something that forces the player to plan around it the way Anderson and Nehring planned around it.

The mechanic you choose says something about what you think mattered most in this campaign. That is your thesis statement as a designer. Be ready to defend it.

Task 8: Playtest (Chapter 16)

Play your game. Solo is fine. Play both sides. Play it at least three times.

What to watch for:

- Does the game end too quickly or drag on?
- Is one side consistently winning? By how much?
- Are there meaningful decisions, or does the game play itself?
- Does your CRT produce reasonable results, or are there columns that never get used?
- Does your signature mechanic work? Does it create interesting decisions, or is it just bookkeeping?

Take notes. Write down what happened, what felt wrong, what you would change. Then change it and play again. This is the development cycle (Chapter 19) in miniature.

What You Should Have When You Are Done

- A playtest map on A4 paper
- 12–14 counters with unit values
- A CRT that produces historically plausible results
- A one-to-two-page rulebook in case format
- A sequence of play
- One mechanic you designed yourself to capture something specific about this campaign
- Notes from at least three playtests documenting what worked and what did not

That is a wargame. Not a published game, and not meant to be. But you have now done everything a wargame designer does: research, mapping, counter design, system design, rules writing, playtesting. Small enough to finish. Large enough to learn from.

Your next design will be better because you did this one first.

Suggested Reading

For additional context on the Race to Tunis: - Rick Atkinson, *An Army at Dawn* (Henry Holt, 2002), Chapters 4–6 cover the advance - Vincent P. O'Hara, *Torch: North Africa and the Allied Path to Victory* (Naval Institute Press, 2015), the naval dimension of the reinforcement race - US Army Center of Military History, *Northwest Africa: Seizing the Initiative in the West*, free online

For examples of small wargames that do a lot with few components: - Mark Simonitch, *Salerno* '43 (GMT, 2023), operational game with clean rules and strong design - Dean Essig, SCS *series* (MMP), proof that simple games reward skill - Ted Raicer, *The Dark Valley* (GMT, 2013), a larger game, but study the CRT and combat system

Appendix D: Historical Movement Rates

This appendix collects march rate data from across three thousand years of military history. It exists because I got tired of hunting through a dozen books every time I needed to convert real-world movement into game movement points. If you are designing a wargame at any scale from ancient to modern, this table and the commentary around it should save you time.

The numbers come from primary sources, military field manuals, and the work of historians who cared enough to do the math. Where figures conflict between sources, I have noted the range. Where a number is an estimate or reconstruction rather than a recorded measurement, I have said so.

The Central Fact

Infantry march rates held steady from antiquity to 1945. A Roman legionary and a World War II rifleman moved at the same speed on foot: fifteen to twenty miles per day on roads under favorable conditions. The Macedonian army's average rate for the full force was about fifteen miles per day. Two thousand years later, German infantry divisions in Russia marched at twelve to eighteen.

The human body set the speed limit for three millennia. Everything else shifted around it. Cavalry opened the first speed gap. Railroads rewired logistics. Mechanization split land forces into foot-speed units and engine-speed units, and that split drives the design of every World War II operational game you will build.

March Rates by Era

Ancient Armies

Alexander's Macedonian Army (from Engels, *Alexander the Great and the Logistics of the Macedonian Army*, 1978)

Engels compiled the most rigorous data set available for any ancient army. His Table 7 records march rates derived from cross-referencing Arrian, Curtius, and Diodorus against measured distances on the actual terrain.

Route	Rate (mi/day)	Force Composition
Therma to Sestos	16.2	Whole army
Sardis to Ephesus	15.0	Whole army
Gaza to Pelusium (Sinai crossing)	19.5	Whole army
Babylon to Susa	12.3	Whole army
Paraetonium to Ammon Oracle	22.5	Small, light force
Ecbatana to Rhagae	22.0	Cavalry, mounted scouts, part of phalanx
Rhagae to Caspian Gates	34.0	Same mounted/light force
Pursuit in Assyria	~46.0	Cavalry only
Sogdiana (Alexandria Eschate to Maracanda)	43–57.5	Half the Companions, archers, lightest phalanx

Route	Rate (mi/day)	Force Composition
Mallians' Territory	~30.6	Bodyguard, archers, Agrianians, mounted archers, half cavalry
Oreitans to Pura (Gedrosian Desert)	10.5–13.0	Whole army minus fleet and Craterus

The full Macedonian army, one of the best-organized forces in ancient history, averaged about fifteen miles per day over sustained marches. Its maximum recorded rate for the entire force was 19.5 miles per day, achieved crossing the Sinai where the fleet handled supply. Small mounted forces reached thirty to fifty miles per day, but those were cavalry and light infantry operating without baggage.

Engels highlights something designers at all scales should internalize: large armies move slower than small ones. Column length creates compounding delays. An army of 65,000 personnel marching ten abreast with 6,000 cavalry five abreast forms a column over sixteen miles long. When the front of the column reaches camp, the rear is still five hours of marching behind. Halts ripple backward through the column, and restarting movement takes time proportional to column length. A one-minute delay at an obstacle costs the rear of the column far more than a minute.

Roman Legions (from Vegetius, *De Re Militari*; Roth, *The Logistics of the Roman Army at War*, 1999)

March Type	Rate (mi/day)	Notes
Standard march (*iter justum*)	~14	15 Roman miles; includes building a fortified camp each night

March Type	Rate (mi/day)	Notes
Full march (*iter magnum*)	~18	20 Roman miles
Forced march (*iter celerissimum*)	22+	24+ Roman miles; unsustainable beyond a few days

Vegetius gives the common military step as twenty Roman miles and the full step as twenty-four; those are training-pace standards, while the rates above are the realistic sustained daily figures a legion actually achieved on campaign (Roth, *The Logistics of the Roman Army at War*).

Caesar's legions in Gaul covered fifteen to twenty miles per day while constructing a fortified camp each evening. Caesar traveling light with a cavalry escort and relay horses could cover eighty to a hundred miles in a day, but that is courier speed, not army speed.

Mongol Cavalry (from May, *The Mongol Art of War*, 2007)

Movement Type	Rate (mi/day)	Notes
Normal advance	30–40	Each warrior maintained 3–5 remounts
Forced march / raid	60–80	Documented during Subotai's campaigns
Strategic sprint	~60	1241 invasion of Hungary: ~180 miles in 3 days through Carpathian passes, in winter

No pre-mechanized force moved faster. The Mongols achieved these rates because their warriors rode, kept three to five spare horses each, and carried almost no baggage. They ate from their herds and from the land. No other military system combined universal horsemanship with such minimal logistics.

Early Modern: The Thirty Years' War through Marlborough

Van Creveld (*Supplying War*, 1977) documents the growth of European armies and the logistics that constrained their movement from the sixteenth through early eighteenth centuries.

By the late seventeenth century, armies had grown so large that moving them became a logistics problem before it was a tactical one. Maurice of Nassau's 1602 campaign in Brabant required 3,000 wagons for 24,000 men, one wagon for every eight soldiers. A typical army of 30,000 trailed a crowd of women, children, servants, and sutlers equal to fifty to one hundred and fifty percent of its combat strength. All of it moved at ox-cart speed.

Force	Rate (mi/day)	Notes
Spanish Army of Flanders, march to Netherlands, 1567	10–12	Along the "Spanish Road" through Alps and Rhineland
Gustavus Adolphus, campaign marches 1630–32	8–12	Limited by siege operations and supply from magazines
Marlborough's march to the Danube, 1704	10–12	250 miles in ~25 marching days, one of the great operational marches of the era
Typical Thirty Years' War army	5–10	Burdened by enormous baggage trains and camp followers

Louvois standardized the magazine system for Louis XIV's armies after 1660, tethering armies to fixed depots by road. March planning worked backward from supply availability. Armies moved from magazine to magazine. The practical operating radius was five days' march, about seventy-five miles, from the nearest depot. Campaigns in this period looked like elaborate chess games between fortified supply points because they were.

Eighteenth Century

Frederick the Great's Prussians (from Duffy, *The Army of Frederick the Great*, 1974; *The Military Experience in the Age of Reason*, 1987)

Movement Type	Rate (mi/day)
Standard march	12–15
Forced march	up to 20

Frederick's campaigns were fought in compact theaters (Silesia, Saxony, Bohemia) where strategic speed mattered less than battlefield maneuver. His army's march rates were typical for the period. Duffy describes the eighteenth-century soldier's daily routine as dominated by the march itself: armies rose before dawn, were on the road by four or five in the morning, and halted by early afternoon after covering twelve to fifteen miles. The rest of the day went to foraging, camp construction, and the administrative overhead of keeping tens of thousands of men fed.

The supply system, not the soldier's legs, set the speed limit. The magazine-based logistics that Van Creveld documents forced eighteenth-century generals to plan campaigns around depots rather than objectives. Frederick's advantage was that he operated in small theaters where he could fight decisive battles within supply range of his magazines.

Napoleonic Era

Marching Cadences by Nation (from Nafziger, *Imperial Bayonets*, 1996)

Nafziger compiled marching cadences from the drill regulations of every major Napoleonic army. These prescribed speeds for different march types show how fast individual soldiers moved on the ground.

Nation	Pace Type	Paces/min	Feet/min
Austrian	*Ordinairschritt* (ordinary)	90–95	188–198
Austrian	*Geschwindschritt* (quick)	105	219
Austrian	*Doublirschritt* (double)	120	250
French	*Pas ordinaire*	76	165
French	*Pas de route* (route march)	85–90	184–195
French	*Pas accéléré*	100	217
French	*Pas de charge*	120	260
Prussian	*Ordinaire Schritt*	75	156
Prussian	*Geschwindschritt*	108	225
British	Ordinary pace	75	188
British	Quick pace	108	270
Russian	*Tchyi szag* (slow)	60–70	150–175
Russian	*Skoryi szag* (quick)	100–110	250–275

At route march speed, most armies covered about 180 to 200 feet per minute, or roughly two miles per hour. In theory, a seven-hour march day produces fourteen miles. In practice, column friction, rest halts, bottlenecks at bridges and defiles, and the time needed to deploy from march column into camp eat heavily into that number. The gap between individual marching speed and army movement rate is where most game designers go wrong.

Column Length and Its Effects (from Nafziger and Engels)

Nafziger recorded the frontage of a single battalion in line for each major army. These numbers determine column length on the march and how long it takes an army to clear a road.

Nation	Year	Battalion Length in Line (ft)
Austrian	1807	684–762
British	1792	640
French	1791	560
French	1808	480
Prussian	1792	360
Prussian	1808	344
Russian	1802	368

A French corps of 20,000 men marching in column of route on a single road could stretch ten to fifteen miles from head to tail. When the vanguard reaches the day's destination, the rear guard is still hours away. Engels identified the same problem for Alexander's army, and it imposes the same speed ceiling: about fifteen miles per day for a full army on a single road. Napoleon's system of corps marching on separate roads within mutual supporting distance was his answer to the column-length problem.

Napoleon's Grande Armee (from Chandler, *The Campaigns of Napoleon*, 1966; Dupuy, *Numbers, Predictions and War*, 1979)

Movement Type	Rate (mi/day)	Notes
Sustained corps march	15	Standard tempo
Forced march	20–25	Sustainable several days
Extreme forced march	30+	See Davout below
Cavalry, strategic	25–30	Up to 50 in pursuit
Artillery train	10–12	The pacing factor

Davout's III Corps marched roughly seventy miles in forty-eight hours to reach Austerlitz on December 1, 1805, the canonical example of an extreme forced march by an entire corps. The artillery train was the slowest element of the column and the factor that paced the rest.

Dupuy's campaign advance rate data, measured as the average daily advance across the full campaign including combat and rest days, shows how much friction eats into raw march speed:

Campaign	Distance (km)	Days	Advance Rate (km/day)
Marengo, 1800	350	31	11
Ulm, 1805	475	22	22
Jena, 1806	140	6	23
Friedland, 1807	210	9	23
Aspern-Wagram, 1809	405	30	14
Russia, 1812	680	57	12
Lutzen-Bautzen, 1813	300	20	15

An army that marches fifteen miles per day advances at a campaign rate closer to eight to fourteen miles per day once you account for rest days, combat, supply halts, and the friction of moving tens of thousands of men across real terrain. Your game's movement allowances should produce results closer to the campaign column than the march column.

Van Creveld (*Supplying War*, 1977) established that the practical radius of supply from a fixed magazine for pre-railroad armies was about seventy-five to a hundred miles, five to seven days of march. Beyond that distance, an army had to either live off the land or establish a new depot. Napoleon's system of foraging from the countryside allowed faster movement in fertile, densely populated regions. In Spain and Russia, where the countryside was sparse, march rates dropped to eight to twelve miles per day because of foraging requirements.

American Civil War

Force Type	Rate (mi/day)	Notes
Standard infantry	10–15	On roads, good conditions
Forced march infantry	20–25	Unsustainable beyond 2–3 days without heavy straggling
Cavalry, mounted	25–35	Stuart's ride around McClellan: ~25/day average
Supply wagons, good roads	10–15	
Supply wagons, bad roads or mud	5–8	

Stonewall Jackson's Valley Campaign, May-June 1862: Jackson's infantry covered roughly 650 miles in 48 days, averaging about 13.5 miles per day including rest days. Peak single-day marches reached 35 to 40 miles during the campaign. Before the Seven Days Battles, his corps marched about 100 miles in five days (from Tanner, *Stonewall in the Valley*, 1996).

Sherman's March to the Sea, November-December 1864: Atlanta to Savannah, roughly 285 miles in about 35 marching days, averaging eight to ten miles per day. The army moved on a front 25 to 60 miles wide, foraging extensively, which limited daily distance.

Franco-Prussian War, 1870–71

The Franco-Prussian War marks the transition from foot-mobile to rail-mobile warfare. Howard (*The Franco-Prussian War*, 1961) documents both sides.

Railroad vs. marching: In 1859, rail moved armies into position within a fortnight that would have required sixty days to march. A four-to-one speed advantage. By 1870, Moltke built his entire war plan around railroad concentration. German forces dispersed over a hundred miles between Karlsruhe and Coblenz converged by rail faster than the French could mobilize.

On foot, the numbers were familiar. German corps on the march covered about fifteen miles in eight hours, roughly 1.9 miles per hour. The advance from Sedan to Paris, about 150 miles, took roughly thirteen days: eleven to twelve miles per day. Cavalry patrols ranged up to forty-five miles ahead of the infantry.

The supply failures are worth studying if you design in this period. The French Intendance had six thousand rations for more than forty thousand men at Froeschwiller. Troops went four days without an adequate meal. The Army of Chalons detoured eighteen miles off its line of march just to reach a railway for resupply. Even the Germans, with their superior organization, were saved from starvation only by capturing French stocks.

World War I

Movement Type	Rate (mi/day)	Notes
Approach march, on road	15–20	Same as Napoleonic rates
Tactical advance under fire	1–5	The core WWI problem
Defender movement by rail	50–100+	The reason breakthroughs failed

Kluck's First Army, August 1914 (from Tuchman, *The Guns of August*): 150 miles in 11 days, approximately 13.6 miles per day on average. On the outer rim of the Schlieffen wheel, forced marches reached 25 to 28 miles per day. Troops marched for sixteen consecutive days without rest before the Battle of the Marne. By early September they were physically spent.

Hausen's Third Army (Saxon): averaged 14.3 miles per day, described as the least demanded of the German armies.

The Spring Offensives, 1918: Ludendorff's *Michael* offensive advanced about 40 miles in 16 days, roughly 2.5 miles per day. This was the practical limit of foot-mobile logistics in the face of organized defense.

The Western Front locked up for four years because of a speed asymmetry. Attackers moved at foot speed, five to ten miles per day through broken ground. Defenders moved at rail speed, fifty to a hundred miles per day, and sealed breaches faster than attackers could exploit them. Mechanization broke the stalemate by giving the attacker speed that matched the defender's rail advantage.

Railroad mobilization data from Tuchman: one German army corps required 6,010 railway cars grouped in 140 trains, plus an equal number for supplies. Moltke scheduled 11,000 trains, each crossing specified tracks at ten-minute intervals. The Schlieffen Plan was a railroad timetable dressed in field gray.

World War II

Movement Type	Rate (mi/day)	Notes
Infantry on roads	15–20	
Infantry, cross-country	10–15	
Armor on roads, light opposition	30–50	
Armor vs. organized defense	5–15	
Combat advance, light resistance	12–15	FM 101-10
Combat advance, heavy resistance	3–6	FM 101-10
Motorized infantry, road	100–150	Admin (non-combat)
Armored units, road	100–150	Admin (non-combat)
Armored units, cross-country	50–75	Admin (non-combat)
Supply convoy by truck	150–200	

Blitzkrieg advance rates:

- **France 1940:** Guderian's XIX Panzer Corps, Sedan to the Channel, roughly 150 miles in six days. About 25 miles per day.
- **Barbarossa 1941:** Army Group Center covered roughly 600 miles (Brest-Litovsk to Smolensk) in four weeks, about 21 miles per day. Peak daily advances of 40–50 miles for armored spearheads. By autumn, rates dropped to 5–10 miles per day as logistics broke down.
- **Patton's Third Army**, August 1944: roughly 600 miles across France in four weeks, about 21 miles per day average, with peak advances of 50 miles per day.

Dupuy's campaign advance rates for WWII:

Campaign	Distance (km)	Days	Advance Rate (km/day)
Flanders, 1940	368	12	31
Barbarossa, 1941	700	24	29
Malaya, 1941–42	515	28	18
Caucasus, 1942	775	34	23
Normandy Breakout, 1944	880	32	28
Manchuria, 1945	300	6	50
Sinai, 1967	220	4	55

German infantry in Russia fell behind the panzers at every stage, opening the gaps between armored spearheads and following foot-mobile divisions that defined Barbarossa. If you design a World War II game at operational scale or above, your movement system must produce this gap. Infantry and armor moving at the same rate is not World War II.

The Mobility Problem: German Forces in Normandy, 1944 (from Zetterling, *Normandy 1944*, 2000)

Zetterling's study of German forces in Normandy demolishes the assumption that mechanized armies move at mechanized speed. Many divisions in the west lacked enough trucks, spare parts, and fuel to move at anything close to their theoretical road speed.

Several panzer divisions committed to Normandy were rebuilding after sustained combat on the Eastern Front. Replacements needed training. Key specialists were missing. When headquarters issued movement orders, units could not execute them because they lacked the vehicles. The slow German response to the invasion owed as much to physical inability to move as to confused command or Allied air interdiction.

Allied air power's effect on movement has been overstated. Zetterling found that direct casualties and equipment losses from air attack were limited. The more significant effects were on decision-making, morale, and the delays from forcing road movement into nighttime hours. The single most effective Allied air contribution was the strategic bombing of the French rail network, which mattered because so many German units depended on rail for long-distance movement.

For the designer, this means vehicle shortages, fuel constraints, and rail interdiction can matter as much as enemy opposition in determining actual movement rates. A panzer division at 60% vehicle strength does not move at panzer speed. Your movement system should degrade with logistics, not just with enemy contact.

Terrain Effects on Movement

Dupuy's Quantified Judgment Model (*Numbers, Predictions and War*, 1979) provides terrain mobility multipliers derived from operational research across dozens of twentieth-century battles. The baseline is rolling, bare terrain, set at 1.0.

Terrain Type	Mobility Multiplier
Flat, bare, hard ground	1.05
Flat, mixed vegetation	0.90
Rolling, bare (baseline)	1.00
Rolling, mixed	0.80
Rolling, heavily wooded	0.60
Flat desert	0.95
Rolling dunes	0.30
Rugged, bare	0.60
Rugged, mixed	0.50
Rugged, heavily wooded	0.40
Swamp, jungled	0.30
Swamp, mixed/open	0.40
Urban	0.70

You can translate these into hex-based movement costs by inverting them. If clear terrain costs one movement point per hex, rugged-wooded terrain should cost about 2.5 (the inverse of 0.40). Rolling-mixed costs about 1.25. Flat desert costs about the same as clear terrain, which catches some designers off guard. Desert ground is firm and open. The hard part of desert warfare is supply, not the surface you are driving across.

Weather multipliers from the same source:

Condition	Mobility Effect
Dry, temperate (baseline)	1.00
Dry, extreme heat	0.90
Dry, extreme cold	0.90
Light rain/snow, temperate	0.80
Light rain/snow, cold	0.80
Heavy rain/snow, temperate	0.60
Heavy rain/snow, extreme heat	0.50
Heavy rain/snow, extreme cold	0.50

Heavy rain cuts mobility close to half. Mud season on the Eastern Front, monsoon in Burma, rasputitsa in Russia: these conditions stalled offensives that had plenty of men and ammunition. If your game covers a period with weather variation, your weather rules need teeth.

Supply Constraints on Movement

An army's theoretical march rate and its actual sustained advance differ, and supply accounts for most of the gap.

Early modern armies (from Van Creveld):

- By the early seventeenth century, a typical army of 30,000 had one wagon for every eight to fifteen men, each drawn by two to four horses. Maurice of Nassau's 1602 campaign: 3,000 wagons for 24,000 men.
- The camp-following population (women, children, servants, sutlers) ranged from 50% to 150% of the fighting strength. All of it marched with the army and consumed supplies.
- Soldiers purchased food from their pay at organized markets in garrison. On campaign, the system collapsed into plunder unless magazines were established along the route.
- The magazine system, formalized under Louvois in the 1660s, solved the plunder problem but imposed rigid geographic constraints. Armies could not outrun their depots.

Pre-railroad armies, ancient through Napoleonic (from Engels and Van Creveld):

- A pack animal carrying 250 pounds of supplies consumes about 20 pounds of its own cargo per day in fodder and grain. After roughly twelve days, it has eaten everything it was carrying.

- Practical supply radius from a fixed depot: 75–100 miles (5–7 days of march).
- Alexander's army consumed roughly 195,000 pounds of grain per day for personnel alone, plus 61,000 pounds for cavalry horses, plus feed for baggage animals. In desert conditions requiring carried water, the total daily requirement exceeded 1,260,000 pounds.
- The army could carry at most a two-and-a-half-day supply of provisions in waterless desert.

Napoleonic era:

- Van Creveld's central finding: a four-horse wagon carried about 1,400 pounds but consumed its own payload in fodder within about five days of travel. The supply system was self-consuming.
- Napoleon's army on the march to Moscow, 1812: 550 miles in about 80 days, an advance rate of under seven miles per day for an army that could march fifteen.

World War II (from Dupuy):

- A WWII-era division required roughly 181 tons of supply per day: 84 tons of ammunition, 47 tons of fuel, 25 tons of food, 15 tons of fodder, 10 tons of equipment.
- A WWI-era division required roughly 117.5 tons per day.
- At the Anzio beachhead, combat effectiveness degraded measurably when supply delivery dropped below 65% of requirements.

Your movement rates should reflect what supply allows. An armored division on paper can cover 100 miles per day on roads. In practice, fuel limits it to 30–50 miles per day in an advance, and far less if the logistics chain stretches. Movement allowances measure how far a unit can advance and still fight when it arrives.

Comparative Summary

Era / Force Type	Sustained March (mi/day)	Forced March (mi/day)	Limiting Factor
Alexander's full army	15–16	19.5	Water, forage, column length
Alexander's light/mounted forces	22–34	43–57	Remount availability
Roman legion	14–18	22+	Camp construction time
Mongol cavalry	30–40	60–80	Remount availability
Crusader armies	8–12	15	Heat, water, combat
Thirty Years' War armies	5–10	12–15	Baggage trains, camp followers
Marlborough's army, 1704	10–12	15	Magazine-based supply
Frederick's Prussians	12–15	20	Supply magazine range
Napoleon's infantry	15	25–30	Food, shoes, straggling
Napoleon's cavalry	25–30	50	Forage for horses
Civil War infantry	10–15	20–25	Roads, weather, straggling
Civil War cavalry	25–35	40+	Forage
Franco-Prussian infantry	12–15	18–20	Rail-based supply
WWI infantry	15–20	25	Rail-mobile reserves negate gains
WWII infantry	15–20	25	Unchanged since Alexander

Era / Force Type	Sustained March (mi/day)	Forced March (mi/day)	Limiting Factor
WWII armor, road	30–50	100+	Fuel supply
WWII armor, combat advance	5–15	25–40	Opposition + fuel
Post-1945 mechanized	25–55	100+	Fuel, air interdiction

Using This Data in Design

These numbers are where you start. Translating historical march rates into movement points requires you to account for hex scale, turn length, and the abstraction level your movement points represent.

A simple method: divide the daily march rate by your hex scale to get a base movement allowance. If your operational game uses 10-mile hexes and weekly turns, and a Civil War infantry division marches about 12 miles per day with one rest day in seven, that division covers roughly 72 miles per week, or 7 hexes. Round to 6 or 8 depending on how much command friction and supply delay you want baked into the movement allowance versus handled by other mechanics.

Then apply terrain multipliers. Dupuy's table gives you the ratios. If clear terrain costs 1 movement point, woods should cost more, roads should cost less, mountains should cost the most. The specific costs will depend on how your combat and supply systems interact with positioning. Playtest and adjust. You can check your results against the historical data. If your mechanics produce movement that matches the tables above, you are in the right range.

Keep march rates and combat advance rates separate. A unit moving through empty territory advances at march rate. A unit advancing against resistance moves at combat advance rate, which Dupuy's campaign data shows rarely exceeds fifteen to twenty-five kilometers per day even in successful offensives. Your movement system should allow high movement in open territory and produce slower net advances in contested areas through zones of control, enemy reaction, and combat results. Reduce the movement allowance to reflect supply and terrain. Let the enemy reduce the advance rate.

Sources

- Chandler, David. *The Campaigns of Napoleon*. New York: Macmillan, 1966.
- Duffy, Christopher. *The Army of Frederick the Great*. Newton Abbot: David & Charles, 1974.
- Duffy, Christopher. *The Military Experience in the Age of Reason*. London: Routledge & Kegan Paul, 1987.
- Dupuy, Trevor N. *Numbers, Predictions and War: Using History to Evaluate Combat Factors and Predict the Outcome of Battles*. Indianapolis: Bobbs-Merrill, 1979.
- Engels, Donald W. *Alexander the Great and the Logistics of the Macedonian Army*. Berkeley: University of California Press, 1978.
- Howard, Michael. *The Franco-Prussian War: The German Invasion of France, 1870–1871*. London: Rupert Hart-Davis, 1961.
- May, Timothy. *The Mongol Art of War*. Barnsley: Pen and Sword, 2007.
- Nafziger, George F. *Imperial Bayonets: Tactics of the Napoleonic Battery, Battalion and Brigade as Found in Contemporary Regulations*. London: Greenhill Books, 1996.

- Roth, Jonathan. *The Logistics of the Roman Army at War* (264 B.C. - A.D. 235). Leiden: Brill, 1999.
- Tanner, Robert G. *Stonewall in the Valley: Thomas J. "Stonewall" Jackson's Shenandoah Valley Campaign, Spring 1862*. Rev. ed. Mechanicsburg: Stackpole Books, 1996.
- Tuchman, Barbara W. *The Guns of August*. New York: Macmillan, 1962.
- U.S. Army. *FM 101-10, Staff Officers' Field Manual: Organizational, Technical, and Logistical Data*. Washington: Department of the Army, 1959.
- Van Creveld, Martin. *Supplying War: Logistics from Wallenstein to Patton*. Cambridge: Cambridge University Press, 1977.
- Vegetius. *De Re Militari*. c. 390 AD.
- Zetterling, Niklas. *Normandy 1944: German Military Organization, Combat Power and Organizational Effectiveness*. Winnipeg: J.J. Fedorowicz, 2000.

Appendix E: The Workshops

The individual contributions of key designers from SPI, GDW, and their orbits to the wargaming hobby. Not company histories but a focused look at specific people who created mechanics, standards, and design philosophies that became industry foundations and persist today.

The goal across all entries is to trace specific, concrete contributions: this person created or refined this mechanic or approach, and here is where you can see it in games today.

James F. Dunnigan — Founded SPI (Simulations Publications, Inc.) in 1969 and built it into the dominant wargame publisher of the 1970s. More than any other individual, Dunnigan industrialized wargame design: he developed systematic, repeatable methods for translating historical research into playable simulations and trained an entire generation of designers in those methods. Designed or co-designed over a hundred games, including *PanzerBlitz*, *Kampfpanzer*, *Jutland*, *War in the East*, and the ground-breaking *Modern Battles* quad series. Created *Strategy & Tactics* magazine as a vehicle for publishing wargames on a regular schedule, establishing the magazine game as a format. Authored *The Complete Wargames Handbook* (later revised as *Wargames Handbook*), which remains the closest thing the hobby has to a foundational textbook on design methodology. Also pioneered the use of commercial wargame design techniques for professional military analysis and consulting, bridging the hobby and defense communities. SPI's case-format rules structure (discussed in Chapter 17), its commitment to historical research standards, and its philosophy that games should be designed around a central historical question all trace back to Dunnigan's influence. The hobby as it exists today, its conventions, its vocabulary, its expectations for what a wargame should be, was shaped more by Dunnigan and the designers he mentored at SPI than by any other single source.

Al Nofi — SPI's Research Director and Associate Editor of *Strategy & Tactics* from 1969 to 1982, over a decade at the center of the hobby's most influential publisher. While Dunnigan was the entrepreneurial force and Simonsen the visual designer, Nofi was the intellectual backbone. He brought genuine academic credentials (a Ph.D. in Military History from CUNY) and a conviction that wargame design is itself a form of historical inquiry. His own words capture the philosophy: "My primary interest in wargames is how they can help understand events." His methodology was meticulous. For the *Imperium Romanum* series, he built a six-foot by two-foot flow chart tracking Roman legion movements during the civil wars following Caesar's crossing of the Rubicon, later converted to a 500 MB computerized file. He researched recruiting pools across different periods, accounting for population demographic shifts caused by epidemics beginning in the Second Century. Designed or co-designed *Renaissance of Infantry*, *Centurion*, *Caporetto*, *Imperium Romanum* (which he refined across three editions spanning 1979 to 2018), *The Great War*, *Salerno*, and numerous other titles, often collaborating with Dunnigan. Published over thirty books of military history independently, including *The Gettysburg Campaign*, *The Waterloo Campaign*, and *To Train the Fleet for War* (which won the John Lyman Book Award in Navy History in 2011). Later became a research analyst at the Center for Naval Analyses and served as field representative to the Chief of Naval Operations Strategic Studies Group. Nofi helped establish the expectation, now taken for granted, that a serious wargame should be backed by serious research. His thirty books of military history confirm that the scholarship came first and the game design followed from it.

John Young (1947–1978) — SPI's accountant by job title, one of its most prolific designers by output. Young worked at SPI's offices in Bay Ridge, Brooklyn, designing games in what was nominally his spare time. He was fired in 1974 when someone had to take the fall for SPI's chronic inability to turn a profit. He died in May 1978 at thirty years old. In the time he had, he designed or co-designed an extraordinary number of games across every scale and period: *Armageddon*, *Austerlitz*, *Borodino*, *Dreadnought*, *Fall of Rome*, *La Grande Armee*, *Lee Moves North*, *Musket & Pike*, *Phalanx*, *Red Star/White Star*, *Rifle & Saber*, *Search & Destroy*, and others. *Strategy I* (1971, co-designed with Dunnigan, Patrick, and Simonsen) was an enormously ambitious modular system containing 36 modules simulating battles from 350 BC to 1984 with over 1,000 counters. Young himself admitted it was "likened to an unwanted orphan child with leprosy" because of its scope. *PRESTAGS* (1975) took five separate ancient and medieval tactical games SPI had published between 1970 and 1972 and unified them under a common ruleset with cross-compatible units. The breakthrough was that strength values derived from the same mathematical base, enabling cross-period scenarios. *The Marne* (1972), a divisional-level simulation of the August–September 1914 campaign, was praised for clean rules and exciting play, and inspired the author's entire World Undone series. Young explicitly rejected over-complexity, preferring games abstract enough to play fast so players could reach strategic decisions. He favored what he called "viciously capricious" mechanics to introduce uncertainty rather than letting calculating players exploit deterministic sequential movement. He is also credited with coining the use of “grognard” to describe veteran wargamers, comparing the grumbling of SPI’s players to Napoleon’s legendary Grognards. The term entered *Strategy & Tactics* from his usage around the office and became permanent wargaming vocabulary.

Eric Smith — Designer at SPI from 1979 to 1982, then co-founder and senior designer at Victory Games when Avalon Hill recruited ex-SPI staff after TSR’s acquisition. At SPI he designed *Pea Ridge*: The Gettysburg of the West (1980, nominated for best wargame of the year) and *The Alamo*: Victory in Death (1981, also nominated, still in print). At Victory Games he produced three designs that each left a mark on the hobby. The Civil War 1861–1865 (1983) simulated the entire war at strategic scale with an innovative Command Point system where players prioritize theaters and dice determine distribution. Mark Simonitch’s 2015 The US Civil War draws from Smith’s design, a measure of its lasting influence. Ambush! (1983, co-designed with John Butterfield) was a solitaire squad-level WWII game where enemy forces had no markers on the board until they appeared, controlled by cross-referenced tables and a paragraph book. It won the 1984 Origins Award and became one of the most successful purpose-designed solitaire wargames ever published. Smith’s most enduring contribution came in *Panzer Command* (1984), a tactical WWII game where he originated the chit-pull activation system. Each player randomly draws a chit to determine which formation activates next, replacing the traditional I-Go-You-Go structure with unpredictable turn sequencing that better simulates fog of war and command friction. The mechanic has since appeared in over 700 published games, making it one of the most widely adopted innovations in wargame design history. Later founded Shenandoah Studio, which brought wargames to iPad, and continues designing with Compass Games (the *Battle Hymn* series).

Kevin Zucker — Operational-level design philosophy and the Napoleonic systems that set the template for an entire generation of operational games. The author's mentor and a recurring reference throughout this book. Zucker worked at SPI in the early 1970s, rising to Production Manager and overseeing 24 issues of *Strategy & Tactics* (plus enclosed games) and 48 boxed games over two years. After leaving SPI in 1976, he founded Operational Studies Group (OSG) to publish *Napoleon at Bay* (1978), a ziplock bag game simulating Napoleon's 1814 campaign. *Napoleon at Leipzig* (1979) won the Charles S. Roberts Award. These games established Zucker's design signature: operational warfare where command control, administration, and supply force players to face conflicting imperatives between maneuvering freely and conserving their army. His concept of "strategic consumption," where great distances absorb manpower through an Attrition Table and lengthening Lines of Communication, modeled logistics as a playable constraint rather than bookkeeping, years before other designers attempted anything similar. The Library of Napoleonic Battles (TLNB) is his life's work: an ambitious project to present 88 battles of the Napoleonic era across a unified system, with 82 published so far through a GMT/OSG partnership. Each box brings together four or five battles of one campaign with detailed historical information. The system has evolved across editions, adding cavalry charges, cards for movement and higher command decisions, and refinements drawn from decades of play. Zucker also published *Wargame Design Magazine* through OSG (38 issues, released as free PDFs), sharing design knowledge with the broader community. Inducted into the Clausewitz Award Hall of Fame in 2003. His four-decade commitment to refining a single operational system is unusual in the hobby and reflects a design philosophy that a good system rewards sustained investment rather than constant reinvention.

Mark Herman — Trained at SPI under Dunnigan and Simonsen starting in 1976, then invented the card-driven game format that reshaped the hobby. His early SPI designs included *The Battle for Jerusalem*, *October War*, *The Next War*, and others across modern and ancient subjects. After TSR acquired SPI, he became Executive Vice President at Victory Games, where *Gulf Strike* (1983, used in US analytic efforts at the start of the Gulf War in 1990) and *Pacific War* (1985) established him as a designer of serious operational simulations. Then came *We the People* (1993, Avalon Hill), the first card-driven wargame. Players alternate playing strategy cards from a 64-card deck, each usable as either a historical event or for operations points to activate leaders, place political control markers, or bring reinforcements. The game replaced hex maps with point-to-point movement and CRTs with card-based resolution. Herman deliberately left the CDG mechanic in the public domain, opening it for other designers rather than protecting it. The result was a new genre: *Hannibal*, *Paths of Glory*, *Twilight Struggle*, and dozens more trace directly to his decision. His subsequent designs continued pushing the format. *Empire of the Sun* (2005, GMT, CSR Award winner) advanced the CDG engine further. *Churchill* (2015, GMT, Golden Geek Award) modeled Big Three negotiations. *Fort Sumter* compressed the CDG format into a short game about the Secession Crisis. *Pacific War* returned in a massive 2022 GMT edition. Beyond design, Herman spent decades in professional wargaming, working at Booz Allen Hamilton leading wargaming and simulation work for the US military, and teaching at Georgetown University and Columbia University. Inducted into the Charles S. Roberts Hall of Fame in 1991. Published *Wargames According to Mark* (GMT, 2024), 200 pages of design writing plus 60 pages of CDG development notes. He trained in SPI's hex-and-counter tradition and then built the format that replaced it.

Richard Berg (1943–2019) — Criminal defense attorney by profession, one of the most prolific and controversial designers in the hobby's history, with 195 credited designs. Trained in Asian History at Union College and held a JD from Brooklyn Law School. Served in the US Army from 1967–1969 as musical director of the Army Theater in Frankfurt. His first published game, *Hooker and Lee* (1975), was part of SPI's *Blue and Gray* II. *Terrible Swift Sword* (SPI, 1976) won the Charles S. Roberts Award for Best Tactical Game and set the standard for "monster" grand-tactical simulations of single battles. *The Campaign for North Africa* (SPI, 1978) was his most infamous design: estimated at 1,500 hours to complete, famous for requiring Italian troops to receive extra water rations to prepare pasta. Berg described it as "wretched excess" by deliberate intent, and it remains the ultimate example of simulation taken to its logical extreme. *The Great Battles of History* series (GMT, starting 1991) was his signature achievement: ancient and medieval tactical combat across fifteen volumes including *The Great Battles of Alexander*, *SPQR* (co-designed with Mark Herman, CSR Award and Origins Award winner), *Caesar*, and *Cataphract*. The system featured leader-driven activation, shock combat, and unit cohesion mechanics. The Men of Iron series (GMT, 1999 onward) covered medieval European warfare with a continuation activation mechanic emphasizing momentum and counterattacks. Inducted into the Charles S. Roberts Hall of Fame in 1987 and the Academy of Adventure Gaming Hall of Fame in 1997. Won the industry award for best game design eleven times. Also published *Berg's Review of Games*, which won Best Amateur Magazine at the Origins Awards three times. Berg was brilliant, prolific, and combative, and the hobby could not ignore him on any count. He died in Charleston, South Carolina in July 2019.

David Isby — SPI's first employee, joining Poultron Press in 1970 before it became Simulations Publications. For nine years he served as copy editor of *Strategy & Tactics* and contributed to the design, development, and research of major SPI products including *War in Europe*, *War in the Pacific*, and Campaign for North Africa. His own designs focused on subjects other designers avoided. *Air War* (1977) simulated jet combat from the 1950s through the 1980s and won a Game Designers' Guild award for showing advantages of position without requiring plotting, a significant advance over prior air combat systems. *To the Green Fields Beyond* (1978), covering the Battle of Cambrai in 1917, won a Charles S. Roberts Award. Inducted into the Charles S. Roberts Hall of Fame in 1979. What set Isby apart from his SPI contemporaries was his parallel career as a Washington, D.C., attorney and national security consultant. He testified before the House Armed Services Committee on tactical air power. His books on the Soviet military and Afghanistan were written while those subjects were current, not retrospective. *Weapons and Tactics of the Soviet Army* (Jane's, 1981) analyzed an army that was actively fighting in Afghanistan. His three books on the Afghan conflict covered an ongoing war. The pre-glasnost Soviet government condemned him for his writings and activities, a measure of how seriously his work was taken outside the hobby. He published or edited twenty books and over 350 articles on national security topics. Where most SPI designers simulated the past, Isby modeled the present, and his professional defense credentials gave that work weight outside the hobby.

Redmond Simonsen (1942–2005) — Invented wargame graphic design as a discipline. Held a BFA from Cooper Union. Before SPI, he designed the book jacket for *Is Paris Burning*?, created album covers for London Records, and sold photographs to TIME and Newsweek. He brought a professional designer's training into a hobby that had been getting by on crude visual standards. Art director and executive art editor at SPI from its early days through 1982, supervising the release of over 400 game titles. Coined the term "physical systems design" for the application of graphic design to board games and is credited with popularizing the term “game designer” as a defined role. Also invented the role of “game developer” as a professional responsible for turning a designer’s prototype into a camera-ready product. His innovations accumulated game by game. He evolved maps from black-and-white to two-color to full-color. He made counters professionally mounted, die-cut, and printed on both sides. He reintroduced the method of conforming rivers to hexsides (originally from Avalon Hill’s D-Day but abandoned) and it became the universal standard. John Prados recalled that “Redmond prided himself on making at least one graphical innovation each game.” His core principle: graphics serve gameplay. “The more graphic engineering the artist can build into the game equipment and rules, the easier and more enjoyable becomes the play of the game.” The map was a functional tool that communicates terrain, movement, and game state to the player. That philosophy is as much a design contribution as any mechanic in this book. He also designed games himself, most notably *StarForce: Alpha Centauri* (1974), the first mass-market science fiction board wargame and an SPI best-seller. Inducted into the Charles S. Roberts Hall of Fame in 1977. After TSR acquired SPI in 1982, he co-founded Ares Development Corporation and later worked in video game development. Died of a heart attack in 2005 at sixty-two. The Redmond A. Simonsen Memorial Award for Outstanding Presentation was created in his honor. The visual language he established at SPI, color-coded terrain on hex maps, standardized counter formats, rivers on hexsides, full-color double-sided counters, remains the foundation of wargame graphic design. Open any wargame published in the last forty years and you are looking at conventions Simonsen established.

GDW and Frank Chadwick — Game Designers’ Workshop was founded June 22, 1973, in Normal, Illinois, by Frank Chadwick, Rich Banner, Marc Miller, and Loren Wiseman. For twenty-three years the company published a new product approximately every twenty-two days, rivaling SPI's output across a wider range of formats: board wargames, miniatures rules, and RPGs. Richard Berg called Chadwick "one of, if not the, finest game designer working today," citing a body of work remarkable for its breadth. Chadwick's contributions started with Drang Nach Osten! (1973, co-designed with Paul Banner), the first commercial "monster" board wargame: 1,792 counters, five maps stretching from Warsaw to Stalingrad, 25 km per hex. It launched the *Europa* series, an ongoing project to simulate the entire Second World *War in Europe* under a unified system. Over a dozen Europa titles followed, and the series was ranked first among 202 wargames in a 1976 player survey. *The Third World War* series (1984) modeled hypothetical NATO versus Warsaw Pact conflict at theater level. *Command Decision* (1988) brought operational-scale thinking to miniatures gaming, with players commanding divisions and corps rather than squads. *Twilight*: 2000 (1984, co-designed with Dave Nilsen, Loren Wiseman, and Lester Smith) put American soldiers in a post-nuclear Europe where survival mattered more than firepower, won the H.G. Wells Award for Best Roleplaying Rules, and spawned over forty supplements. Marc Miller's *Traveller* (1977) became the defining science fiction RPG. *Space: 1889* predated the term "steampunk." Where SPI was survey-driven and magazine-centric, GDW was designer-driven with Chadwick as the creative engine across formats. Their house style favored operational and strategic scale with realistic orders of battle and logistical modeling. Inducted into the Charles S. Roberts Hall of Fame and the Origins Hall of Fame, both in 1984. GDW closed on February 29, 1996. After its closure, Chadwick became a fiction author through Baen Books. His *Gulf War Fact Book* hit the New York Times bestseller list. Free League Publishing relaunched *Twilight*: 2000 in 2021. GDW's influence on the hobby stands alongside SPI's.

Jack Radey — Self-published *Korsun Pocket* through his own People's War Games in 1979, proving that a machinist with no game industry connections could produce a professional-quality monster game (2,400 counters, artwork by Rodger B. MacGowan). Taught himself German and Russian to access primary sources that other Western designers could not read, and assembled a network of translators and archivists to get at Soviet records when most of the hobby relied uncritically on Wehrmacht memoirs. His research-driven methodology set the standard for Eastern Front historical rigor: when Russian archives were digitized decades later, he went back and corrected errors in the original *Korsun Pocket*, removing units (JS-2s, Ferdinands, T-34-85s) that turned out not to have been present at the battle. The 43-year gap between *Korsun Pocket* (1979) and *Korsun Pocket 2* (2022), during which he continued researching and publicly correcting his own work, is almost unprecedented in the hobby. His design philosophy that games should reward historically-organized play rather than counter-pushing, and his insistence that wargame designers are historians first, influenced the seriousness with which subsequent designers approached research. Also notable as the hobby's self-described "house red": an open Communist who designed games about Soviet military operations, wore a Red Army star to a professional wargaming conference at a U.S. Air Force base, and challenged speakers on the ethics of precision weapons. His games are respected for fairness and accuracy regardless of his politics, demonstrating that political conviction and historical objectivity can coexist. Co-authored *The Defense of Moscow*, 1941 (Pen & Sword Press).

Rick Barber (1954–2021) — Landscape painter by training who became the principal map artist at Clash of Arms Games, working with founder Ed Wimble starting in the early 1980s. Barber used a hand-etched terrain style that he never abandoned in favor of computer graphics, producing maps with the quality of landscape paintings applied to wargame cartography. His work on *The Emperor Returns*, *Napoleon at Leipzig*, *The Six Days of Glory*, and Summer Storm (which he also designed) established Clash of Arms' visual identity and raised expectations across the hobby for what a wargame map could look like. Where Redmond Simonsen at SPI established a clean, functional graphic language for wargames, Barber showed that maps could be beautiful objects in their own right while remaining playable. He knew the Gettysburg battlefield particularly well and rendered it multiple times for different games. A signature touch: he hid a small terrain feature labeled "Le Chat Noir" (or the local-language equivalent) on every map, a nod to his Black Cat Studio workspace. His daughter Nikki has preserved his remaining artwork through blackcatstudio.live.

Jack Greene — Founded Quarterdeck Games in 1979, building it into a publisher focused on naval simulations. His empirical approach to ship ratings, shell weight multiplied by rate of fire rather than shell size, became a foundational method for quantifying naval firepower (referenced in Chapter 13 of this book). *Ironbottom Sound* won the 1981 Charles S. Roberts Award for Best Initial Release. Other notable designs include *Destroyer Captain*, *The Royal Navy* (which introduced a staggered movement phase system for handling initiative in naval combat), and TOGO: *Dawn of the Dreadnought* covering the Russo-Japanese War. Also a published historian, co-authoring *Rommel's North Africa Campaign*, *Hitler Strikes North*, and *Ironclads at War*. Inducted into the Charles Roberts Awards Hall of Fame in 1990. Greene's work gave naval wargaming a rigor and depth that encouraged other designers to take the subject seriously.

Lee Brimmicombe-Wood — The most important air combat wargame designer working today. The *Wing Leader* series (GMT Games) introduced a side-view map perspective where the X-axis represents the raid path and the Y-axis represents altitude. This was not a visual gimmick but a genuine insight about WWII air combat, where the tactical problem for fighter controllers was managing approach axis and altitude advantage. The system operates at squadron and flight scale, modeling formation management, cohesion, and situational awareness. Downtown modeled the interaction between American strike packages and North Vietnam's integrated air defense system. Nightfighter used a blind play/umpire system for WWII night interception with asymmetric information. *Bomber Command* covered the RAF night bombing campaign. Won the James F. Dunnigan Award for Outstanding Achievement in wargame design. Also has a parallel career in video game design (Supermassive Games, Rebellion, Eidos, among others). Across his body of work, Brimmicombe-Wood has covered air warfare from every angle, strike packages, night interception, strategic bombing, tactical air combat, and proved that air warfare can sustain deep, innovative game systems rather than being bolted onto ground games as an afterthought.

Mark Simonitch — Graphic artist and designer, inducted into the Charles S. Roberts Hall of Fame in 2002 on the strength of either career independently. Created maps, counters, and art for over 100 wargames across GMT Games, *Strategy & Tactics*, Avalon Hill, and others. His maps for *Twilight Struggle*, *Paths of Glory*, *Empire of the Sun*, and *Wilderness War* established the modern visual standard for wargame cartography, doing for the GMT era what Redmond Simonsen did for SPI. As a designer, created the ZOC Bond mechanic: when two friendly units sit two hexes apart, they project a bond along the connecting hex spines that enemy units cannot enter or cross, also blocking retreats and supply lines. A defensive line does not require a unit in every hex. Two units two hexes apart seal the gap between them, forcing the defender to choose between a dense line (strong everywhere, thin reserves) and a bond line (gaps filled by bonds, freeing reserves but vulnerable if an anchor is destroyed). The ZOC Bond system drives his operational WWII series: France '40, Normandy '44, Ardennes '44, Holland '44, *Salerno* '43, Stalingrad '42, and others, all published by GMT. Also designed *Hannibal: Rome vs. Carthage* (originally Avalon Hill), one of the foundational card-driven games alongside *We the People*.

Dean Essig (1961–2024) — Founded The Gamers in 1988 and built it into the most series-driven wargame publisher in the hobby before its acquisition by Multi-Man Publishing in 2001, where he continued designing until his death from cancer at 63. Created seven major game series spanning 99+ titles: the Civil War Brigade Series, Tactical Combat Series, Operational Combat Series, Standard Combat Series, Napoleonic Brigade Series, Line of Battle, and Battalion Combat Series. His single most important contribution was the series rules concept itself. Before Essig, most wargames had bespoke rule sets and every new game meant learning from scratch. Essig organized games into series with shared core rules and game-specific "exclusive" rules (what he called the "gimmick"), which reduced learning curves, enabled efficient production, and let players build expertise in a system rather than starting fresh each time. This approach became an industry standard that publishers from GMT (COIN series) to Columbia adopted. The OCS made logistics a central, playable mechanic rather than bookkeeping. The SCS proved that simple, fast-playing hex-and-counter games could still reward skill and capture historical flavor. Fourteen Charles S. Roberts Awards across his series. He also gave the author the opportunity to design Rostov '41 for the SCS, which helped launch the author's career in the business. Also served as artist for many of his company's games and hosted Homercon, an annual gaming convention in Homer, Illinois, from 1990 to 2012. After his death, Carl Fung assumed management of The Gamers' game lines at MMP.

Mark McLaughlin — Designing consistently good games since the Avalon Hill era, with a catalog of 27+ published designs spanning five decades. *War and Peace* (1980), his Napoleonic Wars game, sold approximately 50,000 copies and remains in print in its sixth edition. A journalist and professional writer by trade (managing editor of *The Wargamer*, AP copy boy, novelist, ghostwriter), McLaughlin pitched *War and Peace* over lunch with Avalon Hill's Tom Shaw and walked out with a signed contract. His design philosophy prioritizes accessibility and sweep: lighter, faster games that capture the grand strategic picture without drowning in tactical detail. *The Napoleonic Wars* and *Wellington* (GMT) brought card-driven grand strategy to the Napoleonic period. *Hitler's Reich* (GMT, 2017) introduced the Card Conquest System, replacing traditional CRTs and counters with card-based conflict resolution, playable in two hours. *Ancient Civilizations of the Inner Sea* (with Christopher Vorder Bruegge) applied similar accessibility principles to the ancient world. *Rebel Raiders on the High Seas* covered Civil War naval warfare at strategic scale. McLaughlin represents the tradition of the accessible wargame, consistently pushing to make complex conflicts playable in shorter timeframes without sacrificing strategic depth, while his background as a professional writer shows in his games' emphasis on narrative and sweep over mechanical crunch.

Amabel Holland — Hollandspiele co-founder. Refuses formulaic design: each game gets mechanics built from scratch for its specific historical situation rather than recycling existing systems. *Table Battles* created a new format (dice-allocation wargame playable in 30 minutes on a small table). Uses games as expressive medium for political and personal themes (*This Guilty Land* on American slavery, *Doubt Is Our Product* on corporate disinformation). Author of *Cardboard Ghosts*: Using Physical Games to Model and Critique Systems (CRC Press). Proves wargames can be art and personal expression while remaining historically grounded. The New Yorker called her "one of today's most innovative game designers." The author submitted designs to Hollandspiele early on and was rejected, but the experience was positive and instructive.

Ty Bomba — Over 160 published designs, one of the two or three most prolific designers in history. Co-founded XTR, edited *Strategy & Tactics*, founded *Command* magazine, edited *CounterFact* (pioneering alternative-history wargaming as a serious subgenre). Eight years in military intelligence (Navy Intelligence, NSA) informed his focus on Eastern Front and Middle Eastern topics. Charles Roberts Awards Hall of Fame inductee. Bomba kept S&T and Command alive as design platforms during periods when the hobby was contracting, and explored more historical topics than almost any other designer. No one has designed more magazine wargames.

Joe Miranda — 185+ published designs, rivaling Bomba for output. Chief wargame designer and editor at Decision Games/*Strategy & Tactics*. Former U.S. Army officer who taught unconventional warfare at JFK Special Warfare Center. Pioneered the solo mini-game format at Decision Games (the *Commando* series), creating affordable solitaire entry points that kept solo wargaming alive during lean years. Designs span from ancient Rome (*Trajan*) to Vietnam (*In Country*) to science fiction (*Phobos Rising*). The workhorse of magazine wargaming: as both editor and chief designer, he ensured subscribers received a playable game every issue across every conceivable period.

Ted Raicer — *Paths of Glory* (1999) took the CDG system pioneered by Mark Herman and expanded its simulation value, applying it to WWI, a war most designers considered too static to make exciting. Before *Paths of Glory*, WWI was a dead-end subject. After it, WWI became one of the hobby's most popular periods. The "Dark" series (*Dark Valley*, *Dark Sands*, *Dark Summer*) demonstrates parallel mastery of traditional hex-and-counter operational design, proving fluency in both CDG and classic formats. *I, Napoleon* showed willingness to experiment beyond traditional formats. Nine Charles S. Roberts Awards. Published more than ten games on WWI alone, making him the hobby's foremost WWI specialist.

Volko Ruhnke — Created the COIN (COunterINsurgency) system, arguably the most important new wargame system of the twenty-first century. COIN uses a shared event card display (not individual hands) to drive initiative and faction selection, with four asymmetric factions carrying completely different capabilities, victory conditions, and strategies. Career CIA intelligence analyst. Designed COIN as an open platform: other designers have created volumes covering topics from Roman Britain to the Troubles in Northern Ireland. The COIN series brought asymmetric multiplayer wargaming to the mainstream hobby and attracted Eurogamers and non-wargamers (Cole Wehrle's Root is directly descended from COIN). Also designed Labyrinth: The War on Terror and *Wilderness War*.

Fred Serval — French designer, one of the newest voices in the hobby. *Red Flag Over Paris* (2021) adapted Mark Herman's card-driven framework for the 1871 Paris Commune with innovations including Political Dynamics, dual asymmetric victory conditions, and Momentum mechanics. Used computational statistics (R scripts testing thousands of card combinations across billions of deck configurations) to balance the game, a data-driven approach to development. *A Gest of Robin Hood* compressed the COIN framework into a two-player game playable in under an hour. Design philosophy prioritizes accessibility without sacrificing historical depth. Hosts the Homo Ludens YouTube channel. The author worked with Serval as developer on With The Hammer (discussed in Chapter 16), and his contributions reshaped that design.

Morgane Gouyon-Rety — French engineer specializing in the ancient and Hellenistic world. *Pendragon* (COIN Vol. VIII) adapted the COIN system for late Roman/post-Roman Britain, modeling civilizational collapse rather than insurgency. *Hubris: Twilight of the Hellenistic World* covers the twilight of the Hellenistic kingdoms (220–165 BCE) using an entirely original system, the "Hubris engine," built around leader-centric gameplay where individual rulers and their ambitions drive events, mirroring ancient historiography. Models Rome as a non-player alien power: threateningly unpredictable, influenceable but never controllable. Design philosophy: complexity should reside in mastering gameplay, not mastering the rules. One of very few designers focusing on the ancient world with the depth and rigor it deserves.

Hermann Luttmann — The hobby's most effective bridge-builder between hardcore wargaming and general gaming audiences. Created the Blind Swords system (chit-pull activation mixing event and unit chits to model fog of war, friction, and fortune) and its evolution, the Black Swan system (card-draw based for larger games like *A Most Fearful Sacrifice*). *Dawn of the Zeds*, a cooperative zombie defense game built on wargame design principles, sold more copies than all his wargame sales combined, three times over. Uses variable die types for combat (better units roll D12, weaker roll D6) and push-your-luck mechanics (*In Magnificent Style* for Pickett's Charge). Focuses on playability and accessibility without dumbing down historical content. When someone asks what wargame to recommend to a first-timer, a Luttmann design is usually the right answer.

Darin Leviloff — Created the State of Siege series at Victory Point Games, one of the most successful and enduring solitaire wargame systems. The system places the player at the center of converging enemy advances, using a track-based structure where threats move inward along multiple axes and the player allocates limited actions to hold them off. Simple enough to learn in minutes, deep enough to model conflicts from the fall of the Roman Republic

(Legions of Darkness) to the Israeli War of Independence (Israeli Independence). The State of Siege format became a template that other designers built on, helping establish Victory Point Games as the home of accessible solo wargaming. A staple of solitaire play and a reliable entry point for new wargamers.

Brian Train — One of the hobby's most important designers of asymmetric and insurgency warfare games, working in this space long before the COIN series brought it mainstream attention. Designs include A *Distant Plain* (co-designed with Volko Ruhnke for the COIN series), *Colonial Twilight* (the first two-player COIN volume), and dozens of smaller games on guerrilla warfare, counterinsurgency, and irregular conflict published through various small publishers. A professional urban planner by training, which informs his systems-level thinking about how populations, infrastructure, and armed groups interact. Has also done serious work in professional military wargaming and education. Train's games model the political and social dimensions of conflict, hearts and minds, population control, legitimacy, with a rigor that pure military simulations often miss. Train has done more to model the political dimensions of conflict than any other designer working today.

Chad Jensen (1967–2019) — Designer of *Combat Commander* (GMT Games), arguably the most significant tactical WWII game since ASL. *Combat Commander* uses no dice. A purely card-driven resolution system where each Fate card carries a random number replacing die rolls while simultaneously serving as orders, actions, and events. No fixed sequence of play: activations are fluid and reactive, with either player able to play Actions at almost any time. Nationality-specific Fate decks model doctrinal strengths and weaknesses through card distribution rather than special rules. Thirty-five different Events trigger at random intervals, injecting narrative chaos that generates emergent war stories.

Players consistently described their CC sessions as producing memorable narratives in a way few other wargames achieved. A random scenario generator provided near-infinite replayability. The net effect was a game that felt chaotic and narrative like real small-unit combat but was far simpler than ASL, making it accessible to a wider audience while satisfying grognards. Jensen was also praised for writing some of the clearest, most readable rulebooks in the hobby. Also designed *Dominant Species* (GMT's third best-selling game of all time, a competitive Eurogame about animal survival during an Ice Age) and *Fighting Formations*. Worked with GMT Games throughout his career. Passed away from pancreatic cancer at 52. The Chad Jensen Memorial Breakthrough Designer Award was established in his honor in 2023 to recognize outstanding new designers entering the wargaming field.

David Thompson — Defense analyst by day, arguably the most commercially successful designer working in the space between Eurogames and wargames. Came to wargame design from Eurogames rather than the other way around, which means his instinct is to ask "how do I make this accessible?" before "how do I simulate this?" The Undaunted series (co-designed with Trevor Benjamin, Osprey Games) uses deck-building as the core engine driving tactical WWII combat. Drawing cards determines what you can do each turn, and casualties remove cards from your deck, simulating attrition through a Eurogame mechanic. Undaunted: Normandy, North Africa, Stalingrad, Battle of Britain, and subsequent volumes have been commercially successful enough to reach mainstream game stores, not just hobby shops. *War Chest* (AEG, also with Benjamin) replaced deck-building with bag-building for abstract tactical combat. *General Orders* fuses worker-placement with hex-map wargaming. *Pavlov's House* (DVG) introduced a triptych battlespace structure for solitaire play: three interconnected boards representing tactical, operational, and strategic scales of the same battle. Thompson's consistent output demonstrates that historical conflict games built on modern Eurogame production values and accessible mechanics can sell at scale. Multiple Charles S. Roberts Awards and Golden Geek Awards. His games function as gateway wargames that teach genuine tactical concepts (suppression, fog of war, combined arms) to audiences who would never touch a hex-and-counter game.

Ryan Heilman — Designer (with Dave Shaw) of Brave Little Belgium, White Eagle Defiant, and Lucky Little Luxembourg (CSR Award winner), all published by Hollandspiele. Also a developer, graphic designer, and manual formatter who has worked behind the scenes on numerous wargames, including formatting most of the author's own rulebooks for Conflict Simulations LLC. A former history and technology teacher whose WWI interest traces back to purchasing Avalon Hill's Guns of August for a school report. Now serves on the Charles S. Roberts Awards Board of Governors and presented the awards at Origins 2025. His designs prioritize asymmetric experiences and storytelling within accessible, light-to-medium complexity frameworks. Co-founded Wharf Rat Games in 2024 with Wes Crawford.

Adam Starkweather — System architect and advocate for modernization in wargame design. Created the Grand Tactical Series at Multi-Man Publishing (The Devil's Cauldron, which reached #1 wargame on BGG), then moved to Compass Games in 2016 where he built the Company Scale System (Saipan, Fulda Gap, The Little Land), the Operational Scale System (Vietnam: Rumor of War, Test of Faith), and the Doomsday Project series covering hypothetical Cold War conflicts with nuclear and chemical escalation modeled as game mechanics. Also developed nearly all titles in MMP's International Game Series, translating Japanese wargame designs (A Victory Lost, Fire in the Sky, Warriors of God) for Western audiences and broadening the hobby's design influences. His design philosophy is purpose-driven: every rule must serve a clear function, and making a game streamlined requires more effort than making it complex. Vocal about the need for modernized graphic design and accessibility in wargaming. Conducts 50–75+ playtests per game using VASSAL. One of the more prolific active designers with four distinct systems across multiple scales, each tailored to its subject rather than forcing a single framework onto different conflicts.

Wes Crawford — Designer of The Pursuit of John Wilkes Booth (with Heilman) and Engine Thieves: The Great Locomotive Chase (Compass Games), both games focused on small-unit raids and manhunts, an underdeveloped area of Civil War gaming. A professional drummer for over a decade (touring with entertainer Jane L. Powell, winning Entertainer of the Year and Jazz Act of the Year in 1990) before spending 28 years teaching at Goucher College, Crawford returned to game design after rediscovering his Avalon Hill collection in a basement tub around 2018. His designs emphasize deep primary-source research and meaningful player agency in situations where historical individuals faced compressed, high-stakes decisions. Co-founded Wharf Rat Games with Heilman in Baltimore, with a business model focused on quality over quantity: one game at a time, light-to-medium weight, solitaire-capable, with faster time-to-market than traditional wargame publishers. Their debut title is A Forlorn Hope by Hermann Luttmann, a WWI push-your-luck game about crossing No Man's Land.

Appendix F: Glossary

Definitions of wargame-specific terms used throughout this book. Intended for readers coming from board gaming, digital gaming, or no wargaming background at all. Entries are kept to one or two sentences; the chapter where each concept is explained in full is noted where applicable.

AAC (Advance After Combat) — Movement into a hex vacated by a retreating or eliminated defender. Most games require at least one attacking infantry unit to advance; artillery and support units may be restricted.

Activation — Selecting a unit or formation to take actions (move, attack, rally) during a turn. Systems vary: some use sequential player turns, others use chit-pull or card-driven activation to randomize who acts when. (Ch. 10)

Area Control — A map system using irregular regions instead of hexes, where units occupy and contest named areas connected by movement paths. Used in games like *Hannibal: Rome vs. Carthage* and the COIN series. (Ch. 5, Ch. 21)

Armor / Armour — Units representing tanks or armored vehicles. Typically faster than infantry, with higher attack values, and eligible for exploitation movement after successful combat.

Artillery — Units representing guns or batteries that fire at range during a bombardment phase. Capabilities and restrictions vary by era and system. (Ch. 6, Ch. 9)

Attrition — Losses suffered by units unable to trace supply or as a result of prolonged operations. Some games use attrition tables; others apply automatic step losses to unsupplied units.

Automatic Victory — A condition that ends the game immediately when triggered, such as capturing a key objective before a certain turn. Distinguished from sudden death, which ends the game when one player's position becomes irretrievable. (Ch. 12)

Battle Formation — A unit state with reduced movement but full combat capability. The opposite of March Formation. Units entering an enemy Zone of Influence flip from March to Battle Formation. (Ch. 8)

Blind Playtest — A playtest where participants play using only the written rules with no guidance from the designer. The definitive test of rules clarity. (Ch. 16)

Bombardment — Ranged artillery fire against enemy hexes during a dedicated phase, separate from regular combat. Results range from disruption to step losses. Can "soak off" an adjacent enemy hex, relieving the obligation to attack it in the combat phase.

C2 (Command and Control) — The system by which leaders direct subordinate units. In game terms, C2 models whether units can act freely (in command) or face restrictions (out of command). Implemented through command points, command range, activation rolls, or leader counters. (Ch. 11)

Card-Driven Game (CDG) — A game format where players use strategy cards, each playable as either a historical event or for operations points, to drive movement and combat. Invented by Mark Herman with *We the People* (1993). (Ch. 14)

Cavalry — Mounted units with higher movement allowance than infantry. Capabilities vary by era: charging, screening, and reconnaissance in horse-era games; typically absent or replaced by mechanized units in modern settings.

Cavalry Charge — A special attack where cavalry units attempt to overrun enemy positions. Implementation varies, but most systems resolve charges on a separate table with high-risk, high-reward outcomes. (Ch. 8)

Chit-Pull — A randomization mechanic where counters are drawn from an opaque cup to determine activation order. Simulates the uncertainty of who moves when. Originated by Eric Smith in *Panzer Command* (1984) and adopted in over 700 published games. (Ch. 10)

COIN (COunterINsurgency) — A game system created by Volko Ruhnke using a shared event card display and four asymmetric factions with different capabilities, victory conditions, and strategies. Has become a genre, with volumes designed by multiple designers. (Ch. 14)

Column Shift — Moving left or right on a Combat Results Table to reflect advantageous or disadvantageous conditions (terrain, combined arms, surprise). A shift left favors the defender; a shift right favors the attacker. (Ch. 9)

Combat Differential — A combat resolution method where the numerical difference between attacker and defender strength determines the outcome, rather than an odds ratio. (Ch. 9)

Combat Phase — The phase of a turn where attacks between adjacent opposing forces are resolved. Games vary on whether combat is mandatory for all adjacent units or voluntary. (Ch. 10)

Combat Results Table (CRT) — A table cross-referencing combat odds or strength with a die roll to produce results: retreats, step losses, exchanges, eliminations, or no effect. The mechanical heart of most hex-and-counter wargames. (Ch. 9)

Combat Strength (CS) — The numerical value representing a unit's fighting power. May be a single number (for both attack and defense) or split into separate attack and defense values.

Combined Arms — A combat bonus granted when an attack includes multiple unit types (infantry, armor, and artillery together). Represents the historical advantage of coordinating different arms.

Command Point (CP) — A value on commander counters representing how many subordinate units or formations can be placed in command during a command phase. (Ch. 11)

Command Range (CR) — The maximum distance in hexes over which a leader can exercise command over subordinate units. Enemy units and their ZOCs may block or extend the path. (Ch. 11)

Counter — A cardboard playing piece representing a military unit, leader, or game marker. Standard size is 1/2 inch or 5/8 inch square. (Ch. 6)

Counter Manifest — The complete list of every counter in a game and what is printed on each one. An essential production document. (Ch. 6)

CRT — See Combat Results Table.

Demoralization (DM) — A negative status from combat losses or failed morale checks. Demoralized units typically fight at half strength, lose their ZOC, and cannot attack. Requires spending time or resources to rally. (Ch. 11)

Depleted — A unit state representing reduced effectiveness after absorbing losses, but not yet reduced to its back side. Used in some systems as an intermediate state between full strength and step loss.

Depot — A supply unit that extends supply range from a source to combat units. Can be captured or destroyed by enemy forces. (Ch. 11)

Die Roll Modifier (DRM) — A positive or negative number added to a die roll result. The most common type of modifier in wargame combat systems. (Ch. 9)

Disrupted (DR) — A temporary negative status from bombardment or other effects. Typically halves combat strength and removes ZOC projection. Recovers during the Rally Phase.

Division (XX) — A military formation indicated by the NATO symbol "XX" on a counter. The standard operational unit in many games, containing brigades and regiments.

DRM — See Die Roll Modifier.

Eliminated — A unit removed from play by combat results or attrition. May be permanent (removed from the game) or temporary (placed in a recovery box with a chance to return).

Engagement — Combat between adjacent opposing units. In some systems, units that begin a phase in an enemy ZOC are "engaged" and cannot move except by retreat or special disengagement rules.

Entry Hex — The specific hex where reinforcements enter the map. If blocked by enemy units, reinforcements may be delayed or rerouted.

Exchange (EX) — A combat result where the defender is eliminated and the attacker must also eliminate units equal to a portion (usually half) of the defender's lost strength.

Exclusive Rules — Game-specific rules that supplement and may override a series' core rules. Cover setup, terrain, special mechanics, and victory conditions unique to each title. The term originates with Dean Essig's series system at The Gamers. (Ch. 17)

Exploitation — Additional movement granted to attacking units (usually armor or mechanized) after winning combat. Represents the historical advantage of mobile forces pressing a breakthrough. (Ch. 7)

EZOC (Enemy Zone of Control) — The ZOC projected by enemy units. Entering an EZOC typically stops movement, may require combat, and can block supply lines. (Ch. 7)

Facing — The direction a unit points toward, determining its front, flank, and rear hexes. Attacks from flank or rear provide combat bonuses. Used in tactical and grand-tactical games. (Ch. 8)

Flanking — Attacking from multiple directions, typically requiring units on opposite sides of a defender or in non-adjacent hexes around the combat hex. Provides significant combat bonuses. (Ch. 8)

Fog of War — The principle that players cannot inspect enemy units beyond what their forces can observe. Implemented through hidden movement, dummy units, or concealed unit strengths. (Ch. 11)

Forced March — Spending additional movement points beyond a unit's normal allowance, risking straggler losses. A die roll determines whether the unit suffers attrition for the extra movement.

Formation — (1) A military organizational grouping (corps, division, brigade). (2) A unit's physical posture (March Formation vs. Battle Formation). Context determines which meaning applies. (Ch. 8)

Fortress — A defensive position that multiplies the combat value of occupying units and requires siege operations to capture. (Ch. 11)

Grand Tactical — A scale covering a single battle at brigade or regiment level, with hexes of 500 meters to 1 kilometer and turns of roughly 1 hour. (Ch. 4)

Grognard — A veteran wargamer. From the French word for "grumbler," originally applied to Napoleon's Old Guard. Attributed to John Young at SPI, who compared the grumbling of SPI's customers to Napoleon's soldiers. The term entered Strategy & Tactics from Young's usage around the office.

Hex — A hexagonal space on the game map, the basic unit of position and distance in hex-and-counter wargames. (Ch. 5)

Hexside — The border between two hexes. Rivers, ridges, slopes, and other linear terrain features are placed on hexsides rather than in hexes. (Ch. 5)

Hexspine — The vertex where three hexes meet. Some games use hexspine facing instead of hexside facing. (Ch. 8)

Hidden Movement — A rule where unit identities are concealed from the opposing player using markers, dummy counters, or separate tracking sheets. Revealed when units enter enemy line of sight. (Ch. 11)

Horse Artillery — Mobile artillery with a higher movement allowance than foot artillery. Historically, guns pulled by horse teams for rapid repositioning alongside cavalry.

HQ (Headquarters) — A counter representing a command focus point. Determines command range, where recovered units return, and may provide combat bonuses to nearby units.

I-Go-You-Go (IGOUGO) — A turn structure where one player completes all movement and combat before the other player acts. The traditional sequence; alternatives include chit-pull and card-driven activation. (Ch. 10)

In Command (IC) — A unit within its parent leader's command range that has received orders. IC units have full freedom of action, in contrast to Out of Command units. (Ch. 11)

Infantry — The basic ground combat unit representing foot soldiers. The most numerous unit type in most games, with moderate movement and combat values. (Ch. 6)

Initiative — The right to act first during a turn. Determined by a preset schedule, die roll, card play, or other mechanic depending on the system. (Ch. 10)

Initiative Rating (IR) — A value on unit or officer counters representing ability to act independently when out of command. Used for activation checks.

Interdiction — An action that disrupts enemy movement or supply lines. In modern-era games, often carried out by air units; in other settings, may be achieved by cavalry or partisan forces.

Line of Communication (LOC) — A path of hexes from a unit to a supply source, free of enemy units and enemy ZOCs (unless occupied by friendly units). Required for units to fight at full effectiveness. Also called Line of Supply (LS) in some systems. (Ch. 11)

Line of Sight (LOS) — A clear path between an artillery unit and its target, unblocked by terrain or intervening units. Required for bombardment. Range and blocking criteria vary by game.

Loss Points (LP) — Strength points or resources lost in combat. Must be absorbed through step reductions, retreats, eliminations, or other means.

Magazine Game — A wargame published as part of a periodical (Strategy & Tactics, Command, CounterFact). Format established by Dunnigan at SPI. Typically a single map, one to two countersheets, and a rulebook included with the magazine.

March Formation — A unit state with higher movement allowance but vulnerability to combat. Must flip to Battle Formation when entering an enemy ZOC or ZOI. (Ch. 8)

Meeting Engagement (ME) — Combat initiated during movement when a unit enters a hex adjacent to or occupied by the enemy. Typically resolved with penalties for the attacker compared to a prepared assault.

Monster Game — A wargame of exceptional physical size, typically with multiple maps and thousands of counters. *Drang Nach Osten!* (GDW, 1973) is considered the first commercial monster game.

Morale Check (MC) — A die roll to determine if a unit maintains cohesion after taking casualties or facing other stresses. Failure results in disruption, rout, or elimination.

Morale Rating (MR) — A value on unit counters representing cohesion and discipline. Higher values pass morale checks more often.

Movement Allowance (MA) — The total number of movement points a unit can spend per activation or turn. Printed on the counter. (Ch. 7)

Movement Points (MP) — The currency spent to move from hex to hex. Each terrain type has a cost in MP. A unit stops when it has spent its MA or enters an EZOC. (Ch. 7)

NATO Symbol — Standardized military map symbols used on counters to indicate unit type: rectangle (infantry), diagonal line (cavalry), dot (artillery), oval (mechanized), among others. (Ch. 6)

No Effect (NE) — A combat result where nothing happens. Neither side takes losses or retreats.

OOB (Order of Battle) — The organizational structure and composition of military forces in a game. Determines which units exist, their strengths, and which formations they belong to. Building the OOB is one of the core design tasks. (Ch. 6)

OOC (Out of Command) — A unit outside its parent leader's command range. OOC units face restrictions: reduced movement, inability to attack, or requirement to pass initiative checks before acting. (Ch. 11)

Odds-Based CRT — A Combat Results Table organized by combat odds ratios (1:2, 1:1, 2:1, 3:1, etc.). The attacker divides total attacking strength by total defending strength to find the correct column. (Ch. 9)

Operational — A scale covering a campaign with division-sized units, hexes of 4 to 25 miles, and turns of days to weeks. (Ch. 4)

Out of Supply (OOS) — A unit that cannot trace a valid line of communication to a supply source. Suffers penalties: halved combat strength, inability to attack, risk of attrition or surrender. (Ch. 11)

Overrun — A combat result where the attacker moves into the defender's hex, eliminating or displacing the defender. Sometimes a separate mechanic from regular combat, initiated during movement.

Permanent Elimination — Removal from the game with no possibility of recovery. Distinguished from temporary elimination, where units may return through reinforcement or recovery mechanics.

Phasing Player — The player whose turn is currently active, as opposed to the non-phasing player.

Player Aid Card (PAC) — A reference card included with a game containing the CRT, Terrain Effects Chart, Sequence of Play, and other frequently referenced information. Designed to minimize rulebook lookups during play. (Ch. 17)

Point-to-Point (PtP) — A map system using named locations connected by lines instead of a hex grid. Units move along connections. Used in CDGs and other systems where specific geographic points matter more than continuous terrain. (Ch. 5)

Prepared Assault (PA) — A deliberate attack by fresh units adjacent to the enemy, with full support from artillery and other arms. Contrasted with Meeting Engagement, which occurs during movement with less preparation.

Rainy Season / Weather — Environmental conditions affecting movement costs, combat, and air operations. May be determined historically (fixed by turn) or randomly (die roll each turn). (Ch. 11)

Rally — Recovering from negative status effects (demoralization, disruption, rout). Requires spending movement points, being in supply, or occupying specific terrain depending on the system.

Reaction — A non-phasing player's attempt to respond to enemy movement by moving forces to counter a threat. Typically requires a die roll and makes the reacting units spent whether or not they succeed. (Ch. 10)

Reconnaissance — Scouting enemy positions to reveal hidden units or gain information before committing to an attack.

Recovery — Returning eliminated units to play, usually through a die roll during an administration phase. Failed recovery rolls may result in permanent elimination.

Reduced / Step Reduction — Flipping a unit from its full-strength front side to its weaker back side, representing significant casualties. Units already on their reduced side that suffer another step loss are eliminated. (Ch. 6)

Reinforcements — Units entering the map on specified turns at designated entry hexes. The reinforcement schedule is usually printed on the scenario setup card. (Ch. 10)

Replacement Points (RP) — Points used to restore strength points lost to combat or attrition. One RP typically equals one SP restored, up to the unit's original printed value.

Retreat — Involuntary movement away from the enemy as a result of combat. Retreating units must move toward a supply source. Retreating into an EZOC typically eliminates the unit.

Road March — Moving along connected road hexes at reduced MP cost. Units in road march are spaced out (no stacking) and vulnerable to enemy contact.

Rout — A severe negative status where a unit has broken and flees the battlefield. Routed units move toward their friendly board edge, cannot attack, and may be permanently eliminated if they cannot rally.

Scale — The relationship between game elements and their real-world equivalents: hex size (meters or miles per hex), turn length (minutes, hours, or days), and unit size (company through army). Choosing scale is one of the first design decisions. (Ch. 4, Ch. 5)

Sequence of Play (SOP) — The ordered list of phases that make up each game turn, defining when movement, combat, supply, and other activities occur. The structural backbone of the game. (Ch. 10)

Series Rules — Core rules shared across multiple games in a series system. Individual games add Exclusive Rules for game-specific mechanics. Pioneered by Dean Essig at The Gamers. (Ch. 17)

Shock Combat — Close-quarters combat between units in the same hex or at adjacent range, resolved separately from regular ranged combat. Common in tactical and grand-tactical games.

Siege — The process of capturing a fortress or fortified position by maintaining forces in position over time. May involve cutting supply, bombardment, and assault.

Solitaire — Rules or modes designed for one player, using prioritized decision lists or automated systems for the non-player side. Some games are designed solitaire-first; others include solitaire variants.

SP (Strength Points) — The primary measure of a unit's combat power. Used to calculate combat odds, determine casualties, and track unit status. One SP usually represents a set number of troops depending on game scale. (Ch. 6)

Spent — A unit that has activated and cannot act again until the next turn or reset phase. Indicated by rotating the counter sideways or flipping to its back side. (Ch. 10)

Stacking — Placing multiple units in the same hex. Stacking limits vary by game (typically one to five units) and may depend on unit type, nationality, or terrain. Exceeding the limit is usually prohibited or penalized. (Ch. 7)

Static — A unit that has been activated and flipped to its back side, indicating it cannot act again this turn. Equivalent to "Spent" in other systems.

Step — One side of a two-sided counter. A full-strength unit has two steps (front and back); losing a step flips the counter to its reduced side; losing both steps eliminates the unit. One-step units go straight from full strength to eliminated. (Ch. 6)

Step Loss — See Reduced / Step Reduction.

Strategic — A scale covering an entire theater or war with army or corps-sized units, hexes of 12 or more miles, and turns of weeks to months. (Ch. 4)

Strategic Movement (SM) — A movement mode that doubles a unit's movement allowance but prohibits moving adjacent to enemy units. Represents organized rear-area movement using roads and rail. Subject to interdiction.

Supply — The logistical ability of a unit to receive ammunition, food, and reinforcements, traced via a Line of Communication to a supply source. Out-of-supply units fight poorly and may be eliminated. (Ch. 11)

Supply Source — The origin point for supply lines: a friendly map edge, a controlled city or port, or a special supply unit.

Sudden Death — A condition that ends the game when one player's position becomes irretrievable, preventing the game from dragging on after the outcome is determined. Functions as a mercy rule rather than a victory condition. Distinguished from automatic victory, which ends the game when a strategic objective is achieved. (Ch. 12)

Tactical — A scale covering a single engagement at company or battalion level, with hexes of 100 to 500 meters and turns of 30 minutes to 1 hour. (Ch. 4)

TEC (Terrain Effects Chart) — A reference chart listing movement costs and combat modifiers for each terrain type in the game. One of the essential components of any wargame. (Ch. 5)

Turn — One complete cycle of the Sequence of Play. A game consists of a fixed number of turns, each representing a set amount of time (hours, days, weeks, or months depending on scale).

Unit Quality Rating (UQR) — A letter grade (A through D, or numerical equivalent) representing a unit's training, morale, and equipment quality. Used to differentiate units beyond raw strength points. (Ch. 6)

Unit Size Symbol — Markings above the NATO symbol indicating organizational level: I (company/battery), II (battalion), III (regiment), X (brigade), XX (division), XXX (corps), XXXX (army). (Ch. 6)

VASSAL — A free, open-source game engine for building and playing online versions of board games and wargames. The standard platform for remote playtesting and online play. (Ch. 15, Ch. 16, Ch. 21)

Victory Conditions — The criteria determining who wins the game. May be territorial (control specific hexes), point-based (accumulate VP), automatic (achieve a threshold condition), or scaled (tiered results from decisive victory to marginal defeat). Designing victory conditions that produce a competitive game while reflecting history is one of the hardest design problems. (Ch. 12)

Victory Points (VP) — A scoring mechanism tracking political, military, and strategic success. Games typically use a VP track; some have automatic victory conditions when VP reaches a threshold.

ZOC (Zone of Control) — The six hexes immediately adjacent to a combat unit. Entering an enemy ZOC typically stops movement, may require combat, and can block supply lines. Some units (demoralized, depleted, reduced) do not project ZOCs. The single most important spatial mechanic in hex-and-counter wargaming. (Ch. 7)

ZOC Bond — A mechanic created by Mark Simonitch where two friendly units two hexes apart project a bond along the connecting hex spines that enemy units cannot cross. Allows a defensive line with fewer units. Used in the *Normandy* '44 / *France* '40 series. (Ch. 7)

ZOI (Zone of Influence) — A broader area of control projected by certain units, extending beyond the standard one-hex ZOC. Typically 2-3 hexes, depending on unit strength. Forces enemy units in March Formation to stop and deploy. (Ch. 7)